Bangladesh AI Policy & Innovation Lab (BAPIL)

TECHNICAL BLUEPRINT

From Framework to Implementation: Institutional Design, Reference Architecture, Governance, Delivery Plan and Strategic Roadmap to 2041 for an AI Policy-Intelligence Capability in Bangladesh

Version 1.0

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This Technical Blueprint turns the BAPIL White Paper (Version 1.0) into a buildable plan. It specifies the institution, the people and skills, the reference architecture, the data and AI design, the governance and assurance system, a tested reference implementation on GitHub, a 36-month delivery plan, a strategic roadmap for 2026 to 2041 and beyond, and the decisions required from the relevant authorities. It is written for ministries, government agencies, technical teams, universities, civil society and development partners.

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Bangladesh AI Policy & Innovation Lab (BAPIL): Technical Blueprint

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This Blueprint is an independent research and policy-design document prepared by the author in an individual capacity. It does not represent the official position of the Government of Bangladesh, any ministry, agency, university or other organisation. Institutions, laws and programmes are named only to describe the context in which BAPIL would operate; naming them does not imply endorsement, partnership or agreement. Legal descriptions are summaries for planning purposes and are not legal advice. Cost, staffing and long-term targets are indicative planning assumptions to be validated by a feasibility study and revised at each strategic review.

How to Read This Blueprint

The Blueprint is organised in five parts. Readers with limited time can use the following routes:

- **Decision-makers:** the Overview at a Glance, Chapter 16 (strategic roadmap to 2041), Chapter 21 (decisions) and the one-page Decision Memo published with this Blueprint.
- **Programme managers:** add Chapters 3, 5, 6, 12, 15, 17, 18 and 19.
- **Technical teams:** Part III (Chapters 7 to 11) with Annexes A, C, K and M.
- **Legal, governance and ethics specialists:** Chapters 8, 13 and 14 with Annexes B, E, F, H, I and J.
- **Researchers and universities:** Chapters 9, 11 and 20 with Annex K.
- **Development partners:** Chapters 2, 16, 17 and 18.

The words **must**, **should** and **may** carry their usual normative meaning: a requirement marked *must* is a condition of the design, *should* is a strong recommendation, and *may* is optional.

Overview at a Glance

The problem. Bangladesh's laws, gazettes, budgets, plans and statistics are published across many websites and formats, often as scanned Bangla PDFs. Officials and researchers spend a large share of their time finding, checking and citing these sources. At the same time, the country is putting in place a new legal framework for data and a national AI policy, but has no operational method for governing AI systems in the public sector and no open Bangla benchmark for policy search.

The proposal. The Bangladesh AI Policy & Innovation Lab (BAPIL) is a university-anchored lab under a Memorandum of Understanding (MoU) with the ICT Division. It builds open, bilingual tools that help officials find and cite official evidence, develops and publishes the methods needed to govern public-sector AI safely, and trains people to use both.

What is built first. In its first year BAPIL delivers a bilingual policy search tool whose answers always cite official sources or abstain (pilot P1), five years of budget allocations as open data (P2), the open BanglaPolicyQA benchmark (P4), a charter with an independent Ethics and Risk Review Committee, a public transparency register of AI systems, and training for 100 officials.

How risk is controlled. Public data first; advice, not decisions; no surveillance. Every system is placed in a risk tier (T0 prohibited to T3 high) and must pass lifecycle gates (G0 to G5) before it moves forward. The Ethics and Risk Review Committee has a binding veto over tier T2 and T3 work.

What already exists. A tested reference implementation of the core components (bilingual normalisation and search, cited answers with abstention, risk-tier assessment, the AI system register, evaluation and a tamper-evident audit log) is published on GitHub and archived with a DOI.

The long view. A strategic roadmap sets five horizons: foundation (Q4 2026), short term (2027 to 2028), medium term (2029 to 2030), long term (2031 to 2035) and vision (2036 to 2041), followed by a charter renewal decision after 2041. It is aligned with the national AI policy cycle and with the 2041 national plans, and it is reviewed at fixed points and whenever major events occur.

What is asked of the authorities. Six decisions (Chapter 21): accept the Blueprint as a technical input to national AI policy drafting; approve a 12-month Phase 1 pilot; authorise bulk access to already-public documents; nominate members of the Steering Board and the independent committee; commission a feasibility and costing study; and confirm the guardrails.

Acronyms

Acronym	Meaning
ADP	Annual Development Programme
ADR	Architecture Decision Record
AIA	Algorithmic Impact Assessment
API	Application Programming Interface
BAPIL	Bangladesh AI Policy & Innovation Lab
BBS	Bangladesh Bureau of Statistics
BCC	Bangladesh Computer Council
BM25	Okapi Best Matching 25 ranking function
BPFM	Bangla Policy Foundation Model
CI	Continuous Integration
CSF	Cybersecurity Framework (NIST)
DCAT	Data Catalog Vocabulary (W3C)
DOI	Digital Object Identifier
DPIA	Data Protection Impact Assessment
EGDI	E-Government Development Index
ERRC	Ethics and Risk Review Committee
FTE	Full-Time Equivalent
GPU	Graphics Processing Unit
KPI	Key Performance Indicator
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MFA	Multi-Factor Authentication
MoU	Memorandum of Understanding
MRR	Mean Reciprocal Rank
nDCG	Normalised Discounted Cumulative Gain
NDMA	National Data Management Act, 2026
NLP	Natural Language Processing
NRDEX	National Responsible Data Exchange
OCR	Optical Character Recognition
PDPA	Personal Data Protection Act, 2026
RACI	Responsible, Accountable, Consulted, Informed
RAG	Retrieval-Augmented Generation
RPO	Recovery Point Objective
RTI	Right to Information
RTO	Recovery Time Objective
SBOM	Software Bill of Materials
SSO	Single Sign-On
SWOT	Strengths, Weaknesses, Opportunities, Threats
TAB	Technical Architecture Board
WCAG	Web Content Accessibility Guidelines

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Part I

Foundations

Why a blueprint is needed now, where Bangladesh stands, who must be involved, and what the system must do.

Chapter 1: Purpose, Scope and Method

1.1 Why a Blueprint Now

The BAPIL White Paper (Version 1.0) set out a vision: an independent research and policy-intelligence platform that helps Bangladesh use artificial intelligence (AI) for evidence-based policymaking, responsible AI governance, Bangla-language technology and public-sector innovation [1, 5, 7, 17, 21]. The Executive Summary condensed that vision for senior policymakers. Both documents deliberately stayed at the level of concepts, principles and illustrative pilots.

This Blueprint answers the next question: **what exactly would be built, by whom, under which rules, in what order, with what resources, and with what evidence of success?** It translates each element of the White Paper into requirements, institutional arrangements, technical components, data rules, assurance procedures, a delivery plan, a long-term strategy and decisions for the authorities.

The timing matters. During 2025 and 2026 Bangladesh enacted a new data-protection and data-management framework and revised its cyber-security law [84, 85, 86, 87]. A draft National AI Policy 2026-2030 was released for public consultation in January 2026 [88, 89], and in August 2026 the government announced that drafting of a national AI policy would begin in the fourth quarter of 2026 with a joint team from government, the private sector, academia and research [90, 91]. A concrete, openly licensed blueprint with working code can serve as a technical input to that process, and a roadmap that looks to 2041 can help connect short pilots to the long-term goals of the Perspective Plan 2041 and the ICT Masterplan 2041 [29, 30].

1.2 The BAPIL Document Family

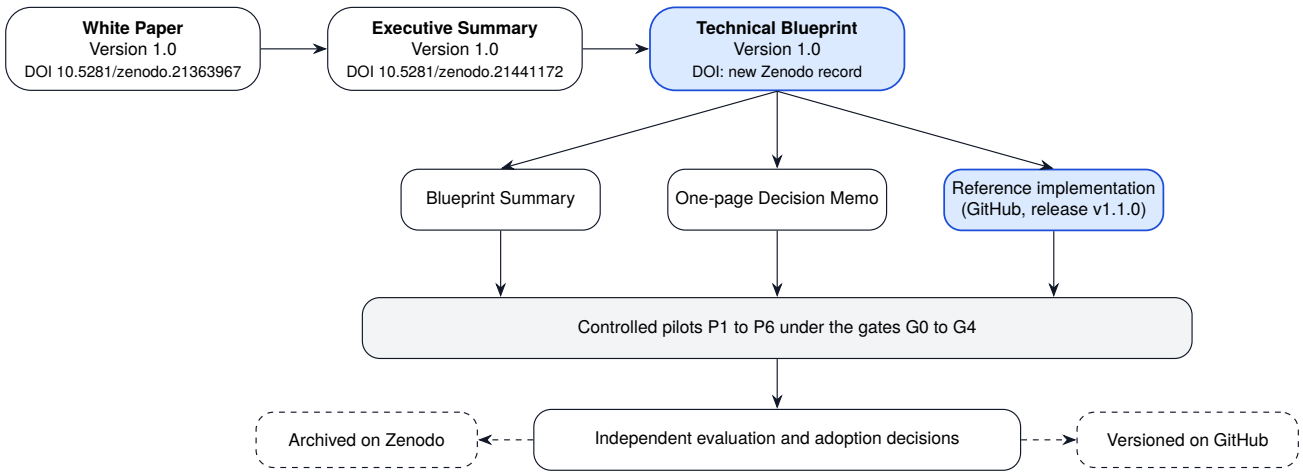


Figure 1. The BAPIL document family: from the White Paper to a tested implementation and controlled pilots.

Table 1. The BAPIL document family and persistent identifiers

Output	Version	Purpose	Primary audience	Identifier
White Paper	1.0 (repository release v1.0.1)	Conceptual framework, international lessons, vision to 2041	All stakeholders	10.5281/zenodo.21363967
Executive Summary	1.0	Concise summary of the White Paper	Senior policymakers	10.5281/zenodo.21441172
Technical Blueprint (this document)	1.0	Buildable design: institution, architecture, data, governance, delivery plan	Ministries, agencies, technical teams, partners	DOI reserved on Zenodo at publication

Output	Version	Purpose	Primary audience	Identifier
Blueprint Summary	1.0	Short summary of the Blueprint	Decision-makers	Same record as the Blueprint
Decision Memo	1.0	One page: decisions requested from authorities	Ministers, secretaries, agency heads	Same record as the Blueprint
Reference implementation	0.1.0 (repository release v1.1.0)	Runnable, tested prototype of core Blueprint components	Engineers, researchers, auditors	GitHub repository; archived with the record

1.3 Maturity Classes

To avoid confusion between experiments and government systems, the Blueprint uses three maturity classes. Each class has its own users, permitted data and approval route.

Table 2. System maturity classes used throughout this Blueprint

Class	Definition	Users	Permitted data	Example	Approval
R Research prototype	Experimental system used to test methods; outputs never used for official decisions	BAPIL researchers and research partners	P0; P1 under a data-sharing agreement	Benchmark experiments with open models	Gates G0 and G1
A Advisory tool	Used by officials as one input among several; a person reviews every output before use	Officials in pilot agencies	P0 and P1; de-identified P2 only	Policy Search pilot in two to three agencies	Gates G2 and G3; ERRC approval for tier T2
O Operational system	Integrated into agency workflows and owned and operated by the agency, not by BAPIL	Host agency staff and, where intended, the public	As authorised by law and the agency	Agency-operated search service	Gate G4 with host-agency legal review

BAPIL builds systems of class R and class A. Class O systems are operated by agencies, never by BAPIL, and only after a G4 adoption decision with the agency's own legal review.

1.4 What This Blueprint Adds to the White Paper

The White Paper described six architectural layers, a Policy Intelligence Dashboard, a Bangla Policy Foundation Model, a governance framework, a five-year roadmap and five illustrative pilots. This Blueprint adds:

1. **A situation analysis** with baseline indicators, a capability gap analysis and a SWOT assessment (Chapter 2).
2. **A stakeholder and engagement plan** (Chapter 3).
3. **Requirements** with identifiers, priorities, personas, user stories and traceability to code and tests (Chapter 4, Annex A).
4. **An institutional design** with an assessed operating model, governance bodies, decision rights and a responsibility matrix (Chapter 5, Annexes E to G).
5. **A people and capacity plan** with staffing, a competency framework and a training curriculum (Chapter 6).
6. **A reference architecture** of twelve named components, an API specification and architecture decision records (Chapter 7).
7. **Data rules**: a four-class data classification, a source inventory, metadata, quality, retention and a public-data-first policy (Chapter 8, Annex I).

8. **AI and language design**: a bilingual, citation-grounded retrieval pipeline, Bangla text rules, an evaluation benchmark, model selection criteria and a staged path for the Bangla Policy Foundation Model (Chapter 9, Annex K).
9. **Infrastructure, security and continuity** specifications (Chapter 10).
10. **A tested reference implementation** published on GitHub and archived on Zenodo (Chapter 11).
11. **Full pilot specifications** for six pilots (Chapter 12).
12. **An operational assurance system**: risk tiers, lifecycle gates, human oversight, documentation and a framework crosswalk (Chapter 13, Annexes B, C, H and J).
13. **Legal alignment** with the Bangladesh instruments in force or in preparation (Chapter 14).
14. **A 36-month delivery plan** and a **strategic roadmap for 2026 to 2041 and beyond** (Chapters 15 and 16).
15. **Financing, procurement, monitoring and evaluation, risk management and a research agenda** (Chapters 17 to 20).
16. **Six decisions** for the authorities (Chapter 21).

Table 3. From White Paper chapters to Blueprint chapters

White Paper chapter (v1.0)	Blueprint chapter	What the Blueprint adds
2. Why Bangladesh needs AI policy intelligence	2, 4	Baseline indicators, gap analysis, user personas and testable requirements
3. Global examples	5, 13, 14	Institutional choices and controls adapted from the cases, with a framework crosswalk
4. Current AI landscape in Bangladesh	2, 3, 14	Situation analysis, stakeholder register and the 2026 legal framework
5. BAPIL vision	1, 16	Scope boundaries, maturity classes and a 2026-2041 strategic roadmap
6. BAPIL architecture	7, 8, 10	Twelve named components, interfaces, data rules, hosting and security zones
7. Policy Intelligence Dashboard	7, 12	API specification and pilot specifications (P1, P2, P5)
8. Bangla Policy Foundation Model	9, 12	Bangla text rules, BanglaPolicyQA benchmark and an evidence-gated model path
9. AI governance framework	13, Annexes B, E to J	Risk tiers, lifecycle gates, charter, terms of reference and checklists
10. National security and cybersecurity	10	NIST CSF 2.0 mapping, AI-specific threats, incident levels and recovery targets
11. AI education and talent	6	Staffing plan, competency framework and training curriculum
12. Startup ecosystem	3, 17	Industry engagement and procurement clauses that keep markets open
13. Resources and funding	17	Cost model, funding mix by horizon and sustainability plan
14. Five-year roadmap	15	36-month workplan with milestones, deliverables and gates
15. Long-term vision	16	Horizons, indicative targets, scenarios and review triggers to 2041 and beyond
16. Risks, ethics and responsible AI	13, 19	Risk register, heat map and operational assurance
17. Strategic recommendations	21	Six concrete decisions for the authorities
Appendix A. Pilot projects	12	Full specifications with users, data, tier, scope and exit criteria

1.5 Scope and Boundaries

BAPIL remains a **research, policy-intelligence and capacity-building** initiative. It does not make policy, does not regulate, does not perform operational security functions and does not operate production government systems. Where a tool proves useful, ownership passes to the responsible agency through a documented adoption decision (Gate G4, Chapter 13).

Three boundaries are designed in from the start:

- **Public data first.** Pilots begin with documents and statistics that are already officially published. Personal data is excluded unless there is a lawful basis, a data protection impact assessment and ethics approval; sensitive personal data is excluded entirely (Chapter 8).
- **Advice, not decisions.** Every AI output is evidence for a person to weigh. No BAPIL system makes or automates decisions about individuals (Chapters 4 and 13).
- **No surveillance.** Social scoring, mass or indiscriminate biometric surveillance and the profiling of people's political or religious opinions are prohibited by charter (tier T0, Chapter 13).

Within these boundaries, the Blueprint covers the full life of the capability: from the first briefing in late 2026, through a three-year pilot programme, to the long-term questions of scale, institutional form and sustainability that arise between 2030 and 2041.

1.6 Method

The Blueprint was prepared in five steps, each of which can be checked against the published sources and the repository:

1. **Review of the BAPIL series.** Every chapter of the White Paper and the Executive Summary was mapped to a design question and to the chapter of this Blueprint that answers it (Table 3).
2. **Review of the context.** The legal and policy changes of 2025 and 2026 were reviewed using official announcements and press reports, and indicators of digital government, cyber security and AI readiness were collected (Chapter 2). Because the gazetted texts of several 2026 Acts were not available in full at the time of writing, legal descriptions are marked as summaries to be confirmed.
3. **Design against reference frameworks.** Requirements and controls were derived from the OECD AI Principles, the UNESCO Recommendation, the NIST AI Risk Management Framework, ISO/IEC 42001, the EU AI Act and Canada's Directive on Automated Decision-Making [1, 4, 5, 6, 7, 68, 92], and adapted to Bangladesh's draft national AI policy categories [88].
4. **Implementation as a test of design.** The core ideas (bilingual search, cited answers with abstention, risk tiers, the register, evaluation and audit) were implemented and tested in code. Where implementation revealed a gap in the design, the design was corrected.
5. **Planning and review.** Delivery, resources, risks and long-term strategy were planned with explicit assumptions, so that each can be tested by the feasibility study and the strategic reviews.

1.7 Summary of the Design

In one paragraph: BAPIL is established as a **university-anchored lab with a government MoU and a multi-stakeholder Steering Board**, an **independent Ethics and Risk Review Committee with a binding veto**, and a **Technical Architecture Board**. It builds an **open, modular platform** whose core is a **bilingual, citation-grounded policy search and answer service** over officially published documents. Six pilots are run under **risk tiers and lifecycle gates** aligned with international frameworks and with Bangladesh's new data laws. A **36-month plan** delivers a policy search pilot and an open budget dataset in Year 1, evaluated adoption decisions in Year 2, and a governed Bangla domain model and Blueprint 2.0 in Year 3. A **strategic roadmap** then carries the capability through medium-term adoption (2029 to 2030), a national policy-intelligence network (2031 to 2035) and a mature, trusted institution by 2041. A **reference implementation** of the core components already exists, is tested, and is archived with a DOI.

Chapter 2: Situation Analysis and Baseline

2.1 Purpose of the Analysis

A blueprint is only as good as its understanding of the starting point. This chapter summarises where Bangladesh stands on the capabilities BAPIL depends on: digital government, data, cyber security, AI readiness, Bangla language technology and the legal framework. It then identifies the gaps that BAPIL is designed to close and assesses the strengths, weaknesses, opportunities and threats that will shape delivery. The indicators also serve as a baseline for the long-term targets in Chapter 16.

2.2 Baseline Indicators

Table 4. Selected baseline indicators for Bangladesh

Indicator	Reported value	Implication for BAPIL	Source
UN E-Government Development Index (EGDI), 2024	Rank 100 of 193, score 0.6570 (rank 111 in 2022); highest EGDI among least developed countries	Digital service foundations exist to build on	[21, 107]
ITU Global Cybersecurity Index, 2024	Reported in the top tier	Frameworks exist; resilience under stress still needs testing	[71, 108]
Oxford Insights Government AI Readiness Index, 2023	Rank 82	Room to improve on governance, data and capacity pillars	[26, 80]
University AI compute, 2025 estimates	2 to 55 GPUs per surveyed department; public universities at the low end	Use shared national compute; keep pilot models small	[80]
Data protection and data management	Personal Data Protection Act, 2026 and National Data Management Act, 2026 passed in April 2026	Clear legal basis for a public-data-first design	[85, 86]
National AI policy	Draft 2026-2030 consulted in January 2026; new drafting from Q4 2026	Timely window for technical input	[88, 91]
Network resilience	A nationwide disruption in July 2024 exposed single points of failure	Plan for offline operation, backups and recovery	[108]

Note. Indices measure different things over different years and should not be compared directly. Values are as reported in the cited sources.

The indicators point in a consistent direction. Bangladesh has a substantial digital government foundation: the United Nations E-Government Survey 2024 placed the country 100th of 193, eleven places higher than in 2022 [21, 107], and the national portal and electronic file system are in wide use [32, 83]. Cyber-security frameworks have been strengthened and Bangladesh has been reported in the top tier of the ITU Global Cybersecurity Index, although commentators note that the index measures the existence of frameworks more than their performance under stress [71, 108]. AI readiness is lower: Bangladesh was ranked 82nd in the 2023 Government AI Readiness Index, and university AI departments report very limited GPU capacity [26, 80]. The legal framework for data has just been renewed [85, 86], and the national AI policy is being drafted [88, 91].

2.3 Digital Government Foundations

BAPIL does not start from nothing. Over the past fifteen years Bangladesh has built a national portal that brings together the websites of ministries, divisions and field offices [83]; an electronic file and workflow system (e-Nothi) used across the administration [32]; a national data centre; an open data portal and regular statistical releases from the Bangladesh Bureau of Statistics [31, 33]; and a large network of digital service centres. These systems matter for BAPIL in three ways. They are the main **sources** of the official documents that P1 indexes. They provide

infrastructure (hosting, identity, networks) that BAPIL should use rather than duplicate. And the people who built and run them form a community of practice with **experience** of digital change in government, which BAPIL's pilots and training can draw on.

The same foundations also show where work is needed. Document formats and metadata vary between sites; many older documents exist only as scans; search functions on government websites are limited and rarely bilingual; and there is no single place where the current, authoritative version of a law or plan can be found and cited. These are exactly the gaps that a governed corpus and a cited search service address.

2.4 The Policy Evidence Problem

The problem BAPIL addresses first is practical and common. A policy analyst who needs to know what the law says on a topic, what the current five-year plan committed to, or how a budget line changed over five years must usually:

- search several government websites, each with its own structure and search function;
- open scanned PDF files whose text cannot be searched or copied reliably, especially in Bangla;
- check whether a document is the current version or has been amended or superseded;
- translate or reconcile terms between Bangla and English versions; and
- record the citation manually for a brief or a file note.

These steps take time and introduce errors. They also make it hard to build reliable AI tools, because those tools depend on clean, versioned, well-described text. Studies of public data in Bangladesh have pointed to inconsistent metadata, licences and quality across portals [78, 79]. The design response is therefore to begin with a governed corpus of official documents and a search service that always shows its sources, before attempting anything more ambitious.

2.5 Bangla Language Technology

Bangla is one of the most widely spoken languages in the world, but it remains under-resourced in AI. Research groups in Bangladesh and abroad have produced strong models and datasets, including BanglaBERT, BanglaT5, Bangla-adapted open LLMs and large sentiment corpora [46, 48, 49, 81]. What is missing for policy work is narrower and more specific: a benchmark built from real policy questions with answers grounded in official documents; consistent rules for normalising Bangla text from scanned and born-digital sources; and evaluation results that show how well open models perform on Bangla policy text compared with English. Chapter 9 and Annex K set out how BAPIL fills these gaps.

2.6 Legal and Policy Context

The legal framework is described in detail in Chapter 14. In summary, the Personal Data Protection Act, 2026 and the National Data Management Act, 2026 establish citizens' rights over personal data, custodial duties for public data and a national data exchange [84, 85, 86]. The cyber-security law was revised in 2026, including an amendment on the removal of harmful AI-generated content [86, 87]. The draft national AI policy proposes risk categories, impact assessment, an oversight committee, a Bangla-based national LLM, centralised compute and an innovation fund [88]. Together these create both a legal basis for a public-data-first design and a policy window in which a tested blueprint can be useful.

2.7 Capability Gap Analysis

Table 5. Capability gap analysis

Capability	Current state	Gap	Blueprint response	Chapter
Access to official documents	Published across many sites, often as scanned PDF	No unified, searchable, cited collection in Bangla and English	Governed P0 corpus and Policy Search (P1)	8, 12
Bangla language technology	Research models and datasets exist [46, 48, 49, 81]	No policy-domain benchmark; uneven OCR and normalisation	BanglaPolicyQA, normalisation rules, staged BPFM	9
AI governance practice	Draft policy categories; new data laws	No operational risk-tiering, gates or transparency register in use	Risk tiers, gates, ERRC, register	13
Data stewardship	Open data portal and statistical releases [31, 33]	Inconsistent metadata, licences and quality [78, 79]	DCAT metadata, quality metrics, NRDEX agreements	8
Compute	Scarce in universities [80]	No shared, governed environment for public-interest AI	Modest pilot hosting; shared national compute for adaptation	10
Skills in government	Strong digital service experience	Limited AI literacy and assurance skills	Six training programmes and fellowships	6
Evaluation culture	Project reporting focused on outputs	Few published, reproducible evaluations of AI tools	Benchmarks, KPIs, independent evaluation, DOIs	18
Coordination	Several national digital and AI programmes	Risk of duplication and unclear roles	Division-of-labour MoU and stakeholder plan	3, 5

Three gaps are critical for BAPIL's first phase. The **document access gap** prevents any reliable AI tool from being built on official text; the **evaluation gap** means there is no shared way to know whether a tool works in Bangla; and the **governance practice gap** means that, even with a draft policy, agencies lack a tested procedure for deciding when an AI system is safe to use. The first year of the roadmap is designed to close these three gaps on a small scale.

2.8 SWOT Assessment

Table 6. Strengths, weaknesses, opportunities and threats (SWOT)

Category	Factors
Strengths	Large digital service footprint (national portal, e-services, e-Nothi); experience with digital transformation at scale; a large, young and digitally active population; active Bangla NLP research community; existing national AI programmes
Weaknesses	Scattered and often scanned official documents; inconsistent metadata and licences; scarce compute; limited AI assurance skills; few independent evaluations; dependence on project funding
Opportunities	National AI policy drafting from Q4 2026; new data laws and the national data exchange; open-weight models that run locally; Bangla as a large-language opportunity; development-partner interest in digital public goods
Threats	Political and institutional change; misinformation and synthetic media; cyber attacks and outages; vendor lock-in; erosion of trust if AI tools fail visibly; talent migration

The SWOT assessment leads to four strategic choices that run through the Blueprint:

1. **Build on strengths:** use the existing digital service experience and the new data laws rather than creating parallel structures.

2. **Address weaknesses directly:** treat document quality and Bangla text processing as core work, not as preparatory chores.
3. **Use the opportunity window:** offer the Blueprint as an input to national AI policy drafting from the fourth quarter of 2026.
4. **Guard against threats:** design for institutional change, misinformation risks and infrastructure disruption from the start, with an institution-neutral design, strict prohibited uses and continuity planning.

2.9 Lessons from International Practice

The White Paper reviewed AI and digital government strategies in Singapore, Korea, Estonia, the United Kingdom, Finland, the United Arab Emirates and Canada [51, 54, 57, 60, 61, 63, 65, 68]. Five lessons are applied in this Blueprint:

- **Start with a concrete service.** Successful programmes began with a specific, visible service rather than a general platform. BAPIL starts with policy search (P1) and open budget data (P2).
- **Make governance operational.** Principles become useful only when turned into procedures with owners and deadlines, such as Canada's Algorithmic Impact Assessment [68, 69] and Singapore's Model AI Governance Framework [51, 53]. BAPIL's tiers and gates follow this approach.
- **Invest in shared infrastructure.** Estonia's X-Road shows the value of a shared, secure data exchange layer [59]; BAPIL should use Bangladesh's national data exchange rather than create its own.
- **Build literacy widely.** Finland's Elements of AI course shows that broad AI literacy is achievable [64]; BAPIL's training programme T1 to T6 applies the same principle to officials.
- **Publish and learn.** The UK's AI Playbook and similar guidance documents are revised as experience grows [61]; BAPIL publishes its evaluations, including failures, and revises the Blueprint at fixed points.

Chapter 3: Stakeholders and Engagement

3.1 Why Engagement Is Part of the Design

AI tools for policy work succeed or fail on the trust and participation of the people who use, govern and are affected by them. Engagement is therefore treated as a design element with its own plan, owners and outputs, not as communication after the fact. The plan has three aims: to ensure that pilots answer real policy needs; to give those who could be affected a voice in governance; and to coordinate with other national programmes so that effort is not duplicated.

3.2 Stakeholder Map

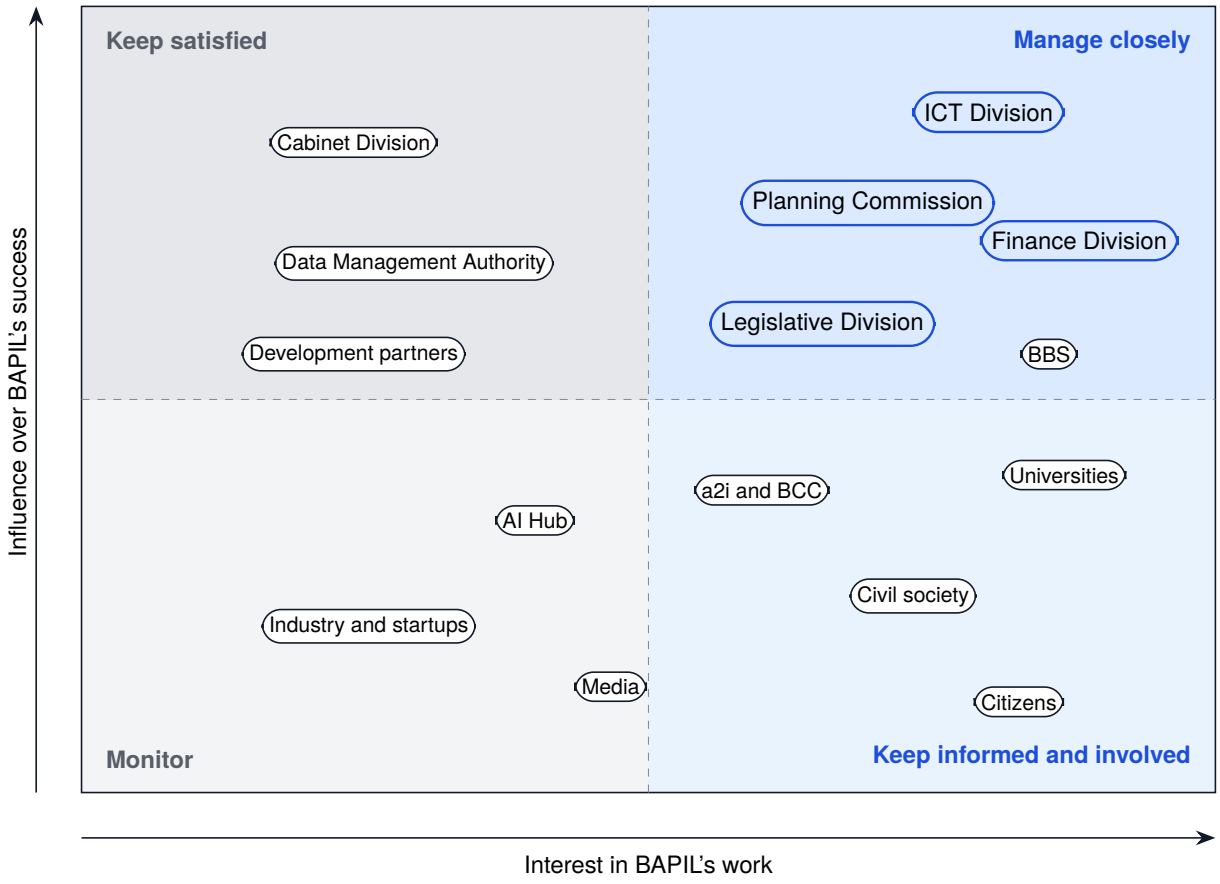


Figure 2. Indicative stakeholder map by influence and interest, used to plan engagement.

The map groups stakeholders by their influence over BAPIL's success and their interest in its work. Those in the upper right quadrant, such as the ICT Division, the Planning Commission, the Finance Division and the Legislative and Parliamentary Affairs Division, are managed closely through the Steering Board, data agreements and product ownership. Those with high influence but less direct interest, such as the Cabinet Division and the data management authority, are kept satisfied through milestone briefings and consultation on data use. Universities, civil society and citizens have high interest and are kept informed and involved through fellowships, ERRC membership, open consultations and the public transparency register.

3.3 Stakeholder Register

Table 7. Stakeholder register

Stakeholder	Main interest	Role for BAPIL	Engagement approach	Frequency
ICT Division (Ministry of Posts, Telecommunications and IT)	National AI policy, digital transformation	MoU partner, Steering Board co-chair	Joint planning, quarterly reviews	Monthly
Cabinet Division	Whole-of-government coordination	Convening across ministries	Briefings at milestones	Semi-annual
Planning Commission	Plans, project appraisal, monitoring	Host agency for P1 and P2; data custodian	Product ownership, co-design	Monthly
Finance Division	Budget transparency and analysis	Data custodian; P2 users	Data agreement, user testing	Quarterly
Legislative and Parliamentary Affairs Division	Authoritative legal texts	Custodian of the laws corpus	Bulk access agreement	Quarterly
Bangladesh Bureau of Statistics	Official statistics and metadata	Data custodian; P5 partner	Metadata alignment	Quarterly
a2i and Bangladesh Computer Council	Digital services, portal, data centre	Platform and hosting partners	Technical working group	Monthly
National Data Management Authority	Data policy and compliance	Regulator and adviser on data use	Consultation on DPIAs and agreements	As needed
Bangladesh AI Hub	Skills, startups, adoption	Complementary programme	Division-of-labour MoU, shared benchmarks	Quarterly
Universities and research institutes	Research, students, publications	Host institution, partners, annotators	Fellowships, joint research	Continuous
Industry and startups	Market for public-interest AI	Technology partners, adopters	Open calls, industry days	Semi-annual
Civil society and media	Rights, transparency, accountability	ERRC members, critical reviewers	Open consultations, register	Quarterly
Development partners	Digital public goods, capacity	Funders and technical advisers	Joint workplans, reviews	Semi-annual
Citizens	Accessible information, trust	Ultimate beneficiaries	Public reports in Bangla, feedback channel	Continuous

Note. Listing an organisation describes its possible role; it does not imply any agreement or endorsement.

3.4 Consultation and Engagement Plan

Table 8. Consultation and engagement plan by phase

Phase	Purpose	Activities	Outputs
Pre-launch (Q4 2026 to Q1 2027)	Test the Blueprint with decision-makers and experts	Briefings with the ICT Division and custodians; expert roundtables; public comment period on the repository	Comment log; Blueprint 1.1 corrections
Launch (months 1 to 6)	Agree roles, data access and safeguards	Charter consultation; ERRC nominations; data custodian workshops	Adopted charter; data access plan

Phase	Purpose	Activities	Outputs
Build (months 4 to 15)	Design with real users	Co-design sessions; usability tests in Bangla and English; monthly demonstrations	User-tested pilots; backlog published
Evaluate (months 12 to 24)	Learn what works	User surveys; time-to-evidence studies; open review of results	Evaluation reports with DOIs
Decide and scale (months 24 to 36)	Decide adoption and next steps	Adoption workshops with host agencies; public hearing on Blueprint 2.0	Adoption memos; Blueprint 2.0

Engagement follows four principles:

- **Bilingual by default.** Consultation materials, summaries and reports are published in Bangla and English.
- **Reach beyond Dhaka.** At least one consultation in each phase is held outside Dhaka or online with participants from divisional universities and local administrations.
- **Inclusion.** Consultations seek a gender balance and include organisations representing persons with disabilities, minorities and young people.
- **Closing the loop.** Each consultation produces a published comment log that records what was raised and how BAPIL responded.

3.5 Engaging Citizens and Communities

Although BAPIL's first users are officials and researchers, its ultimate beneficiaries are citizens, and some of its work (such as P3 on public consultations and P6 on information integrity) touches citizens directly. Citizen engagement therefore has three elements. First, **transparency**: the public register, plain-language summaries of every pilot in Bangla, and an annual public report. Second, **voice**: open consultations on the charter and on Blueprint 2.0, civil-society members on the ERRC, and a public channel for reporting problems and complaints. Third, **benefit**: open datasets such as the budget data from P2, and, in the long term, public-facing search over official documents so that citizens, journalists and civil-society organisations can find and cite the same authoritative texts as officials.

3.6 Coordination with National Programmes

BAPIL complements, and must not duplicate, existing programmes. The Bangladesh AI Hub (implemented by the ICT Division with KOICA support) focuses on skills, startups and adoption [35, 36]; BAPIL focuses on policy intelligence, governance methods and open evaluation. The draft national AI policy envisages a Bangla-based national LLM, centralised compute at the National Data Center and an AI Innovation Fund [88]; BAPIL's benchmark and governed corpus can serve those efforts, and BAPIL should use shared national compute rather than build its own large cluster. A short division-of-labour MoU between BAPIL, the AI Hub and other relevant programmes should be signed at Milestone MS2 and reviewed annually.

Chapter 4: From Vision to Requirements

4.1 Approach

Requirements were derived from three sources: the White Paper's architecture, dashboard, foundation-model and governance chapters; the international frameworks the White Paper adopted as reference points [1, 4, 5, 6, 7, 68]; and the Bangladesh legal context summarised in Chapter 14. User needs were captured as personas and user stories. Each requirement is stated once, given an identifier and a priority, and traced to the components, code modules and tests that satisfy it (Annex A).

4.2 Users and Personas

Six personas represent the main user groups. They are composites built from the White Paper's analysis and the stakeholder register; they will be validated and refined through co-design sessions in the first six months.

Table 9. User personas

Persona	Goals	Pain points today	Most important requirement
A. Policy analyst in a planning wing	Find the governing text and past commitments quickly; cite them in briefs	Searches many websites; scanned PDFs; unsure which version is current	FR-01, FR-02, FR-03
B. Budget analyst or journalist	Compare allocations across years and ministries	Tables locked in PDFs; inconsistent headings between years	FR-04, NFR-07
C. Consultation owner in a ministry	Understand thousands of submissions fairly	Manual reading; risk of missing minority views	FR-05, GR-03
D. Researcher in a university	Benchmark Bangla methods on real policy text	No open domain benchmark or governed corpus	FR-08, NFR-06
E. Senior official	Trust the evidence behind a recommendation	Cannot see sources or limitations of AI outputs	NFR-08, GR-01
F. Citizen or civil-society monitor	Know how AI is used in public research	No public register of AI systems	FR-09, GR-06

4.3 Core Requirements

Requirements are grouped as functional requirements (FR), non-functional requirements (NFR) and governance requirements (GR).

Table 10. Core requirements (functional, non-functional and governance)

ID	Requirement	Priority
FR-01	Bilingual (Bangla and English) search over policy documents, robust to spelling variants and to Bangla or ASCII digits	Must
FR-02	Answers grounded only in retrieved sources, with passage-level citations; the system abstains when evidence is insufficient	Must
FR-03	Ingestion records provenance for every document: source URL, publisher, date, checksum and reuse terms	Must
FR-04	Indicator and budget dashboards with downloadable data and methods notes	Should
FR-05	Aggregate analysis of consultation feedback (topics, sentiment) with no profiling of individuals	Should
FR-06	Register of all AI systems with owner, purpose, risk tier, model card and status	Must
FR-07	Risk-tier assessment applied before any build and at every lifecycle gate	Must
FR-08	Evaluation harness with versioned benchmarks and reproducible reports	Must

ID	Requirement	Priority
FR-09	Public transparency register generated from the AI system register	Should
FR-10	Documented, versioned API for programmatic access	Should
NFR-01	Language parity: Bangla quality within 5 points of English on core metrics	Must
NFR-02	Data residency: P1 to P3 data hosted in Bangladesh in accredited facilities	Must
NFR-03	Security by design aligned with NIST CSF 2.0 and zero-trust access	Must
NFR-04	Availability of 99% for pilots (working hours) and 99.5% at scale	Should
NFR-05	Accessibility to WCAG 2.2 level AA; usable on mobile and low-bandwidth connections	Must
NFR-06	Open standards and open source by default; no single-vendor lock-in	Must
NFR-07	Reproducibility: versioned code, data snapshots and evaluation reports archived with DOIs	Must
NFR-08	Explainability: every output shows its sources, confidence and known limitations	Must
NFR-09	Performance: search p95 at most 3 s; answers p95 at most 10 s	Should
NFR-10	Cost transparency: cost per 1,000 queries reported each quarter	Could
GR-01	Human review of consequential outputs; AI is never the final decision-maker	Must
GR-02	No build without tier screening (G0) and no pilot without gate approval (G2)	Must
GR-03	Data minimisation: public data first; DPIA before any P2 or P3 processing	Must
GR-04	Tamper-evident audit logs of queries, sources consulted and outputs	Must
GR-05	Incident reporting and response within defined time limits	Must
GR-06	Prohibited uses (tier T0) excluded by the BAPIL charter	Must

Note. Priority follows the MoSCoW convention (Must, Should, Could). Requirement IDs are traced to components, code modules and tests in Annex A.

4.4 Priority User Stories

User stories express requirements from the user's point of view and carry measurable acceptance criteria. They are the basis of the pilot backlogs and of the acceptance tests at gates G2 and G3.

Table 11. Priority user stories with acceptance criteria

ID	User story	Acceptance criteria	Req.
US-01	As a policy analyst, I search in Bangla and get relevant English and Bangla passages	Top 10 results include the gold passage for at least 85% of benchmark questions in both languages	FR-01
US-02	As an analyst, I get a short answer where every sentence links to its source	100% of answer sentences carry a passage identifier; citation precision at least 0.95	FR-02
US-03	As an analyst, I am told when the system does not know	At least 90% correct abstention on unanswerable questions; reason shown	FR-02
US-04	As a reviewer, I can see where a document came from and whether it changed	Every passage resolves to source URL, date, checksum; checksum mismatch raises an alert	FR-03
US-05	As a budget analyst, I download five years of allocations in one file	CSV and Parquet downloads with a methods note; values match source on a 99% sample	FR-04
US-06	As a consultation owner, I see topics without exposing individuals	Outputs aggregate at least 20 submissions per topic; no names or contact data in outputs	FR-05
US-07	As the ERRC, I see every AI system, its tier and gate status	Register lists 100% of systems; export generated automatically each release	FR-06, FR-09
US-08	As an auditor, I can verify that logs were not altered	Hash-chain verification passes; any edit, deletion or reordering is detected	GR-04

ID	User story	Acceptance criteria	Req.
US-09	As a developer, I call a documented API	Versioned endpoints with examples; breaking changes only in a new version	FR-10
US-10	As a mobile user on a slow connection, I can still search	Search page under 500 KB; usable on 3G; WCAG 2.2 AA checks pass	NFR-05

4.5 Design Principles Applied

The White Paper's principles (human-centred AI, transparency, responsibility, inclusiveness, evidence-based decision support and collaboration) become five engineering rules in this Blueprint:

1. **No citation, no claim.** A generated statement that cannot be traced to a retrieved passage is not shown (FR-02).
2. **Bangla is a first-class language.** Every feature is specified, tested and evaluated in Bangla and English, with a measured parity target (NFR-01).
3. **Open by default.** Code, benchmarks, evaluation reports and non-sensitive data are published with persistent identifiers; closed components need a written justification (NFR-06, NFR-07).
4. **Smallest data that works.** Each pilot uses the least sensitive data class that meets its purpose (GR-03).
5. **Gate before scale.** Nothing moves from experiment to advisory use, or from advisory to operational use, without passing a documented gate (GR-02).

4.6 Requirements Management

Requirements are kept in the repository as versioned text, alongside the code. A change to a requirement is proposed through an issue, reviewed by the Technical Architecture Board (and by the ERRC where it affects a governance requirement), and recorded in the change log. The traceability matrix in Annex A is regenerated at every release so that each requirement can be followed to the components, modules and tests that satisfy it.

Part II

Institution and People

How BAPIL is constituted and governed, who does the work, and how national capacity is built.

Chapter 5: Institutional Design and Governance

5.1 Operating-Model Options

Four operating models were assessed against five criteria: independence, data access, policy uptake, speed to start and sustainability.

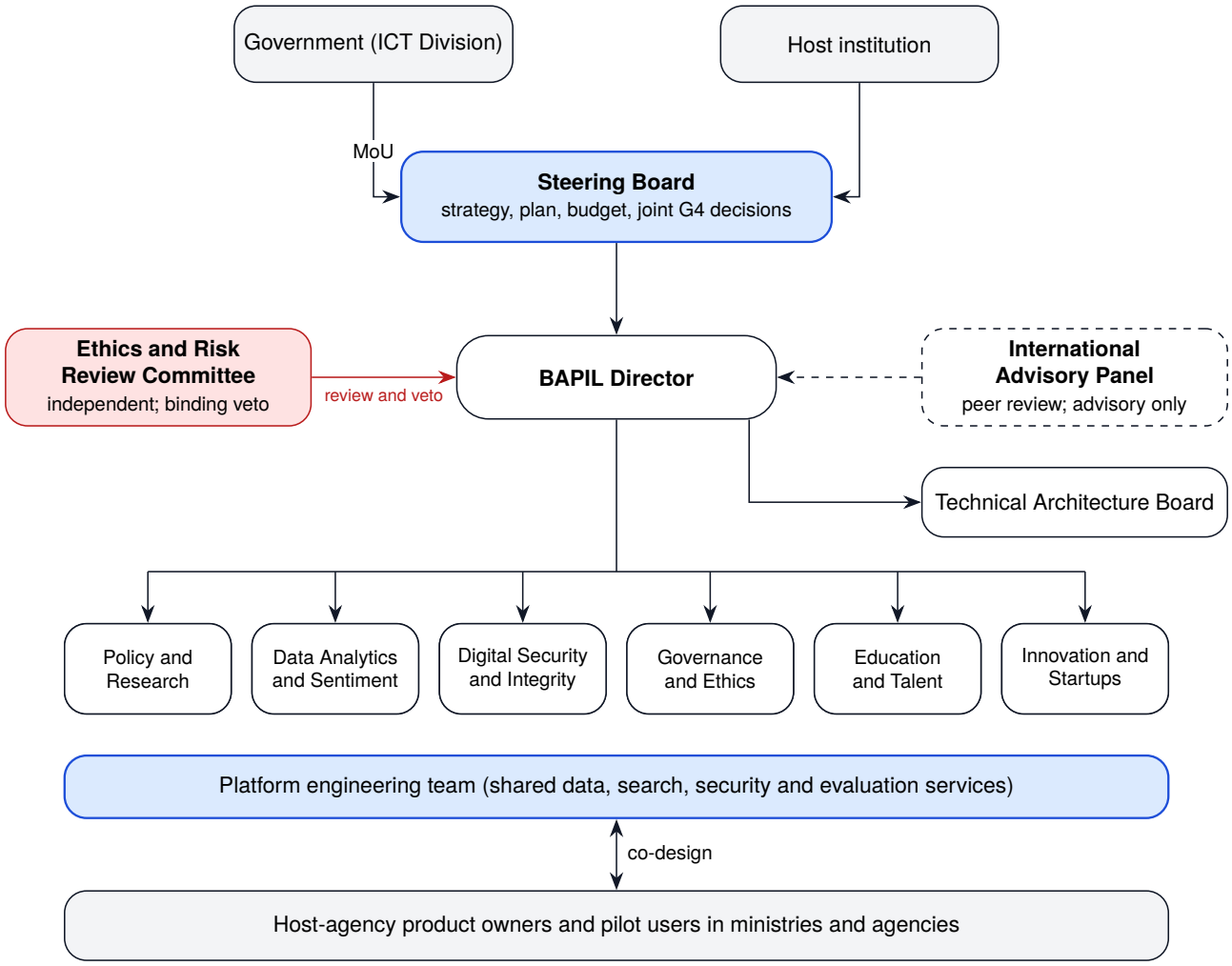


Figure 3. Proposed institutional and governance model (Option C: university-anchored lab with a government MoU).

Table 12. Assessment of institutional operating-model options

Option	Independence	Data access	Policy uptake	Speed to start	Sustainability	Verdict
A. Unit embedded in a government agency	Low	High	High	Medium	Medium	Not recommended for Phase 1
B. University research centre	High	Low	Low	High	Medium	Partial fit
C. Hybrid: university-anchored lab with a government MoU and a multi-stakeholder board	High	Medium to high	Medium to high	High	Medium to high	**Recommended**

Option	Independence	Data access	Policy uptake	Speed to start	Sustainability	Verdict
D. Independent non-profit foundation	High	Low to medium	Medium	Low	Medium	Option for a later stage

Note. Qualitative assessment by the author against five criteria; to be validated in the Phase 1 feasibility study (Deliverable D1.3).

Option C is recommended for Phase 1. A lab anchored in a public or private university gives academic independence, the ability to hire researchers and engage graduate students, and a credible home for open research. A Memorandum of Understanding with the ICT Division (Ministry of Posts, Telecommunications and Information Technology) provides a channel for policy questions, data access agreements and the eventual transfer of successful tools. A multi-stakeholder Steering Board keeps both sides accountable. Option D, an independent non-profit foundation, corresponds to the "mature research ecosystem" stage in Appendix C of the White Paper and is reconsidered at the end of the 36-month plan and again in the long-term horizon (Chapter 16).

5.2 Selecting the Host Institution

The host institution should be selected through an open expression-of-interest process against published criteria and weights. An independent panel scores the proposals; the Steering Board, once constituted, confirms the result. Until then, the ICT Division and an interim panel of academic and civil-society members may run the selection.

Table 13. Host-institution selection criteria and weights

Criterion	Weight	Evidence requested
Research strength in AI, NLP and Bangla language technology	20%	Publications, datasets, active projects, faculty profiles
Research ethics and data protection capacity	15%	Functioning ethics committee, data protection officer, past DPIAs
Commitment to the BAPIL charter and open publication	15%	Signed letter of commitment; open-access and open-source policies
Financial management and audit record	15%	Audited accounts; experience with grants from public and development partners
Infrastructure and secure facilities	10%	Server rooms, network, physical security, access to national compute
Interdisciplinary breadth (law, economics, public policy)	10%	Relevant departments and joint programmes
Inclusion and access	10%	Gender balance plans; outreach to universities outside Dhaka
Location and convening ability	5%	Ability to host government, industry and civil-society meetings

Note. Scores are assigned by an independent panel; the Steering Board confirms the result. Weights may be adjusted before the call is published.

The call should be open to public and private universities and to consortia led by a university. A consortium can combine, for example, a strong computer science department with a law faculty and a policy institute, and can include a university outside Dhaka to support national reach.

5.3 Governance Bodies

Table 14. Governance bodies, composition and decision rights

Body	Composition	Key responsibilities	Decision rights
Steering Board	7 to 9 members: ICT Division (co-chair), host institution (co-chair), Planning Commission, a data custodian agency, academia, civil society, industry; BAPIL Director as secretary	Strategy, annual plan and budget; partnership approvals; joint G4 adoption decisions with host agencies	Approves plans and budgets
Ethics and Risk Review Committee (ERRC)	5 to 7 members, majority external: law and human rights, data protection, ethics, domain experts, civil society, technical reviewer	Reviews tier T2 and T3 systems at G1, G2 and G4; reviews incidents; publishes decision summaries	Binding veto on pilots and deployment
Technical Architecture Board (TAB)	Lead engineers, security lead, two external technical reviewers	Architecture and security standards; design reviews; model and dependency approvals	Approves designs at G1
International Advisory Panel	Invited experts serving in a personal capacity	Peer review of methods, benchmarks and governance; international linkages	Advisory only
Director and units	Director, six thematic units from the White Paper, and a cross-cutting platform engineering team	Delivery of research, pilots, training and publications	Operational decisions within approved plans
Host-agency product owners	Named officials in each agency piloting a tool	Define needs, recruit pilot users, accept or reject at G3 and G4	Acceptance of pilots in their agency

Two features deserve emphasis. First, the **Ethics and Risk Review Committee (ERRC)** has a binding veto over pilots and deployments of tier T2 and above, a majority of external members, and an obligation to publish summaries of its decisions (terms of reference in Annex F). Second, **host-agency product owners** are named officials inside each agency that pilots a tool. They define the need, recruit users and accept or reject the result, so that pilots answer real questions rather than technical curiosity.

5.4 Decision Rights and Accountability

Clear decision rights prevent both paralysis and unchecked action. The responsibility matrix below shows who is responsible, accountable, consulted and informed for the main decisions. Accountability for each decision rests with exactly one body, except at G4, where the Steering Board and the host agency share it because the agency owns the operational decision.

Table 15. Responsibility matrix (R responsible, A accountable, C consulted, I informed)

Decision or activity	Steering Board	ERRC	TAB	Director	Units	Host agency
Annual plan and budget	A	C	C	R	C	I
Charter amendments	A	C	I	R	I	I
Tier screening at G0	I	I	C	A	R	C
Design approval at G1	I	C	A	R	R	C
Pilot approval at G2 (tier T2)	C	A	C	R	R	C
Adoption decision at G4	A	C	C	R	C	A
Suspension after a serious incident	I	A	C	R	R	C
Model and dependency selection	I	I	A	C	R	I
Data-sharing agreements	I	C	C	A	R	C
Transparency register publication	I	C	I	A	R	I

Decision or activity	Steering Board	ERRC	TAB	Director	Units	Host agency
Senior staff appointments	A	I	I	R	I	I
Public communication of results	I	C	I	A	R	C

Note. At G4 the Steering Board and the host agency are jointly accountable: the agency owns the operational decision.

5.5 The BAPIL Charter

The Steering Board adopts a charter at Milestone MS1. A full draft is provided in Annex E. It must at least:

- state the mission and the boundaries in Section 1.5;
- list the prohibited uses of tier T0 and commit never to build or support them;
- establish the ERRC, its independence, its veto and its publication duty;
- commit to open publication of code, benchmarks and evaluation reports, subject only to security and privacy exceptions recorded in writing;
- require registration of every AI system in the AI system register before work begins;
- set conflict-of-interest rules for board members, staff and partners, including vendors;
- define how successful tools are transferred to agencies and how BAPIL withdraws.

5.6 Units and Ways of Working

The six thematic units of the White Paper (AI Policy and Research; Data Analytics and Public Sentiment; Digital Security and Misinformation; AI Governance and Ethics; AI Education and Talent; Innovation and Startup Collaboration) remain the organising structure for research. A cross-cutting **platform engineering team** builds and runs the shared services (data, search, security, evaluation) so that units do not duplicate infrastructure.

Work is organised in **product teams around pilots**: each pilot has a product owner from the host agency, a BAPIL lead, engineers, a governance analyst and, for Bangla features, a language editor. Teams work in short iterations, publish their backlog and evaluation results, and present at a monthly demonstration open to partner agencies.

5.7 Independence, Integrity and Conflicts of Interest

A policy-intelligence lab is valuable only if its evidence is trusted. The following safeguards protect independence:

- **Editorial independence.** BAPIL researchers decide the methods and publish the results of evaluations, including results that are unwelcome to a partner. Partners may comment before publication but cannot veto.
- **Funding rules.** No single non-government funder provides more than 30% of the annual budget, and funding agreements cannot give a funder control over research results or governance decisions (Chapter 17).
- **Declarations.** Board members, ERRC members, staff and contractors declare interests annually and before each relevant decision; declarations are kept in a public register in summary form.
- **Vendor neutrality.** BAPIL does not endorse products. Evaluations of commercial tools follow a published protocol and are reported in full.
- **Protection for staff.** Staff can raise concerns about safety, integrity or misuse with the ERRC chair directly, without fear of reprisal.

5.8 Transfer to Agencies and Exit

Successful tools must not depend on BAPIL forever. When a tool passes Gate G4, a transfer plan is agreed with the host agency covering: ownership of code, data and infrastructure; staff training and handover; service levels and support during a transition period (typically six to twelve months); the budget line from which the agency will

operate the service; and BAPIL's continuing role in benchmarks and independent evaluation. Code remains open, so that other agencies can reuse it. If BAPIL itself must close, the charter requires that open archives, benchmarks and code are preserved, that data held under agreements is returned or deleted, and that users of any remaining services are given notice and alternatives.

5.9 Institutional Evolution

The institutional form is expected to change as BAPIL matures. Option C is appropriate for proving value; once tools have been adopted and funding has stabilised, the Steering Board will consider whether BAPIL should become an independent public-interest foundation (Option D) or a statutory centre attached to government, or whether its functions should be absorbed by agencies. The criteria for that decision (independence, sustainability, policy uptake and public trust) and the review points are set out in Chapter 16.

Chapter 6: People, Skills and Capacity

6.1 The Capacity Challenge

Talent is both BAPIL's most important resource and one of its main risks (R7 in Chapter 19). Experienced AI engineers, Bangla NLP specialists and AI governance practitioners are scarce and in demand by industry and international organisations. At the same time, the long-term value of BAPIL depends less on its own staff than on the skills it builds in government, universities and civil society. This chapter therefore covers both the core team and the national capacity programme.

6.2 Core Team

Table 16. Indicative core staffing plan (full-time equivalents)

Role	Months 1-12	Months 13-24	Months 25-36
Director and principal investigator	1	1	1
Policy research lead and analysts	1	2	3
Machine learning and Bangla NLP engineers	2	3	4
Data engineers	1	2	2
Software and platform (DevSecOps) engineers	1	2	2
Security engineer	0.5	1	1
Governance, legal and ethics analysts	1	1	2
User experience and accessibility designer	0.5	1	1
Programme and partnership coordinator	1	1	1
Training and outreach officers	0	1	2
Total core staff	**9**	**15**	**19**

Note. Excludes part-time ERRC members, advisers, visiting fellows and graduate students. Costs are to be set in the feasibility study (D1.3).

The team grows from about nine to nineteen full-time equivalents over three years. Bangla language expertise is embedded in the machine learning team and complemented by part-time language editors. Graduate fellows from partner universities work on benchmark creation, evaluation and research papers, which also builds national talent. Diaspora researchers can contribute remotely through visiting fellowships.

6.3 Recruitment and Retention

Recruitment follows open, merit-based procedures with published job descriptions, diverse interview panels and a commitment to gender balance. To compete for scarce skills, BAPIL should offer:

- salaries benchmarked against the university's research positions and adjusted for scarce skills, within the host's rules;
- meaningful work on public-interest problems with open publication and credit;
- time for research and professional development, including conference attendance;
- flexible and remote work arrangements, which also allow diaspora and out-of-Dhaka participation; and
- joint appointments with universities and secondments from agencies.

Key-person dependency (R16) is reduced by documenting systems and decisions in the repository, naming deputies for critical roles and sharing ownership of code and data with the host institution.

6.4 Competency Framework

The competency framework sets out what different groups need to know at three levels: aware, practitioner and expert. It is used to design training, to write job descriptions and to assess progress.

Table 17. Competency framework by role family and level

Role family	Aware	Practitioner	Expert
Public leaders	What AI can and cannot do; risk tiers; when to ask for evidence	Sponsor pilots; read evaluation reports; apply procurement clauses	Set strategy; chair reviews; balance innovation and safeguards
Policy analysts	Use cited search; verify sources; recognise abstention	Design evidence questions; spot bias and gaps; brief on limitations	Co-design tools; define benchmarks; train peers
Governance, legal and ethics staff	Data classes; prohibited uses; DPIA basics	Run tier assessments, DPIAs and gate reviews	Lead ERRC work; audit systems; shape national guidance
Data stewards	Metadata and licences; data classes	Quality metrics; agreements through the data exchange	Design catalogues and data-sharing frameworks
ML and NLP engineers	Bangla text basics; evaluation metrics	Build retrieval and evaluation pipelines; write model cards	Adapt models; red-team; lead benchmark design
Platform and security engineers	Zones; secrets handling; logging	Infrastructure as code; SBOMs; incident response	Zero-trust design; threat modelling for AI systems

6.5 Training Curriculum

Table 18. Training curriculum

Programme	Audience	Core modules	Duration	Format	Year 1 target
T1 AI literacy for public leaders	Secretaries, heads of agencies	AI capabilities and limits; risk tiers; procurement; reading evaluations	1 day	In person	40
T2 Responsible AI for policy analysts	Pilot users and analysts	Cited search; verification; abstention; bias; data protection basics; prompt-injection awareness	3 days	Blended	100
T3 AI governance practitioner	Governance and legal staff, ERRC	Tier assessment; DPIA; impact assessment; gate reviews; audits; transparency	5 days	In person	30
T4 Data stewardship	Data custodians, chief data officers	Metadata; licences; quality; classification; data exchange agreements	3 days	Blended	40
T5 Bangla NLP and MLOps	Engineers in agencies and universities	Bangla text processing; retrieval; evaluation; model cards; secure deployment	10 days	Blended	30
T6 BAPIL fellowships	Graduate fellows	Research placement on a pilot or benchmark with mentoring	6 to 12 months	Placement	10

Note. All course materials are published in Bangla and English under CC BY 4.0. Year 1 targets add up to the KPI of 100 trained officials plus specialist and fellowship cohorts.

The curriculum follows three design rules. First, **training uses the real tools and real documents**, so that officials practise on the policy questions they face. Second, **every programme teaches limitations**: participants

learn how and why AI systems fail, how to recognise abstention and uncertainty, and when not to rely on an output. Third, **materials are open**: slides, exercises and recorded sessions are published in Bangla and English under CC BY 4.0 so that other institutions can reuse them.

6.6 Fellowships and Research Capacity

The BAPIL fellowship programme places graduate students and early-career researchers on pilots and benchmark work for six to twelve months, under joint supervision by BAPIL staff and their home institution. Fellows contribute to datasets, evaluations and publications and are credited as authors where their contribution warrants it. At least half of the fellowships should go to students from universities outside Dhaka, and the programme should aim for gender balance. Over the long term, fellows form a network of people who can carry the methods into government, academia and industry.

6.7 Gender Balance and Inclusion in the Workforce

AI teams in Bangladesh, as elsewhere, are often unbalanced by gender and concentrated in Dhaka. BAPIL sets explicit goals: at least 40% women among staff, fellows and trainees by the end of the 36-month plan; at least half of fellowships for students from universities outside Dhaka; accessible workplaces and materials; and flexible work arrangements that support carers. Progress is reported in the annual report. Diversity is not only a matter of fairness: teams with varied backgrounds are more likely to notice where a tool fails particular users or regions.

Part III

Technology

The architecture, data, AI and language design, infrastructure and security of the platform, and the reference implementation that tests them.

Chapter 7: Reference Technical Architecture

7.1 From Six Layers to Twelve Components

The White Paper's six layers (data sources, data management, policy intelligence, knowledge management, decision support, governance and security) are refined here into twelve components grouped into four horizontal layers and one cross-cutting governance and security column.

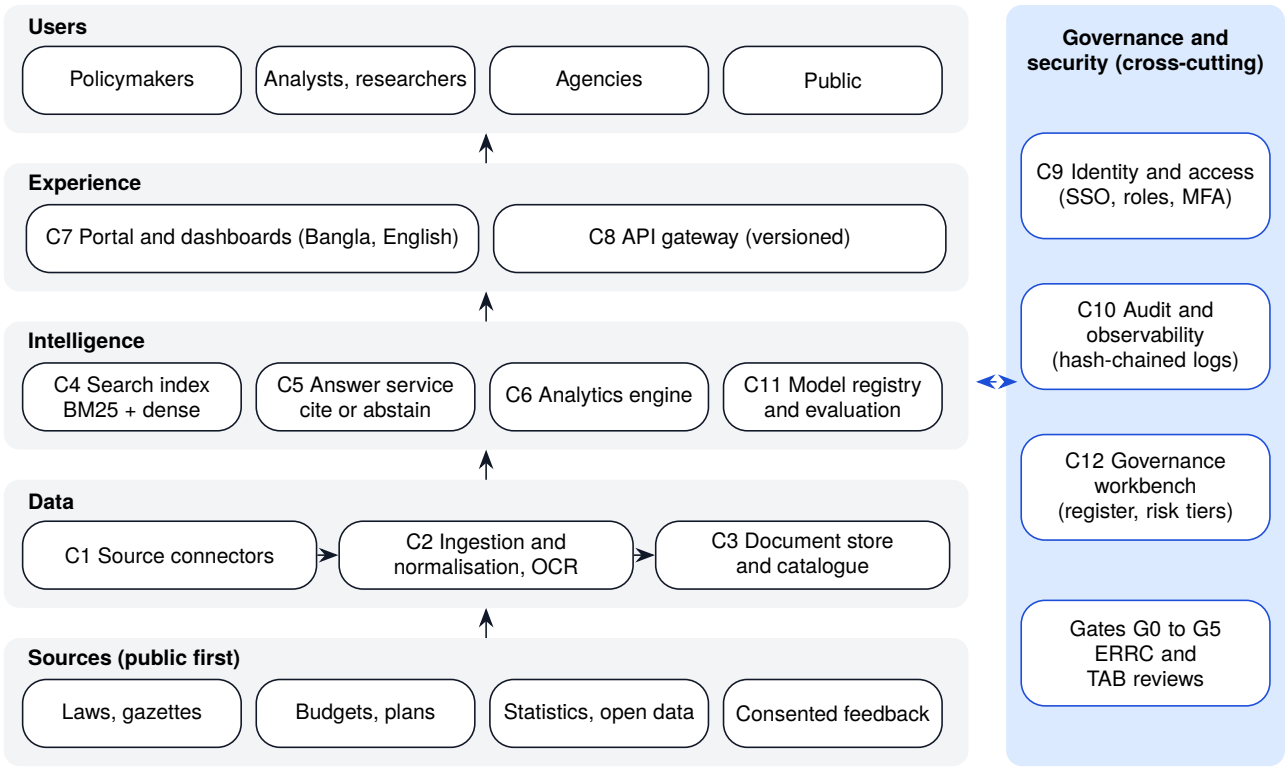


Figure 4. BAPIL reference architecture: twelve components in four layers, with cross-cutting governance and security.

Table 19. Reference architecture components and candidate technologies

ID	Component	Function	Candidate open technologies	Reference implementation
C1	Source connectors	Harvest public documents on a schedule, respecting site terms; record provenance	Python workers, Scrapy, cron or Airflow	Planned
C2	Ingestion and normalisation	Extract text from PDF and HTML, Bangla OCR, Unicode normalisation, chunking	Apache Tika, Tesseract (ben, eng), custom normaliser	bapil.text, bapil.corpus
C3	Document store and catalogue	Originals, extracted text and DCAT-aligned metadata with versioning	PostgreSQL, MinIO (S3-compatible), CKAN	JSONL corpus with checksums
C4	Search index	Lexical BM25 plus dense vectors with hybrid ranking	OpenSearch, or PostgreSQL with pgvector	bapil.search (BM25, bilingual)
C5	Answer service	Retrieve, cite, abstain; optional LLM generation with citation verification	Open-weight LLMs served with vLLM or llama.cpp	bapil.answer (extractive, cited)

ID	Component	Function	Candidate open technologies	Reference implementation
C6	Analytics engine	Budget extraction, indicators, topic models	Python, pandas, DuckDB	Planned
C7	Portal and dashboards	Bilingual, accessible user interface	Web framework; Apache Superset	Planned
C8	API gateway	Versioned REST API, quotas, rate limits	FastAPI, Kong	bapil.api (standard library)
C9	Identity and access	Single sign-on, role-based access, MFA	Keycloak	Planned
C10	Audit and observability	Tamper-evident logs, metrics, tracing	OpenTelemetry, Prometheus, Grafana	bapil.audit (hash-chained JSONL)
C11	Model registry and evaluation	Model cards, versioned evaluations, benchmarks	MLflow	bapil.evaluate
C12	Governance workbench	AI system register, risk assessments, incidents, transparency register	Lightweight web application	bapil.risk, bapil.registry

Note. Technologies are candidates, not mandates; final choices are made by the Technical Architecture Board against requirements NFR-02, NFR-03 and NFR-06.

7.2 How a Request Flows

The core service is a cited answer to a policy question. The sequence below shows how a request passes through the components, and where controls are applied.

Three properties of this flow matter for trust. First, the answer service never answers without evidence: if the top passages do not cover the query well enough, it abstains and shows the closest sources with the reason. Second, every answer sentence carries the identifier of the passage it came from, so a user can open and check the source. Third, every request, the passages retrieved and the response are written to a hash-chained audit log, so that what the system did can be verified later.

7.3 Interfaces and API Specification

The first public interface is a versioned REST API (`/v1`). The minimum endpoints are already implemented in the reference implementation (Chapter 11). Every response includes the corpus version and the identifiers of the passages used, so results can be reproduced later.

Table 20. API specification (version 1)

Endpoint	Method	Parameters	Response	Errors
/v1/health	GET	None	status, version, corpus_version, documents, passages	None
/v1/search	GET	q (1 to 500 characters); k (1 to 50, default 10)	query, corpus_version, results: pid, doc_id, heading, lang, score, matched_terms, text, provenance	400 missing or invalid parameter
/v1/answer	GET	q (1 to 500 characters)	abstained; confidence; evidence (top_score, coverage); answer sentences with citation; sources; corpus_version	400 missing parameter
/v1/risk/assess	POST	JSON questionnaire (Annex B schema)	system_id, tier, tier_name, total, reasons, controls, approver	400 invalid questionnaire; 400 body over 64 KB

Endpoint	Method	Parameters	Response	Errors
All endpoints			JSON in UTF-8; every request logged in the audit chain	404 unknown path

API rules: versions are part of the path; breaking changes are introduced only in a new version, with the old version kept for at least six months; all text is UTF-8; errors return a JSON object with a message; and request bodies are limited in size. In production the API sits behind the gateway (C8), which adds authentication, quotas and rate limits.

7.4 Architecture Decision Records

The Technical Architecture Board records each significant choice as an architecture decision record (ADR) in the repository, with its context, the decision, the alternatives considered and the consequences. The initial decisions are:

Table 21. Initial architecture decision records

ADR	Decision	Rationale	Alternatives considered
001	Modular services with open interfaces	Components can be replaced without rewriting others; avoids lock-in	Monolithic application; single vendor suite
002	Retrieval before generation	Answers stay traceable to official text; reduces invented content	Pure generative assistant; fine-tuned model without retrieval
003	Open-weight models hosted in Bangladesh	Data residency; control over versions; no data sent abroad	Commercial API models for all data classes
004	Hybrid lexical and dense retrieval	Lexical search is transparent and strong on exact terms; dense search handles paraphrase	Dense only; lexical only
005	Append-only, hash-chained audit log	Independent verification of what the system did	Ordinary application logs
006	Containers and infrastructure as code	Reproducible, auditable environments	Manually configured servers
007	Accessibility and low bandwidth first	Reach users outside Dhaka and on mobile	Desktop-first interface
008	Public data first (P0)	Value without privacy risk while governance matures	Start with administrative records

Lexical BM25 retrieval [101] is combined with dense multilingual embeddings [102] (ADR-004); lexical retrieval remains available as a transparent fallback. The portal meets WCAG 2.2 level AA [104] (ADR-007). ADRs are revisited when evidence changes, for example if a model selection review (Table 29) shows that a different approach performs better.

7.5 Accessibility and Inclusive Design

Many potential users work outside Dhaka, use mobile devices and face slow or unreliable connections; some have visual or other impairments. The portal is therefore designed to WCAG 2.2 level AA [104] from the first release, with these specific commitments:

- **Bangla first in the interface.** Menus, help texts and error messages are written in Bangla and English, and the user's language choice is remembered.
- **Light pages.** The search page stays under 500 KB and works on 3G connections (US-10); heavy features such as dashboards load only on request.

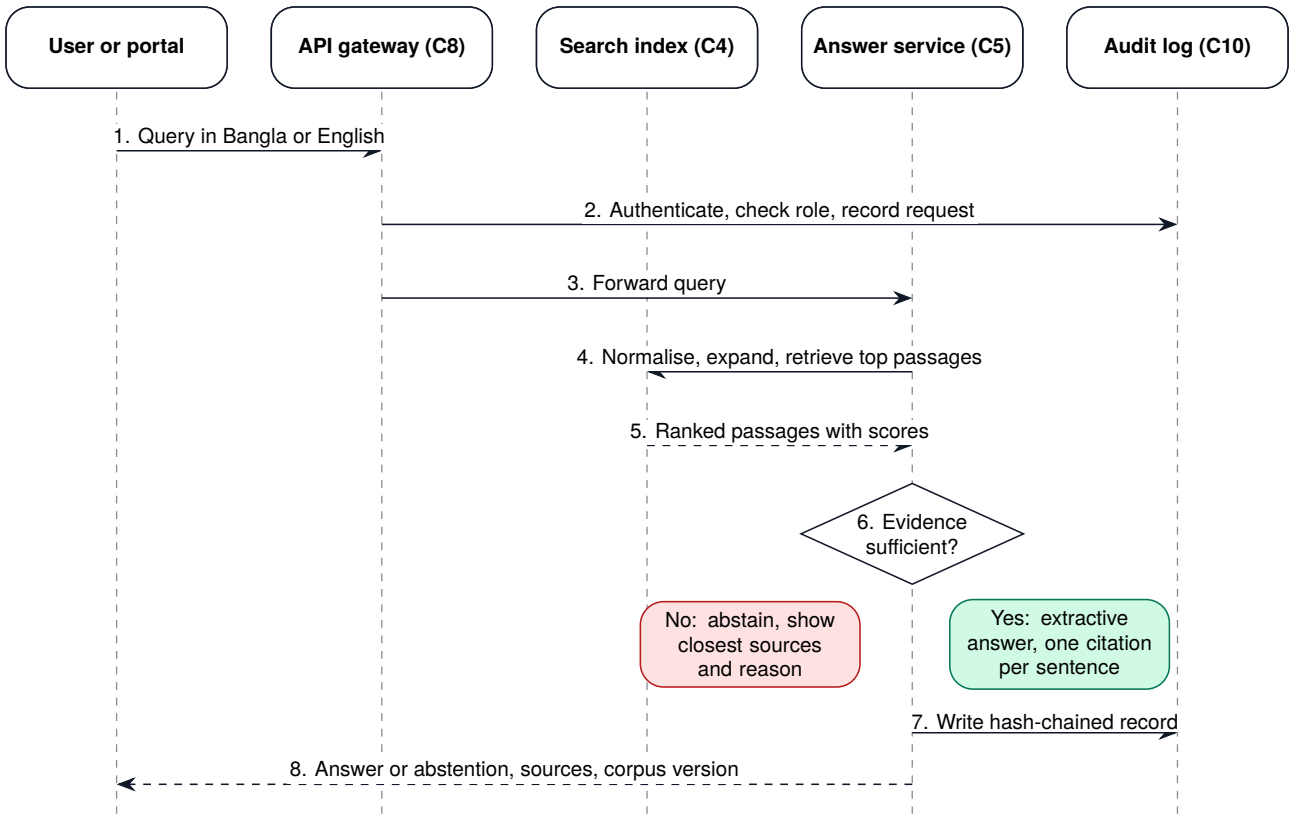


Figure 5. Request sequence for a cited answer, from the user's query to the audit record.

- **Screen-reader support.** Proper headings, labels and alternative text, and correct language tags so that screen readers pronounce Bangla and English text correctly.
- **Bangla input.** Search accepts Unicode Bangla typed with any common keyboard layout, as well as romanised Bangla, which is matched through transliteration candidates in later versions.
- **Plain language.** Answers and limitations are explained in plain words, and technical terms are linked to the glossary.

Accessibility is tested with automated tools at every release and with users, including users with disabilities, at least once in each pilot.

7.6 Technology Choices

The candidate technologies in Table 19 are open-source and widely used, which keeps costs low, supports local skills and avoids dependence on a single vendor. They are candidates, not commitments: the feasibility study (Deliverable D1.3) confirms choices against the host's infrastructure, skills and support arrangements. Where a commercial product is considered, it is assessed with the procurement clauses in Chapter 17, including the right to evaluate on BAPIL's benchmark and the right to exit with all data.

Chapter 8: Data Architecture and Data Governance

8.1 Public Data First

The safest and fastest route to value is to begin with information the state has already chosen to publish: laws, gazette notifications, budget documents, development plans, statistical releases and parliamentary papers. This material is large, important for policy work, under-used because it is scattered across sites and formats, and free of the privacy risks of personal data. The Right to Information Act, 2009 already establishes the principle of proactive disclosure [105].

8.2 The Data Lifecycle

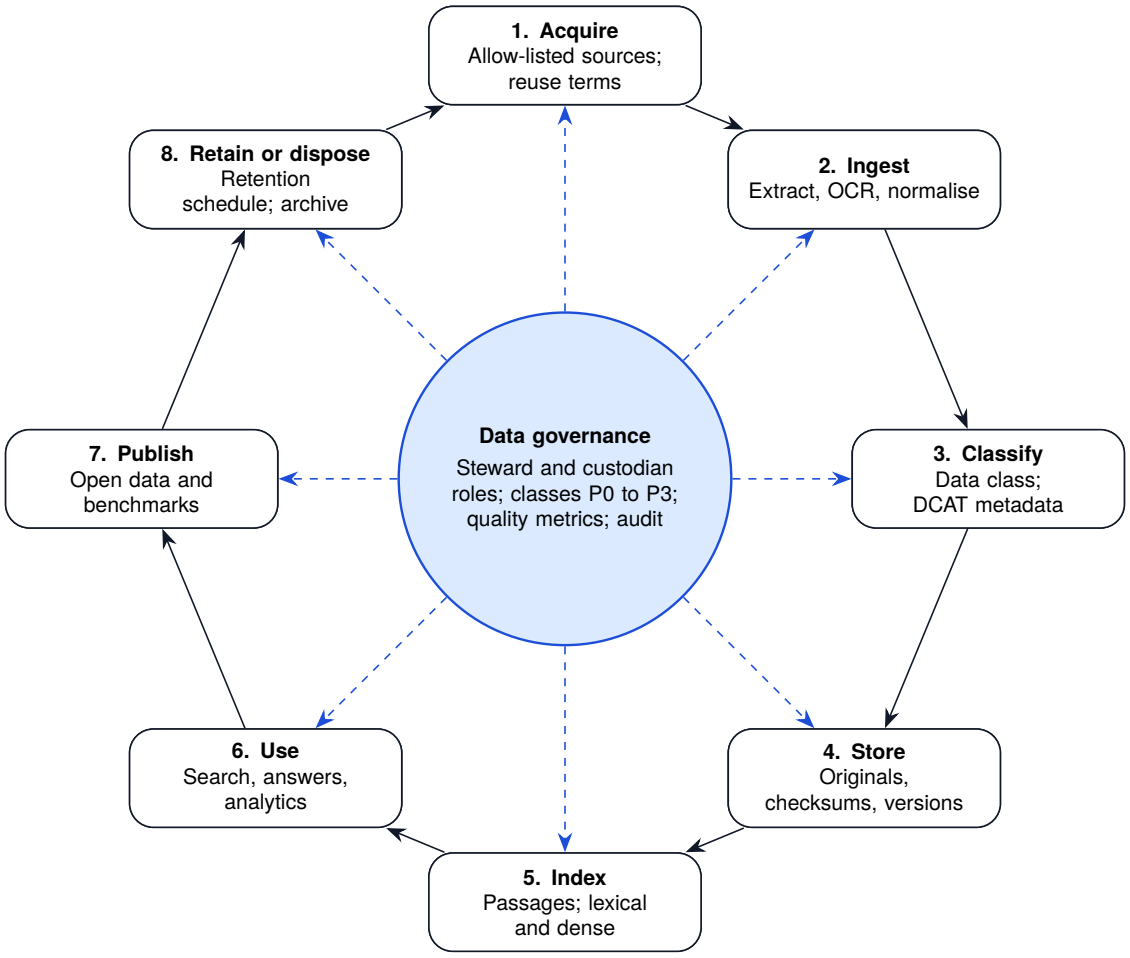


Figure 6. Data lifecycle under data governance, from acquisition to retention or disposal.

Each stage of the lifecycle has an owner and a control. Documents are **acquired** only from allow-listed sources whose reuse terms are recorded. They are **ingested** with text extraction, OCR where needed, and normalisation. They are **classified** into a data class and described with DCAT-aligned metadata. Originals are **stored** with checksums and version history. Text is **indexed** as passages for search. Data is **used** for search, answers and analytics under access control. Non-sensitive results are **published** as open data and benchmarks. Finally, records are **retained or disposed of** according to the retention schedule (Table 26).

8.3 Data Classification

Table 22. Data classification and handling rules

Class	Description	Examples	Handling rules	Use in pilots
P0 Public	Officially published and intended for the public	Laws, gazettes, budget documents, plans, statistical releases	Provenance and integrity checks; reuse subject to source terms	Yes
P1 Internal	Non-personal information not yet published	Draft reports shared under agreement, administrative metadata	Data-sharing agreement; role-based access; hosted in Bangladesh	Limited, under agreement
P2 Personal	Information identifying a person	Named consultation submissions, survey microdata	Lawful basis under the PDPA 2026; DPIA; de-identification before analysis; ERRC approval	Only consented and de-identified
P3 Sensitive or restricted	Sensitive personal data or security-classified information	Health, biometric, financial or children's data; security data	Excluded from BAPIL work	No

The class of every dataset and document is recorded in the catalogue (component C3) and enforced at query time by the access-control layer (C9). A dataset takes the class of its most sensitive element. Moving a dataset to a lower class, for example by de-identification, requires a documented method and approval by the governance analyst and, for P2 data, by the ERRC.

8.4 Initial Source Inventory

Table 23. Initial inventory of public data sources for the pilot corpus

Source	Custodian	Content	Format	Class	Access note
Laws of Bangladesh (bd-laws.minlaw.gov.bd)	Legislative and Parliamentary Affairs Division	Acts and ordinances in Bangla and English	HTML	P0	Confirm reuse terms; request bulk export (Decision D3)
Bangladesh Gazette (dpp.gov.bd)	Department of Printing and Publications	Gazette notifications	PDF, often scanned	P0	OCR quality is a key risk (R3)
National budget documents (mof.gov.bd)	Finance Division	Budget speeches, MTBF, allocations	PDF, spread-sheets	P0	Basis of Pilot P2
Development plans (plancomm.gov.bd)	Planning Commission	Five-year plans, perspective plan, ADP	PDF, spread-sheets	P0	Versioning across plan revisions
Official statistics (bbs.gov.bd)	Bangladesh Bureau of Statistics	Yearbooks and survey reports	PDF, spread-sheets	P0	Aggregates only; microdata excluded
Open Government Data (data.gov.bd)	ICT Division and a2i	Machine-readable datasets	CSV, spread-sheets, API	P0	Check dataset-level licences
National Portal (bangladesh.gov.bd)	ICT Division and a2i	Office information, services, notices	HTML	P0	High volume; prioritise by ministry
Parliament (parliament.gov.bd)	Bangladesh Parliament Secretariat	Bills, debates, committee reports	PDF	P0	Mixed scanned and digital
e-Nothi records	a2i and ministries	Administrative files	Internal system	P1 to P3	Out of scope for pilots

Source	Custodian	Content	Format	Class	Access note
BAPIL publications (this repository)	BAPIL	White Paper, Executive Summary, Blueprint, studies	Markdown, PDF	P0	Demonstration corpus of the reference implementation

Note. Custodians and access terms must be confirmed with each agency before ingestion. Listing a source does not imply any agreement with its custodian.

8.5 Metadata, Provenance and Integrity

Each document in the corpus carries a metadata record aligned with the W3C Data Catalog Vocabulary (DCAT) version 3 [97] and the FAIR principles [98]. The fields are listed below; the JSON schema is published in the repository.

Table 24. Document metadata fields (policy_document schema)

Field	Description	Required	Example
id	Persistent identifier inside the corpus	Yes	bd-act-2009-20
title	Title in Bangla and English where available	Yes	Right to Information Act, 2009
publisher	Organisation that published the document	Yes	Legislative and Parliamentary Affairs Division
issuing_body	Body that issued the instrument	No	Parliament
document_type	Act, ordinance, rule, gazette, budget, plan, statistics, parliamentary paper, policy, report	Yes	act
date_issued	Date of issue	Yes	2009-04-06
version	Version or amendment state	No	as amended to 2026
language	Languages of the text	Yes	bn, en
source_url	Where the original was obtained	Yes	http://bdlaws.minlaw.gov.bd/…
retrieved	Date retrieved	Yes	2027-02-01
reuse_terms	Licence or reuse conditions	No	Government publication; terms confirmed
data_class	P0, P1 or P2	Yes	P0
sha256	Checksum of the original file	Yes	64 hexadecimal characters
extraction	Method (born digital or OCR), quality score, tool	Yes	ocr, 0.93, tesseract 5
supersedes	Identifier of an earlier version	No	bd-act-2009-20-v1

Note. Examples are illustrative. The machine-readable schema is implementation/schemas/policy_document.schema.json.

Integrity is protected in three ways: checksums detect any change to a stored original; a change log records every new version of a document; and the source allow-list (Chapter 10) limits where documents can come from.

8.6 Data Quality

Data quality is measured, not assumed. Each dimension below has a metric and a target for the pilot phase, and results are reported in the corpus release notes.

Table 25. Data quality dimensions, metrics and targets

Dimension	Metric	Pilot target
Provenance	Share of documents with complete source, date, checksum and reuse terms	100%
Text accuracy	Character error rate of OCR on a hand-checked sample	Below 3% for indexed pages
Completeness	Share of a defined collection ingested (for example, all Acts in force)	At least 90% of the target collection
Timeliness	Median days from publication to availability in search	Under 7 days for monitored sources
Consistency	Share of records valid against the metadata schema	100%
Uniqueness	Share of duplicate documents detected and merged	Duplicates under 1%
Language parity	Share of documents with both Bangla and English titles where both exist	At least 95%

8.7 Bangla Text Quality

Much official Bangla text is published as scanned PDF. OCR errors, inconsistent Unicode encodings (for example, the letter য় written as one code point or as two) and mixed Bangla and ASCII digits all damage search quality. The ingestion component therefore (a) prefers born-digital sources, (b) applies Unicode normalisation and digit folding, (c) records an OCR quality score for every page, and (d) routes low-quality pages to a human correction queue. These corrections become a public dataset in their own right, useful to the wider Bangla NLP community [46, 48]. The detailed rules are set out in Chapter 9 (Table 27).

8.8 Data Roles

Clear roles make data governance work in practice:

- **Data custodians** are the agencies that publish or hold the source data. They decide on access and remain responsible for the authoritative version.
- **The BAPIL data lead** is accountable for the corpus, the catalogue, quality metrics and data agreements.
- **Data stewards** in each pilot team maintain metadata, check quality and handle correction queues.
- **The governance analyst** checks data classes, approves reclassification with the ERRC where required, and maintains the records of processing.
- **Chief data officers** in partner agencies, where appointed under the new data-management framework, coordinate agreements and the use of the national data exchange [84].

8.9 Retention and Disposal

Table 26. Indicative retention schedule

Record type	Retention	Disposal or archive
Original public documents (P0)	Indefinite	Archived with checksums; superseded versions kept
Extracted text and index	Until superseded	Rebuilt from originals; old index versions kept for one year for reproducibility
Audit logs (queries, sources, outputs)	5 years	Archived read-only; then deleted
Query text in logs	2 years	Truncated to aggregate statistics
Raw consultation submissions (P2)	Deleted within 90 days of de-identification	Secure deletion with certificate
De-identified analysis data	Duration of the pilot plus 2 years	Deleted or archived under agreement
Model artefacts and evaluation reports	Life of the system plus 3 years	Archived on Zenodo where licences allow

Record type	Retention	Disposal or archive
Incident records and ERRC decisions	7 years	Summaries published; full records archived

Note. Subject to legal review against the Personal Data Protection Act, 2026, the National Data Management Act, 2026 and any rules issued under them.

8.10 Open Data Publication

Data that BAPIL creates from public sources (for example, the budget dataset from P2, corrected OCR text and the benchmark) is published as open data when quality checks are complete. Each release has a DOI, a datasheet, a licence (CC BY 4.0 unless the source requires otherwise), a methods note and a version number. Files are provided in open formats (CSV and Parquet for tables, JSON Lines for text and annotations), with a data dictionary in Bangla and English. Errors reported by users are logged publicly and fixed in the next version, and earlier versions remain available so that published analyses can be reproduced.

8.11 Data Access Agreements

Where data is not already public, access is governed by a written data-sharing agreement that states purpose, data class, retention, security controls, permitted outputs and audit rights (outline in Annex I). Where the national data exchange (NRDEX) provided for under the new data-management framework is available, BAPIL should use it rather than bilateral transfers [84]. Agreements are listed in the transparency register, without confidential details.

8.12 Personal Data

The Personal Data Protection Act, 2026 recognises citizens' rights over their personal data, requires consent and gives enhanced protection to sensitive and children's data; data subjects may restrict automated decisions based on their data [84, 85]. BAPIL's design responds conservatively: P2 data is used only in pilot P3, only with consent, only after a DPIA (template in Annex H), only in de-identified form and only for aggregate results; P3 data is never used. Commentators have argued that the new framework leaves broad room for state access to data [106]; BAPIL's own safeguards therefore do not rely on legal minimums alone.

Chapter 9: AI and Language Technology Design

9.1 Design Goal

The core technical goal is a **trustworthy bilingual question-answering and search service over official documents**: a user asks in Bangla or English and receives the most relevant passages and, where the evidence is sufficient, a short answer in which every sentence points to its source. When the evidence is insufficient, the service says so.

9.2 The Retrieval Pipeline

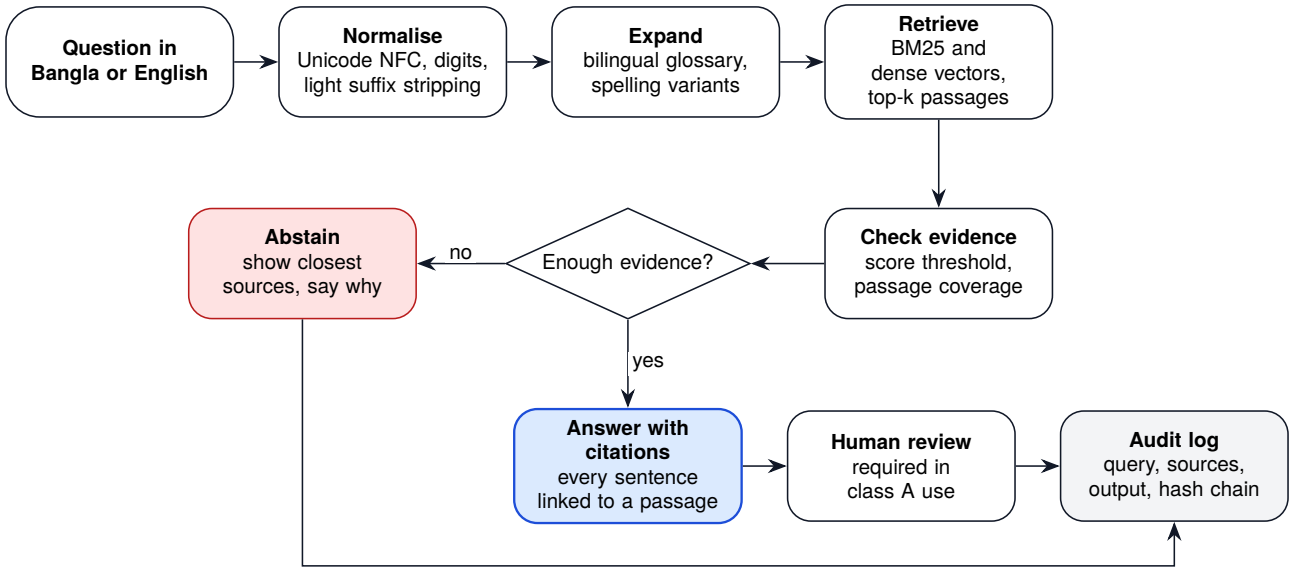


Figure 7. Bilingual, citation-grounded retrieval pipeline: the system answers with citations or abstains.

The pipeline has six stages:

1. **Normalise.** Apply Unicode NFC; unify alternative encodings of Bangla letters; fold Bangla digits to ASCII for matching while keeping the original for display; remove zero-width characters that break matching; apply light stripping of common Bangla inflectional suffixes (for example -এর, -কে, -তে, -গুলো, -সমূহ) and simple English plural forms.
2. **Expand.** Add cross-lingual equivalents from a curated bilingual glossary of policy terms (for example *budget* and বাজেট, *data protection* and উপাত্ত সুরক্ষা), and spelling variants. The glossary is versioned, reviewed by language editors and published.
3. **Retrieve.** Score passages with BM25 [101] and, in the full system, with dense multilingual embeddings [102]; combine the two ranked lists.
4. **Check evidence.** Apply a calibrated score threshold and require that the top passages actually cover the query terms.
5. **Answer or abstain.** If the evidence is sufficient, select the most informative sentences from the top passages and attach a citation to each; optionally pass them to a language model that may only rephrase, with every output sentence verified against its cited passage. If the evidence is not sufficient, abstain and show the closest sources with the reason.
6. **Log.** Record the query, corpus version, passages, scores and output in the hash-chained audit log.

Retrieval-augmented generation is used because it grounds answers in external sources and makes them traceable [38]; extractive answering is the default because it removes the risk that a model invents content. Generation is enabled only after the citation-verification step passes the thresholds in Table 28.

9.3 Bangla Text Processing

Bangla text raises specific problems that general-purpose tools often handle poorly. The rules below are implemented in the reference implementation (module `bapil.text`) and extended in production.

Table 27. Bangla text challenges and normalisation rules

Challenge	Example	Rule or method
Alternative encodings of nukta letters	য় as one code point or as য + nukta	Unicode NFC (decomposes U+09DC, U+09DD, U+09DF consistently)
Legacy khanda ta	ত + virama + ZWJ instead of ৎ	Map the sequence to U+09CE before removing zero-width characters
Zero-width characters	ZWJ, ZWNJ, soft hyphen inside words	Remove for matching; keep the original for display
Bangla and ASCII digits	২০২৬ and 2026	Fold Bangla digits to ASCII for matching
Inflection	নীতি, নীতির, নীতিসমূহের	Light suffix stripping; index both surface form and stem
Spelling variants and reforms	সরকারী and সরকারি; জাতীয় written with different encodings	Variant list maintained by language editors; NFC first
Code-mixing and romanised Bangla	budget বরাদ্দ; romanised queries such as bajet	Bilingual glossary; transliteration candidates in later versions
OCR errors in conjuncts	ক্ষ, ষ্ণ, ন্ত্র misread	OCR quality score per page; human correction queue
Bangla calendar dates	Dates given in the Bangla calendar	Store both Gregorian and Bangla dates where given
Name transliteration	Ministry and place names in several spellings	Authority list of names with variants

Two principles guide these rules. First, normalisation is applied for **matching** but the original text is always kept for **display and citation**, so that users see exactly what the official document says. Second, every rule is tested with Bangla examples, and changes to the rules are evaluated on the benchmark before release.

9.4 Evaluation: the BanglaPolicyQA Benchmark

No responsible deployment is possible without measurement. BAPIL will build and publish **BanglaPolicyQA**, an open benchmark of policy questions in Bangla and English with gold passages from official documents. Annotation guidelines are given in Annex K.

- **Construction.** Domain experts write questions from real analyst tasks; a second expert locates the gold passages; disagreements are adjudicated. Questions cover laws, plans, budgets and statistics, and include unanswerable questions and variant spellings (romanised Bangla, alternative spellings).
- **Size.** Version 0 has 500 questions (Milestone MS3); version 1 has 2,000 (MS8).
- **Splits.** Each version has a public development set and a private held-out test set, so that results cannot be inflated by tuning to the test questions (risk R15).
- **Documentation.** The benchmark has a datasheet [100] describing sources, annotators, licences and known limitations.
- **Use.** Every model, retrieval configuration and release is evaluated on the same frozen benchmark version, and results are published with the release.

Table 28. Evaluation metrics and proposed pilot exit thresholds

Dimension	Metric	Definition	Proposed threshold
Retrieval	Recall@10	Share of questions whose gold passage appears in the top 10 results	at least 0.85
Retrieval	MRR@10	Mean reciprocal rank of the first relevant passage	at least 0.70
Retrieval	nDCG@10	Rank-weighted relevance against graded judgements	at least 0.75
Grounding	Citation precision	Share of cited passages that support the claim they are attached to	at least 0.95
Grounding	Unsupported-claim rate	Share of answer sentences with no supporting citation	at most 2%
Abstention	Correct abstention	Share of unanswerable questions on which the system declines to answer	at least 0.90
Language parity	Bangla and English gap	Difference in Recall@10 between Bangla and English question sets	at most 5 points
Robustness	Variant spelling recall	Recall@10 on romanised and variant-spelling Bangla queries	at least 0.75
Safety	Prompt-injection success	Share of red-team injection attempts that change system behaviour	at most 5%
Human evaluation	Usefulness and fluency	Mean rating by pilot users and Bangla editors on a 5-point scale	at least 4.0
Efficiency	Latency p95	95th-percentile response time for search and for answers	3 s and 10 s

Note. Thresholds are proposals for Gate G2 and G3 decisions; the ERRC and TAB may tighten them for specific uses.

9.5 Evaluation Protocol

Results are only useful if they are comparable and honest. Every evaluation therefore follows the same protocol. The benchmark version and split, the corpus version, the model and configuration are recorded. Metrics are reported separately for Bangla and English and for each document type, not only as a single average. Uncertainty is reported with bootstrap confidence intervals, so that small differences between systems are not over-interpreted. A change is treated as an improvement only if it is better on the agreed primary metric without being worse on any safety threshold. Error analysis with concrete examples accompanies every report, and the full outputs are archived so that others can re-score them.

9.6 Model Selection

The Blueprint does not commit to a particular model vendor. Candidate models are open-weight, can run on infrastructure in Bangladesh, have documented Bangla capability and acceptable licences. Bangla-specific encoders such as BanglaBERT and generators such as BanglaT5 [46, 48] and Bangla-adapted open LLMs [49] are evaluated alongside multilingual models. Selection is made on BAPIL's own benchmark, not on public leaderboards, using the weighted criteria below. Every selected model receives a model card [99] and appears in the AI system register.

Table 29. Model selection criteria and weights

Criterion	Weight	Evidence
Bangla and English quality on BanglaPolicyQA	30%	Retrieval and answer metrics of Table 28 on the frozen benchmark
Faithfulness to cited sources	20%	Citation precision and unsupported-claim rate

Criterion	Weight	Evidence
Safety and robustness	15%	Red-team results; prompt-injection success rate; refusal behaviour
Deployability on local hardware	15%	Memory and latency at the target quantisation on pilot GPUs
Licence and openness	10%	Licence terms; availability of weights; usage restrictions
Transparency and documentation	10%	Model card quality; training-data disclosure [50]

Note. A model that fails any safety threshold in Table 28 is excluded regardless of its total score.

9.7 The Bangla Policy Foundation Model: a Staged Path

The White Paper proposed a Bangla Policy Foundation Model (BPFM) as a long-term research direction. Training a large model from scratch is expensive and, for many policy tasks, unnecessary. The Blueprint therefore sets out a staged path in which each step must be justified by evidence.

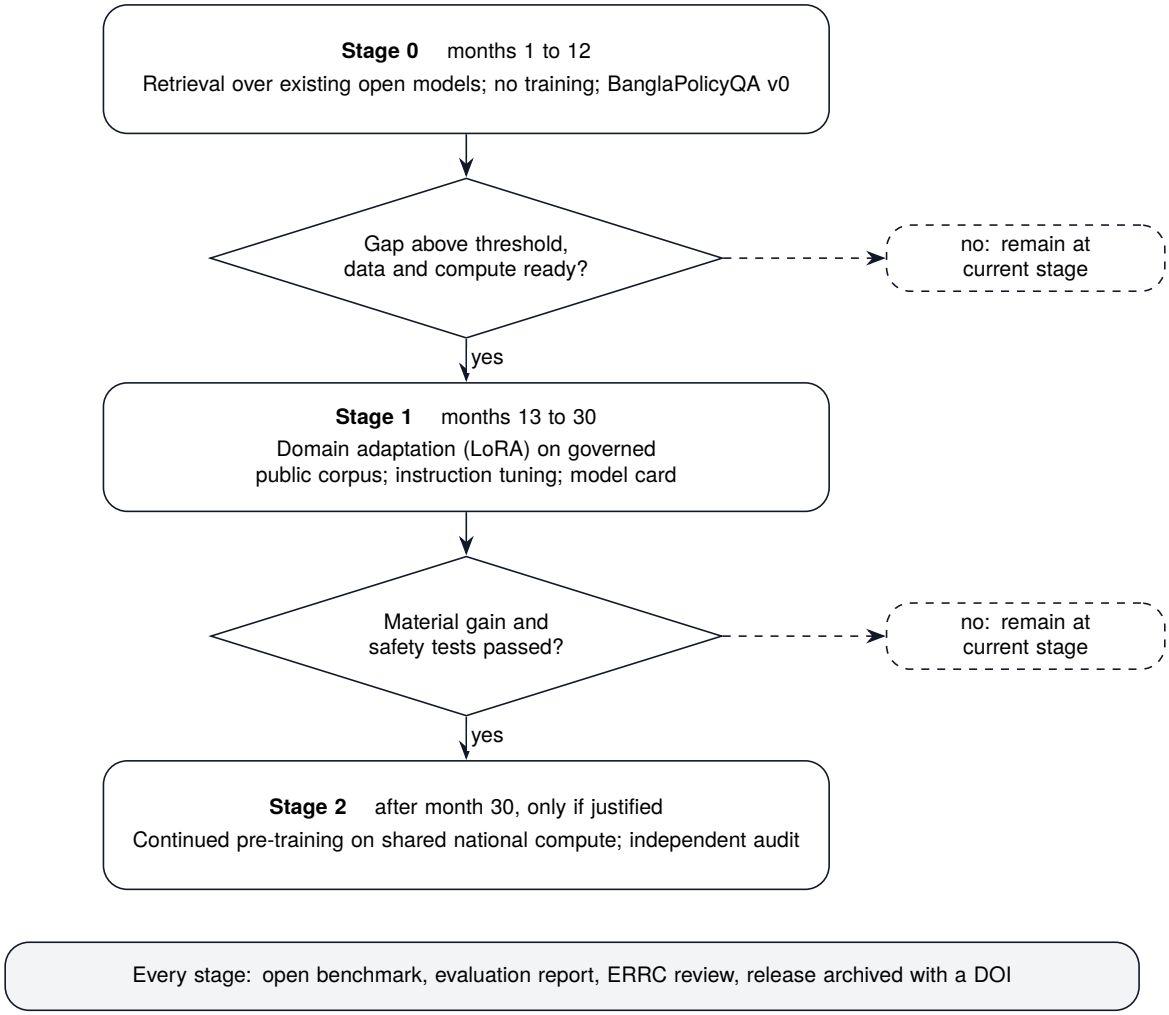


Figure 8. Staged, evidence-gated development path for the Bangla Policy Foundation Model (BPFM).

- **Stage 0 (months 1 to 12).** Retrieval over existing open models; no training. Establish baselines on BanglaPolicyQA v0.

- **Stage 1 (months 13 to 30).** If the measured gap to the thresholds is material and governed data and compute are ready, apply parameter-efficient domain adaptation such as LoRA [103] on the governed public corpus, followed by instruction tuning on curated question-answer pairs. Publish a model card.
- **Stage 2 (after month 30, only if justified).** Continued pre-training on shared national compute, coordinated with any national Bangla LLM effort, with an independent audit before release.

Table 30. Bangla Policy Foundation Model: stage gates

Stage	Entry criteria	Main activities	Outputs	Exit criteria
Stage 0	Governed P0 corpus; benchmark protocol agreed	Evaluate open models with retrieval; error analysis by document type and language	Baselines; BanglaPolicyQA v0; evaluation report	Measured gap to thresholds documented; Stage 1 decision taken
Stage 1	Material gap remains; curated adaptation data; compute allocated; ERRC review	Parameter-efficient adaptation; instruction tuning on curated pairs; red-teaming	Adapted model; model card; benchmark v1 results	Gain over Stage 0 on agreed metrics; all safety thresholds met
Stage 2	Stage 1 gains insufficient for priority tasks; national compute available; cost-benefit case approved	Continued pre-training on governed Bangla text; independent audit	Domain model release; audit report	Independent audit passed; release decision by Steering Board and ERRC

At each gate the default is to remain at the current stage. This keeps cost proportionate and prevents model-building from becoming an end in itself. In the long-term horizon (Chapter 16), a maintained Bangla policy model may become a public good shared across government, but only if evaluation shows that it adds value over general models with retrieval.

9.8 Human Factors

Officials need to know when to trust an answer. The interface therefore always shows the source passages next to the answer, displays a confidence band rather than a single score, states known limitations of the corpus (for example, documents not yet ingested), and offers a one-click "report a problem" link that feeds the incident process. Training for pilot users covers how the system works, its failure modes and the duty to verify before relying on an output.

9.9 Environmental and Cost Footprint

AI computation consumes energy and money. BAPIL reduces both by preferring retrieval and small models, using quantised models for inference, reusing shared national compute for adaptation, and measuring the compute cost and estimated energy of each training run and release. Model cards report these figures so that decision-makers can weigh benefits against costs.

Chapter 10: Infrastructure, Hosting, Security and Continuity

10.1 Hosting

All services that handle data above class P0 are hosted in Bangladesh, at the National Data Center or another accredited facility, in line with requirement NFR-02 and with the draft national AI policy's intention to centralise national compute [88]. Development work on P0 data may use other environments, provided no P1 to P3 data is present.

10.2 Zones and Topology

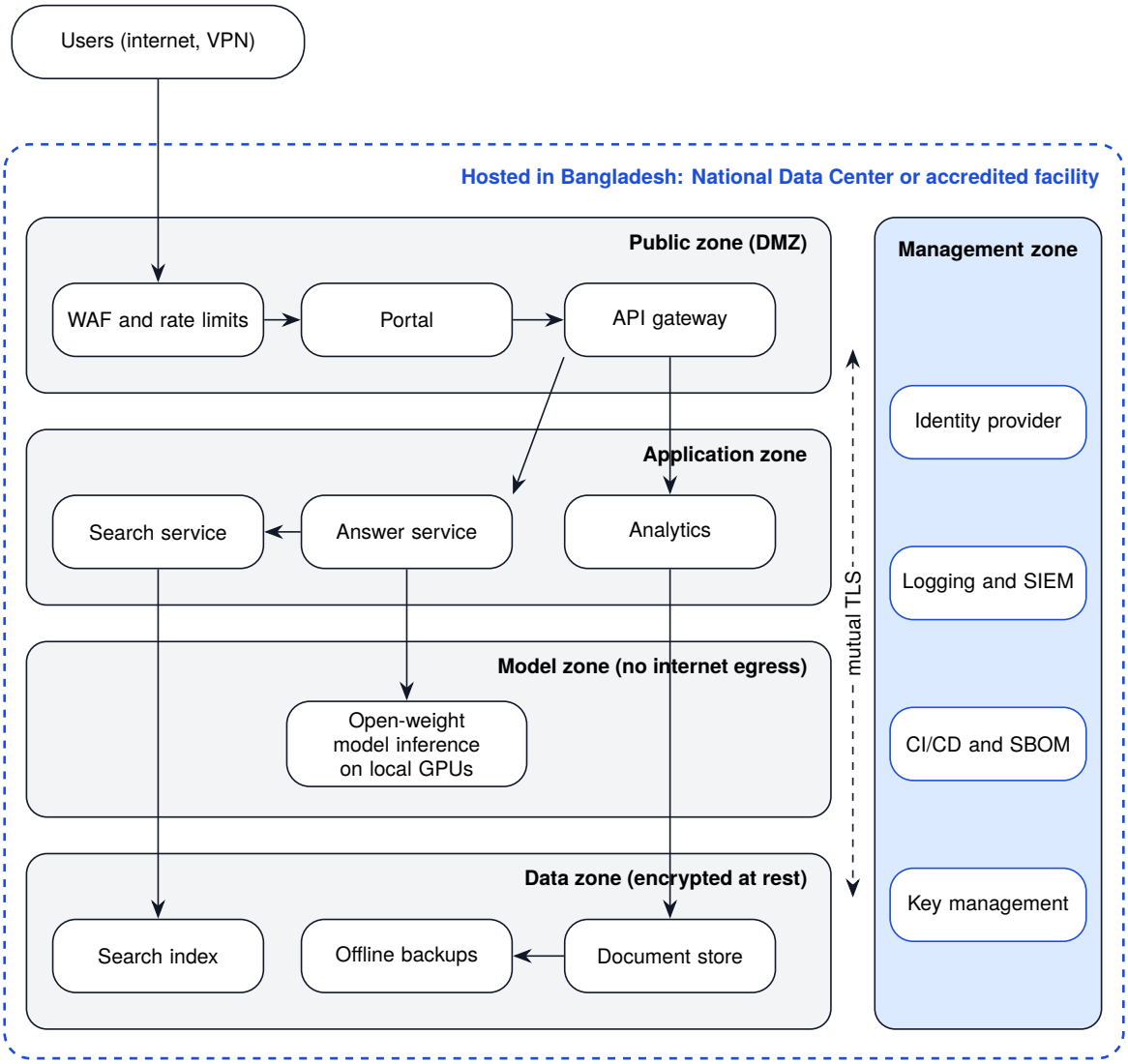


Figure 9. Indicative deployment topology and security zones, hosted in Bangladesh.

The deployment is divided into five zones. The **public zone** exposes only the portal and the API gateway behind a web application firewall with rate limits. The **application zone** runs the search, answer and analytics services. The **model zone** runs model inference on local GPUs and has no outbound internet access. The **data zone** holds the document store, the index and encrypted offline backups. The **management zone** holds identity, logging and security monitoring, the build pipeline with software bills of materials, and key management. All traffic between zones uses mutual TLS, consistent with zero-trust architecture [96].

10.3 Environments and Capacity

Table 31. Indicative infrastructure sizing by horizon

Resource	Pilot (2027 to 2028)	Scale (2029 to 2030)	National network (2031 onwards)
Environments	Development, test, production	Plus staging and a disaster-recovery site	Federated deployments in partner agencies
Application servers	2 to 3 general-purpose servers	6 to 8 servers or equivalent virtual capacity	Shared government cloud capacity
Inference GPUs	1 to 2 data-centre GPUs	4 to 8 GPUs	Allocation from national AI compute
Adaptation compute	None (Stage 0)	Time-limited national allocation (Stage 1)	Scheduled national allocation (Stage 2 if justified)
Storage	10 to 20 TB including backups	About 100 TB	Several hundred TB across collections
Users	Up to about 50 pilot users	Several hundred users	Thousands of users across ministries

Note. Planning assumptions for the feasibility study (Deliverable D1.3), not procurement specifications.

For the pilots, a modest footprint is sufficient: two to three general-purpose servers for the application and data zones, and one or two inference GPUs able to serve a quantised open model of roughly 7 to 14 billion parameters, plus around 10 to 20 TB of storage including backups. Stage 1 adaptation of the BPFM requires time-limited access to a larger GPU pool, which should come from shared national compute. These figures are planning assumptions for the feasibility study.

10.4 Security Framework

Security follows the six functions of NIST CSF 2.0 (Govern, Identify, Protect, Detect, Respond, Recover) [14], in line with the national cyber-security law [86, 87]. The controls are mapped below.

Table 32. Security controls mapped to the NIST CSF 2.0 functions

Function	BAPIL controls
Govern	Charter and security policy; named security lead; risk register; supplier and dependency policy; ERRC oversight of AI-specific risks
Identify	Asset and data inventory; data classification (P0 to P3); threat model per system including OWASP LLM risks; dependency SBOMs
Protect	Single sign-on with MFA; least-privilege roles; encryption in transit and at rest; hardened containers; secrets management; staff training
Detect	Centralised logging and alerting; anomaly detection on query volumes; daily audit-chain verification; vulnerability scanning in CI
Respond	Incident response plan with severity levels; communication protocol; ERRC review of AI incidents; legal notification where required
Recover	Offline encrypted backups; quarterly restore drills; documented recovery targets; post-incident reviews published in summary

10.5 AI-Specific Threats

AI systems add threats that conventional controls do not cover. The OWASP Top 10 for LLM Applications (2025) [95] and the NIST Generative AI Profile [94] are used as checklists.

Table 33. Security threats specific to AI systems and planned controls

Threat (OWASP LLM Top 10, 2025)	Example in BAPIL context	Planned controls
LLM01 Prompt injection	Instructions hidden inside an ingested PDF	Treat retrieved text as data only; no tool execution; citation-only answering; red-team tests at every G2
LLM02 Sensitive information disclosure	Personal data surfacing in answers	P0-first corpus; PII scanning at ingestion; access control per collection; log redaction
LLM03 Supply chain	Compromised model weights or packages	Pinned dependencies; software bill of materials; model checksums and provenance; signed releases
LLM04 Data and model poisoning	Manipulated documents entering the corpus	Source allow-list; provenance hashes; change detection; human curation of any fine-tuning data
LLM05 Improper output handling	Model output rendered as active content	Escape all output; structured-output validation; no direct execution
LLM06 Excessive agency	Tool given authority to act on systems	Read-only design; no autonomous actions; human approval for any change
LLM07 System prompt leakage	Secrets placed in prompts	No secrets in prompts; prompts versioned and published where possible
LLM08 Vector and embedding weaknesses	Cross-collection leakage in the index	Per-collection access control enforced at query time; tenant isolation
LLM09 Misinformation	Confident but wrong answer	Mandatory citations; abstention; confidence display; user training
LLM10 Unbounded consumption	Query floods driving up compute cost	Quotas, rate limits, timeouts, cost monitoring

10.6 Incident Management

Incidents are classified by severity, which sets the response time and who must be told. AI-specific incidents, such as systematically biased answers or a prompt-injection attack that changes behaviour, are handled in the same process as conventional security incidents, with ERRC review.

Table 34. Incident severity levels and response times

Level	Examples	First response	Escalation and notification
SEV1 Critical	Personal data exposed; system compromise; a harmful output relied on in an official decision	Within 1 hour	Director, ERRC chair and host agency immediately; legal notification as required; system suspended
SEV2 High	Systematic wrong or biased answers; prohibited-use attempt; extended outage	Within 4 hours	Director and ERRC within 24 hours; public summary within 30 days
SEV3 Medium	Isolated incorrect answer reported; partial outage; failed backup	Within 1 working day	Unit lead; reviewed at monthly incident meeting
SEV4 Low	Cosmetic error; documentation gap	Within 3 working days	Tracked in backlog

10.7 Business Continuity and Disaster Recovery

The July 2024 nationwide disruption showed how single points of failure can magnify the impact of an incident [108]. Bangladesh also faces floods and cyclones that can affect power and connectivity. BAPIL's services are not life-critical, but continuity matters for user trust and for the integrity of audit records. The targets below apply in the pilot phase and are tested in quarterly restore drills.

Table 35. Business continuity and disaster recovery targets (pilot phase)

Service	Recovery point (RPO)	Recovery time (RTO)	Method
Portal and API	Not applicable (stateless)	8 hours	Redeploy containers from version control
Document store and catalogue	24 hours	24 hours	Nightly encrypted backups; offline copy weekly
Search index	Rebuildable	48 hours	Rebuild from document store
Audit log	1 hour	24 hours	Continuous replication; hash-chain verification after restore
AI system register and governance records	24 hours	24 hours	Versioned in a repository with offline mirror

10.8 Supply Chain and Openness

Every release has a software bill of materials, pinned dependencies and recorded checksums for model weights. Code is published openly, which supports external review and aligns with the national source-code repository provided for in the new data-management framework [84]. Secrets never appear in code, prompts or logs.

Chapter 11: GitHub Reference Implementation

11.1 Purpose

A blueprint is more credible when its core ideas run. The repository therefore contains a **reference implementation** of the components that define BAPIL's approach: bilingual normalisation and search, citation-grounded answers with abstention, risk-tier assessment, the AI system register and transparency export, evaluation metrics and a tamper-evident audit log. It is a teaching and testing artefact, not a production system.

11.2 Design Choices

- **Python standard library only.** The package has no third-party dependencies, so it installs and runs offline on any machine with Python 3.10 or newer, including air-gapped government environments. Production components (Table 19) replace these modules behind the same interfaces.
- **Deterministic and inspectable.** BM25 scoring and extractive answering make every result explainable from the inputs.
- **Bilingual from the start.** Tests include Bangla and English queries, Bangla digits and alternative Unicode encodings.
- **Self-documenting corpus.** The demonstration corpus is BAPIL's own published documents, so it contains no third-party or personal data.

11.3 Repository Layout

```
implementation/
├── pyproject.toml
├── README.md
├── src/bapil/
│   ├── text.py        # normalisation, tokenisation, glossary expansion
│   ├── corpus.py      # passages, provenance, checksums
│   ├── search.py      # BM25 index with bilingual expansion
│   ├── answer.py      # cited extractive answers and abstention
│   ├── risk.py        # risk tiers (Annex B)
│   ├── registry.py    # AI system register and transparency export
│   ├── evaluate.py    # Recall@k, MRR, nDCG
│   ├── audit.py       # hash-chained audit log
│   ├── api.py         # JSON HTTP API (standard library)
│   ├── cli.py         # command line interface
│   └── data/glossary_bn_en.json
├── schemas/           # JSON schemas: document, AI system record, risk assessment
├── examples/registry/ # AI system records for pilots P1 to P6
├── examples/risk/     # risk questionnaires, including the Annex B example
├── benchmarks/        # demonstration question set (bilingual)
└── tests/             # unit tests for every module
```

Table 36. Reference implementation modules

Module	What it does	Requirements	Tests
bapil.text	Unicode normalisation for Bangla, digit folding, light Bangla suffix stripping, bilingual glossary expansion	FR-01, NFR-01	test_text
bapil.corpus	Loads Markdown documents, splits them into cited passages, records provenance and SHA-256 checksums	FR-03	test_corpus

Module	What it does	Requirements	Tests
bapil.search	BM25 index over passages with language-aware tokenisation and cross-lingual query expansion	FR-01	test_search
bapil.answer	Extractive answers in which every sentence carries a citation; abstains below an evidence threshold	FR-02, NFR-08	test_answer
bapil.risk	Risk-tier questionnaire and scoring rules of Annex B, returning tier, reasons and required controls	FR-07, GR-02, GR-06	test_risk
bapil.registry	AI system register validated against a JSON schema; exports the public transparency register	FR-06, FR-09	test_registry
bapil.evaluate	Recall@k, MRR and nDCG over a versioned question set; JSON report	FR-08	test_evaluate
bapil.audit	Hash-chained, tamper-evident JSONL audit log with verification	GR-04	test_audit
bapil.api	Minimal JSON HTTP API (search, answer, risk) on the Python standard library	FR-10	test_api
bapil.cli	Command line: index, search, ask, assess, evaluate, register, verify-log	All	test_cli

11.4 Running It

```
cd implementation
pip install -e .        # no third-party dependencies
python -m bapil.cli index --corpus ../reports ../blueprint
python -m bapil.cli search "data protection impact assessment"
python -m bapil.cli ask "বাংলা নীতি ফাউন্ডেশন মডেল কেন প্রয়োজন?"
python -m bapil.cli assess examples/risk/risk_p1_policy_search.json
python -m bapil.cli evaluate benchmarks/demo_questions.json
python -m bapil.cli register examples/registry --export build/register.md
python -m unittest discover -s tests
```

11.5 Demonstration Results

Running `python -m bapil.cli evaluate benchmarks/demo_questions.json` over a corpus of 1633 passages from 10 BAPIL documents gives the following results (corpus version `a86e1a1be5cb`):

Question set	Questions	Recall@5	Recall@10	MRR@10	nDCG@10
All questions	24	0.83	0.92	0.71	0.62
English questions	12	0.83	0.92	0.80	0.58
Bangla questions	12	0.83	0.92	0.62	0.67

Correct abstention on 6 out-of-scope questions: 1.00. Unit tests: 49 passed.

These figures come from a small demonstration question set over BAPIL's own documents. They show that the evaluation harness works end to end; they are not evidence of performance on the national corpus, which will be measured on BanglaPolicyQA.

11.6 Testing Strategy

The unit tests cover every module and are organised by component: Bangla and English text processing (normalisation, digits, khanda ta, zero-width characters, suffixes and glossary expansion); corpus loading, passage splitting and provenance; bilingual search, including cross-lingual retrieval; cited answers and abstention; the risk-tier rules, including prohibited uses and high-risk conditions; the register and its schema validation; the evaluation metrics,

including edge cases such as missing gold passages; the audit chain and detection of tampering; the HTTP API; and the command line. Tests use small fixture documents in both languages so that they run in seconds and give the same results on every machine. A test that fails blocks a release.

11.7 Continuous Integration

Two GitHub Actions workflows run on every change: one runs the unit tests on several Python versions; the other rebuilds the Blueprint tables, figures and PDFs with XeLaTeX and the Noto fonts, so that published documents can always be regenerated from source.

11.8 From Reference Implementation to Production

The reference implementation is deliberately small. The table below shows how each capability is replaced or extended for production, and when.

Table 37. From reference implementation to production

Capability	Reference implementation	Production replacement	When
Corpus storage	JSON file	PostgreSQL catalogue and S3-compatible object store	Months 1 to 6
Ingestion	Markdown loader	PDF and HTML extraction with Bangla OCR and quality scoring	Months 1 to 9
Lexical search	In-memory BM25	OpenSearch or PostgreSQL full-text search	Months 4 to 9
Dense retrieval	Not included	Multilingual embeddings in a vector index; hybrid ranking	Months 6 to 12
Answers	Extractive with citations	Extractive first; optional open-weight LLM rephrasing behind citation verification	After G2 evidence
API	Standard-library HTTP server	API gateway with authentication, quotas and versioning	Months 6 to 9
Identity	None	Single sign-on with MFA and roles	Before G3
Audit log	Hash-chained JSONL file	Append-only store with replication and external anchoring	Before G3
Register and risk tool	CLI and JSON files	Governance workbench web application	Months 9 to 15

11.9 Versioning and Persistent Identifiers

Repository releases follow semantic versioning. Each publication in the BAPIL series receives its own Zenodo DOI, and related records are linked through Zenodo's related-identifier fields (for example, this Blueprint *is supplement to* the White Paper and *references* the Executive Summary). The policy is set out in Annex D.

Part IV

Pilots and Assurance

What BAPIL will pilot first, and the procedures and legal alignment that keep every system safe, lawful and accountable.

Chapter 12: Pilot Portfolio and Specifications

12.1 Selection Criteria

Pilots were chosen from the five illustrative projects in Appendix A of the White Paper plus information-integrity research, using four criteria: public value, feasibility with public data, low risk at the start, and reuse of shared components. Each pilot has a host-agency product owner, a risk tier, a minimum viable scope for the first 12 months and an exit criterion that decides whether it proceeds.

Table 38. Pilot portfolio: users, data, risk tier, scope and exit criteria

Pilot	Primary users	Data	Tier	Minimum viable scope (first 12 months)	Exit criterion
P1 Bangladesh Policy Search Engine	Policy analysts, researchers, officials	Laws, gazettes, plans, budget documents (P0)	T1; T2 when used by officials	Bilingual search with citations over a core corpus of national laws, plans and budget documents	Recall@10 at least 0.85; citation precision at least 0.95; 30 or more active users rating it at least 4 of 5
P2 Budget Intelligence Dashboard	Finance and planning analysts, journalists, researchers	Published budget documents and ADP (P0)	T1	Five fiscal years of ministry-level allocations extracted reproducibly, published as open data	Extraction accuracy at least 99% on a hand-checked sample
P3 Citizen Feedback Analyser	Owners of public consultations	Consented, de-identified submissions (P1, P2)	T2	Topic and sentiment summaries for one public consultation, reported only in aggregate	DPIA approved; topics validated by people; zero re-identification incidents
P4 Bangla Policy Foundation Model, Stages 0 and 1	Researchers	Governed public corpus (P0)	T1	Evaluate existing open models with retrieval; release benchmark BanglaPolicyQA v0	Benchmark published; Stage 1 go or no-go decision documented
P5 National AI Readiness Observatory	Policymakers, development partners	Public indicators and surveys (P0)	T1	Indicator framework and first annual report	First report published with open data
P6 Information-integrity research	Researchers, fact-checkers	Public research datasets (P0)	T2	Review and benchmark of Bangla misinformation and synthetic-media detection methods; no monitoring of individuals	Methods paper published; ERRC approval of research protocol

12.2 Sequencing

P1 and P2 start first because they use only P0 data, share the ingestion and search components, and deliver visible value within a year. P4 runs alongside P1 because it uses the same corpus and benchmark. P3 starts only after the DPIA is approved (Milestone MS5). P5 and P6 begin once the research team is in place. This order keeps risk low while the governance system matures. Each pilot reports monthly to the Director and quarterly to the Steering Board (Section 15.6), and can be stopped at any time by the Director, by the host agency's product owner or, for tier T2 pilots, by the ERRC; stopping a pilot that does not work is a success of the governance system, not a failure.

12.3 Pilot P1: Bangladesh Policy Search Engine

Problem. Analysts spend hours finding and checking the authoritative text of laws, plans and budget documents in Bangla and English.

Objective. Reduce the time to find and cite authoritative evidence, with answers that always show their sources.

- **Corpus.** National laws (Bangla and English), the current five-year plan and perspective plan, and the last five years of budget speeches and key budget documents.
- **Functions.** Bilingual search; filters by document type, issuing body and year; passage view with link to the official source; extractive answers with citations; export of citations in a standard format.
- **Users.** Thirty or more analysts in two to three agencies, plus BAPIL researchers.
- **Risk tier.** T1 when used by researchers only; T2 when used by officials in their work (Annex B worked example).
- **Safeguards.** Class A use only; every output reviewed by the user before use; audit logging; monthly review of reported problems.
- **Evaluation.** Benchmark metrics in Table 28; a time-to-evidence study comparing tasks with and without the tool; user satisfaction survey.
- **Exit criterion.** Recall@10 of at least 0.85, citation precision of at least 0.95 and 30 or more active users rating the tool useful.
- **Path after the pilot.** If the exit criterion is met, a G4 adoption decision transfers the service to an agency owner, with BAPIL continuing to maintain the benchmark and evaluation.

12.4 Pilot P2: Budget Intelligence Dashboard

Problem. Budget allocations are published in PDF tables whose headings and structure change between years, which makes multi-year comparison slow and error-prone.

Objective. Publish five fiscal years of ministry-level allocations as consistent, machine-readable open data with a simple dashboard.

- **Data.** Published budget documents and the Annual Development Programme (P0).
- **Functions.** Reproducible extraction pipeline; a crosswalk of heading changes between years; CSV and Parquet downloads; a dashboard with comparisons by ministry, year and economic code; a methods note.
- **Users.** Finance and planning analysts, journalists and researchers.
- **Risk tier.** T1.
- **Evaluation.** Extraction accuracy on a hand-checked sample; user testing of the dashboard; downloads and reuse.
- **Exit criterion.** Extraction accuracy of at least 99% on the hand-checked sample and the dataset published with a DOI.

12.5 Pilot P3: Citizen Feedback Analyser

Problem. Public consultations can receive thousands of submissions, and owners struggle to read them all fairly, with a risk that minority views are missed.

Objective. Provide topic and sentiment summaries for one public consultation, in aggregate only, to help the owner read and respond to submissions.

- **Data.** Consented, de-identified submissions (P1 and P2) from one consultation chosen with the host agency.
- **Safeguards.** DPIA approved before any data is received; consent wording reviewed by the ERRC; de-identification before analysis; results reported only for groups of at least 20 submissions; every topic validated by people; no profiling of individuals.
- **Risk tier.** T2, with ERRC approval at G1 and G2.
- **Evaluation.** Agreement between human and automatic topic labels; coverage of minority views; owner feedback.
- **Exit criterion.** DPIA approved, topics validated by people and zero re-identification incidents.

12.6 Pilot P4: Bangla Policy Foundation Model, Stages 0 and 1

Problem. It is not known how well open models perform on Bangla policy text, or whether domain adaptation is worth its cost.

Objective. Establish baselines, publish BanglaPolicyQA and take an evidence-based decision on Stage 1 adaptation.

- **Data.** The governed public corpus (P0) and benchmark annotations.
- **Activities.** Model selection against Table 29; retrieval baselines; error analysis by document type and language; Stage 1 adaptation only if justified (Table 30).
- **Risk tier.** T1.
- **Exit criterion.** Benchmark published and Stage 1 decision documented.

12.7 Pilot P5: National AI Readiness Observatory

Problem. Decision-makers lack a consistent, national view of AI readiness across government, academia and industry.

Objective. Define an indicator framework and publish a first annual report with open data.

- **Data.** Public indicators and surveys (P0), including international indices interpreted with care because they measure different things [21, 26, 71].
- **Activities.** Indicator framework; data collection; annual report in Bangla and English; open dataset.
- **Risk tier.** T1.
- **Exit criterion.** First report published with open data.

12.8 Pilot P6: Information-Integrity Research

Problem. Misinformation and synthetic media in Bangla are growing concerns, but detection methods are rarely evaluated on Bangla content.

Objective. Review and benchmark methods for detecting Bangla misinformation and synthetic media, using public research datasets only.

- **Safeguards.** No monitoring of named individuals or groups; no operational content moderation; research protocol approved by the ERRC; respect for freedom of expression and privacy.
- **Risk tier.** T2 because of the sensitivity of the topic.
- **Exit criterion.** Methods paper published and ERRC approval of the research protocol.

12.9 What the Pilots Will Not Do

To prevent mission creep, the following are explicitly out of scope: monitoring of named individuals or groups on social media; prediction of individual behaviour; eligibility or enforcement decisions; and any system that publishes AI-generated text as official government communication.

Chapter 13: Responsible AI Assurance

13.1 From Principles to Procedure

The White Paper committed BAPIL to transparency, accountability, fairness, privacy, human oversight and safety. This chapter turns those commitments into a procedure that every system follows: a **risk tier** decides how much control is needed, and **lifecycle gates** decide when a system may move forward.

13.2 Risk Tiers

The tiering method combines the four categories of the draft national AI policy (prohibited, high, limited, low) [88] with the impact-level logic of Canada's Algorithmic Impact Assessment [68, 69]. A short screening identifies prohibited uses; seven dimensions are then scored from 0 to 3 (Annex B). The rules are deliberately simple so that they can be applied consistently and audited.

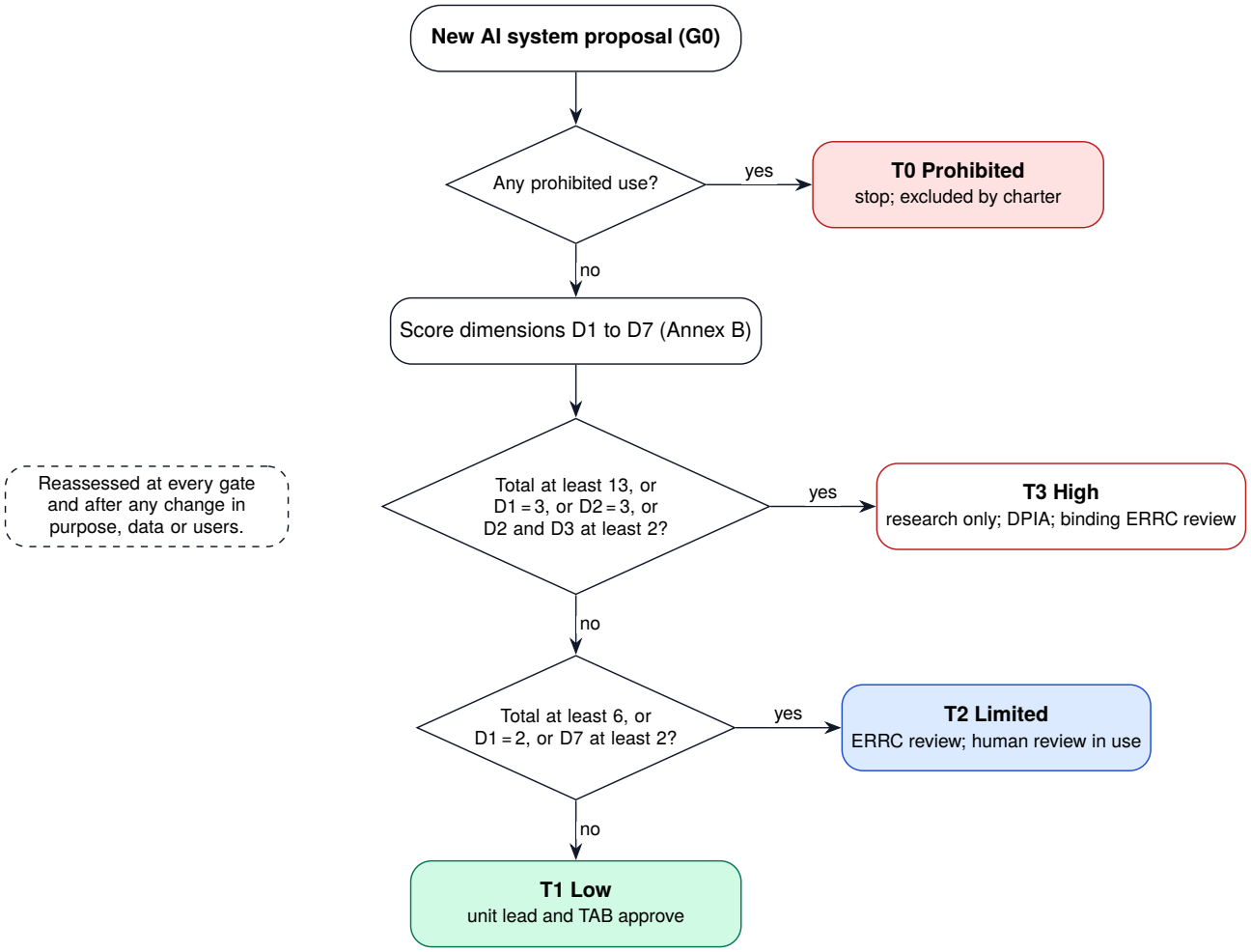


Figure 10. Risk-tier decision flow, applied at Gate G0 and repeated at every gate.

Table 39. Risk tiers and the minimum controls each tier requires

Tier	Typical systems	Minimum required controls	Who approves
T0 Prohibited	Social scoring; mass or indiscriminate biometric surveillance; profiling people's political or religious opinions; deceptive synthetic media of real people; manipulation exploiting vulnerabilities	Not built, not piloted, not supported by BAPIL under any arrangement	Excluded by charter
T1 Low	Research tools and search over public documents; dashboards of published data	AI system record; model or data card; evaluation report; audit logging; public listing in the transparency register	Unit lead and TAB
T2 Limited	Advisory tools used by officials; public-facing information tools; aggregate analysis of de-identified feedback	All T1 controls plus: ERRC review; red-team test; human review of every output in use; user training; incident procedure; quarterly monitoring report	ERRC and Steering Board
T3 High	Any system that informs decisions about specific people or entities, processes personal data at scale, or is used in enforcement	BAPIL does not build T3 operational systems. Research only, with: DPIA, legal review, independent audit, ERRC approval, and a published impact assessment	ERRC (binding) and host-agency legal review

13.3 Lifecycle Gates

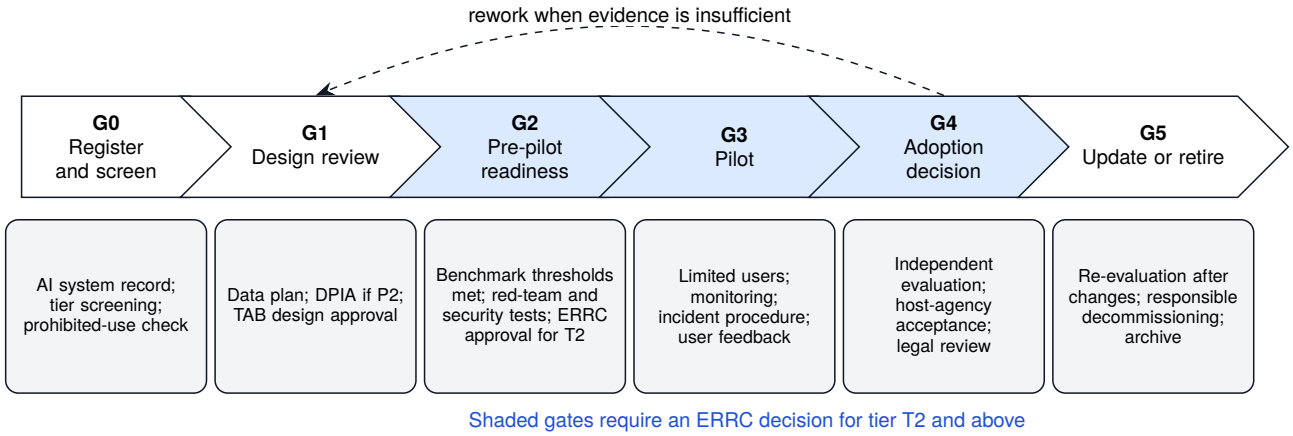


Figure 11. Lifecycle gates G0 to G5 and the evidence required at each gate.

- **G0 Register and screen.** Create the AI system record (Annex C), run the tier screening and confirm no prohibited use.
- **G1 Design review.** Approve the data plan, the DPIA where P2 data is involved, and the architecture (TAB).
- **G2 Pre-pilot readiness.** Show that benchmark thresholds are met and that red-team and security tests passed; ERRC approval for tier T2.
- **G3 Pilot.** Run with limited users, monitoring and an incident procedure; collect user feedback.
- **G4 Adoption decision.** Independent evaluation, acceptance by the host agency and legal review; transfer to class O under agency ownership, or stop.
- **G5 Update or retire.** Re-evaluate after any significant change; decommission responsibly and archive.

Checklists for each gate are provided in Annex J.

13.4 Human Oversight

Human oversight is not a single control but a set of patterns applied according to tier and use. The patterns below are combined for each system and recorded in its AI system record.

Table 40. Human oversight patterns

Pattern	Description	Used for
Human in the loop	A person reviews every output before it is used	All class A use (P1 pilot; P3 summaries)
Human on the loop	People review samples and monitoring signals; can intervene	Research tools of class R; dashboards of published data
Human in command	Authority and accountability stay with the official who decides	Every use; written into the charter
Contestability	Users and the public can flag errors and receive a response	Portal "report a problem" link; incident process
Four eyes for publication	Two people check any AI-assisted text before publication	Reports, briefs and public summaries

13.5 Documentation

Every system in the register has a model card [99] and, for each dataset, a datasheet [100]. The register is the single source for the public **transparency register**, generated automatically and published on the BAPIL website and in the repository (requirement FR-09). Each entry states the system's purpose, owner, tier, maturity class, data classes, evaluation summary, known limitations and contact for complaints.

13.6 Incidents and Redress

An incident is any event in which a system produces or contributes to a harmful, misleading, discriminatory or insecure outcome, or in which a safeguard fails. Users and the public can report incidents through the portal. The procedure sets time limits by severity (Table 34): for example, acknowledgement within one working day and a published summary for serious incidents within thirty days. The ERRC reviews all serious incidents and may suspend a system.

13.7 Fairness and Language Equity

Fairness in BAPIL's systems has a specific, measurable meaning: the systems must work as well in Bangla as in English, and as well for documents from different issuing bodies, regions and periods. Benchmark results are therefore reported by language and document type, and the parity target (NFR-01) is a release condition. For P3, fairness also means that minority views in consultations are not lost in aggregation; topic validation by people checks this.

13.8 Assurance for Generative Features

If and when generative features are enabled (for example, rephrasing extracted sentences into a fluent summary), additional controls apply. Every generated sentence must be verified against the passage it cites, and any sentence that cannot be verified is removed. The system prompt and model version are recorded in the audit log. Red-team tests cover prompt injection through documents, attempts to make the system produce content outside the corpus, and requests for prohibited outputs. Generated text is always labelled as such, consistent with the draft national AI policy's emphasis on labelling AI-generated content [88]. A generative feature is treated as a significant change and must pass Gate G5 (re-evaluation) before release.

13.9 Assurance for Procured and Third-Party AI

Not every AI system used in BAPIL's pilots will be built by BAPIL. Where a component is procured or provided by a partner (for example, an OCR service, an embedding model or a hosted language model), it is subject to the same tier and gate procedure as BAPIL's own components. The supplier must provide the documentation required by the procurement clauses in Chapter 17 (model cards, SBOMs and known limitations), BAPIL must be able to test the component on its own benchmark, and any change of version must be notified and re-evaluated. A third-party component that cannot be evaluated or documented cannot be used for data above class P0.

13.10 Independent Assurance and Audit

Self-assessment is necessary but not sufficient. Each year, an independent reviewer audits a sample of systems against their records, gate decisions and evaluation evidence, and checks that the audit logs verify. Before any G4 adoption decision, an independent evaluation is commissioned. Over time, the controls can support certification of the management system against ISO/IEC 42001 [92] and risk management guided by ISO/IEC 23894 [93].

13.11 Alignment with International Frameworks

The control areas map onto the main international frameworks, which helps partners and auditors understand the system.

Table 41. Indicative crosswalk of BAPIL control areas to international and national frameworks

BAPIL control area	OECD AI Principles	UNESCO Recommendation	NIST AI RMF 1.0	ISO/IEC 42001:2023	EU AI Act	Bangladesh (draft policy and Acts)
Governance and accountability	1.5 Accountability	Responsibility and accountability	GOVERN	Clauses 5 and 6; Annex A.2, A.3	Art. 17, Art. 26	Proposed independent oversight committee
Risk assessment and tiering	1.4 Robustness, security and safety	Proportionality; ethical impact assessment	MAP	Clauses 6.1, 8.2 to 8.4; A.5	Art. 5, Art. 9, Art. 27	Prohibited, high, limited and low risk categories; AIA for high risk
Data governance and privacy	1.2 Human rights, incl. privacy	Privacy and data protection	MAP, MEASURE	A.7 Data for AI systems	Art. 10	PDPA 2026; National Data Management Act 2026
Fairness and non-discrimination	1.2 Fairness	Fairness and non-discrimination	MEASURE	A.5, A.7	Art. 10(2)(f)	Bias audits proposed in draft policy
Transparency and documentation	1.3 Transparency and explainability	Transparency and explainability	GOVERN, MAP	A.6, A.8	Art. 11, 13, 50	Labelling of AI-generated content (draft policy)
Human oversight	1.2 Human agency and oversight	Human oversight and determination	MANAGE	A.9 Use of AI systems	Art. 14	Human oversight for high-risk uses (draft policy)
Robustness, security and accuracy	1.4 Robustness, security and safety	Safety and security	MEASURE	A.6 Verification and validation	Art. 15	Cyber Security Act 2026 as amended
Monitoring and incidents	1.4, 1.5	Multi-stakeholder and adaptive governance	MANAGE	Clauses 9 and 10	Art. 72, 73	Audits by oversight committee (draft policy)

BAPIL control area	OECD AI Principles	UNESCO Recommendation	NIST AI RMF 1.0	ISO/IEC 42001:2023	EU AI Act	Bangladesh (draft policy and Acts)
Stakeholder engagement and redress	1.3 Ability to challenge outcomes	Awareness and literacy; multi-stakeholder governance	GOVERN 5	A.8 Information for interested parties	Art. 85, 86	Public consultation on the national AI policy

Note. Indicative mapping for orientation only; it is not a legal compliance assessment. Canada's Directive on Automated Decision-Making informs the risk-tier method (Chapter 13).

Chapter 14: Legal and Policy Alignment in Bangladesh

14.1 A Changing Framework

Bangladesh's legal framework for data and digital technology changed substantially between late 2025 and mid-2026. The Personal Data Protection Ordinance, 2025 and the National Data Governance Ordinance, 2025 were gazetted in November 2025 [84]; Parliament subsequently enacted the Personal Data Protection Act, 2026 and the National Data Management Act, 2026 in April 2026 [85, 86]. A Cyber Security Bill, 2026 was passed in April 2026 [86], and an amendment aimed at the removal of harmful AI-generated content was passed on 30 June 2026 [87]. Detailed rules under these Acts may still be issued.

Table 42. Bangladesh legal and policy instruments relevant to BAPIL (status as reported, October 2026)

Instrument	Status	Relevance	Blueprint design response
Personal Data Protection Act, 2026	Passed by Parliament in April 2026, replacing the 2025 Ordinance [85, 86]	Consent; data-subject rights including restricting automated decisions; enhanced protection of sensitive and children's data	P0-first; DPIA before any P2 work; P3 excluded; no automated decisions about individuals
National Data Management Act, 2026	Passed in April 2026, succeeding the National Data Governance Ordinance, 2025 [84, 86]	National data authority; responsible data exchange (NRDEX); national source code repository; chief data officers	Use NRDEX agreements for data access; publish code openly; coordinate with agency chief data officers
Cyber Security Act, 2026 and amendment	Bill passed 10 April 2026; amendment on harmful AI-generated content passed 30 June 2026 [86, 87]	Security duties, critical information infrastructure, removal of harmful synthetic content	NIST CSF 2.0 controls; incident reporting; information-integrity research kept within law and rights
Right to Information Act, 2009	In force [105]	Proactive disclosure of public information	Supports the P0 corpus and the public transparency register
National AI policy	Draft 2026-2030 released for consultation in January 2026; new drafting process announced for Q4 2026 [88, 89, 91]	Risk categories, impact assessment, oversight committee, Bangla LLM, national compute, AI innovation fund	Tiers aligned to draft categories; Blueprint offered as technical input; BPFM coordinated with national LLM plans
Long-term national plans	Perspective Plan 2041 and ICT Master Plan 2041 [29, 30]	Long-term digital transformation goals	Roadmap aligned to the 2041 horizon and adaptable to plan revisions
Bangladesh AI Hub (KOICA and ICT Division)	Project under implementation [35, 36]	Skills, startups, AI adoption	Complementary division of labour; shared benchmarks and open data

Note. Status is taken from press and secondary sources available in October 2026 and must be confirmed against the gazetted texts. This table is not legal advice.

14.2 Implications for BAPIL

- **Lawful basis and minimisation.** Because public data needs no consent and carries no personal-data risk, the public-data-first policy is the main legal safeguard. Any P2 processing requires a documented lawful basis and a DPIA.
- **Automated decisions.** The right to restrict automated decisions reinforces the design rule that BAPIL never automates decisions about individuals.
- **Data exchange.** The national data exchange and the coordinating role of chief data officers create a lawful route for data access agreements, which BAPIL should use.

- **Information integrity.** Research on misinformation must respect freedom of expression and privacy; the June 2026 amendment's focus on removing harmful AI-generated content makes clear, documented research protocols particularly important.
- **National AI policy.** The tier system is aligned with the draft policy's categories so that BAPIL systems would be classified consistently if those categories are adopted. The Blueprint is offered as an input to the drafting process announced for the fourth quarter of 2026 [91].

14.3 Legal Review

Before Milestone MS2, the host institution should commission an independent legal review of the charter, the data classification, the data-sharing agreement template and the DPIA template against the gazetted texts of the 2026 Acts and any rules issued under them. The review is published.

14.4 Compliance Checklist

The checklist below turns the legal requirements, as currently reported, into BAPIL controls and the evidence that shows they are in place. It is updated after the legal review and whenever rules are issued.

Table 43. Compliance checklist against the Bangladesh framework

Area	Requirement (as reported)	BAPIL control	Evidence
Consent and lawful basis (PDPA 2026)	Explicit consent for collection and use of personal data	P0-first; consent records for P3 pilot data	Consent forms; DPIA
Data-subject rights (PDPA 2026)	Access, correction, deletion; restriction of automated decisions	No automated decisions on individuals; rights-request procedure	Procedure; request log
Sensitive and children's data (PDPA 2026)	Enhanced protection; restrictions on profiling minors	P3 data excluded by charter	Charter; data inventory
Data custody and exchange (NDMA 2026)	Custodian duties; approved data exchange	Data-sharing agreements; use of NRDEX where available	Signed agreements
Source code (NDMA 2026)	National source code repository for government software	Open-source code; mirror to national repository when available	Repository links
Security duties (Cyber Security Act 2026)	Protection of systems and data; incident handling	NIST CSF 2.0 controls; incident plan	Security reviews; drills
Harmful AI-generated content (2026 amendment)	Removal of harmful synthetic content	No generation of synthetic media; research protocols for P6	ERRC approvals
Proactive disclosure (RTI Act 2009)	Publication of public information	Open publications; transparency register	Published register
Draft national AI policy	Risk categories; impact assessment; oversight	Tier system aligned; impact assessments for T2 and T3	Assessments; register

Note. Requirements summarise press and secondary descriptions [84, 85, 86, 87, 88] and must be confirmed against the gazetted texts by the legal review (Section 14.3).

14.5 Right to Information and Proactive Disclosure

The Right to Information Act, 2009 gives citizens a right to information held by public authorities and encourages proactive disclosure [105]. BAPIL supports this principle in two ways. First, by making already-published information easier to find and cite, it strengthens the practical value of proactive disclosure. Second, BAPIL itself practises openness: its transparency register, evaluation reports, ERRC decision summaries and annual financial summary

are published without waiting for requests. Where BAPIL holds information on behalf of a public authority under a data-sharing agreement, requests for that information are referred to the authority that owns it.

14.6 Intellectual Property and Licensing

Government publications are generally reusable for research and public-interest purposes, but reuse terms differ between sources and are not always stated. BAPIL records the reuse terms of every source (Table 24), seeks written confirmation from custodians for bulk use, and respects any restrictions. BAPIL's own outputs are licensed openly: documents and datasets under CC BY 4.0 and code under the MIT License. Benchmarks that contain extracts of official documents carry the source attribution required by the custodians.

Part V

Delivery and Long-Term Strategy

How BAPIL is delivered in its first 36 months, how it grows to 2041 and beyond, and how it is financed, monitored, protected against risk and held to account.

Chapter 15: Delivery Plan (36 Months)

15.1 Overview

The delivery plan covers the 36 months after the Phase 1 MoU takes effect, with January 2027 as an indicative start. It implements Phases 2 to 4 of the White Paper's five-year roadmap (2026-2030) in concrete terms and corresponds to horizons H1 and the first year of H2 in the strategic roadmap (Chapter 16).

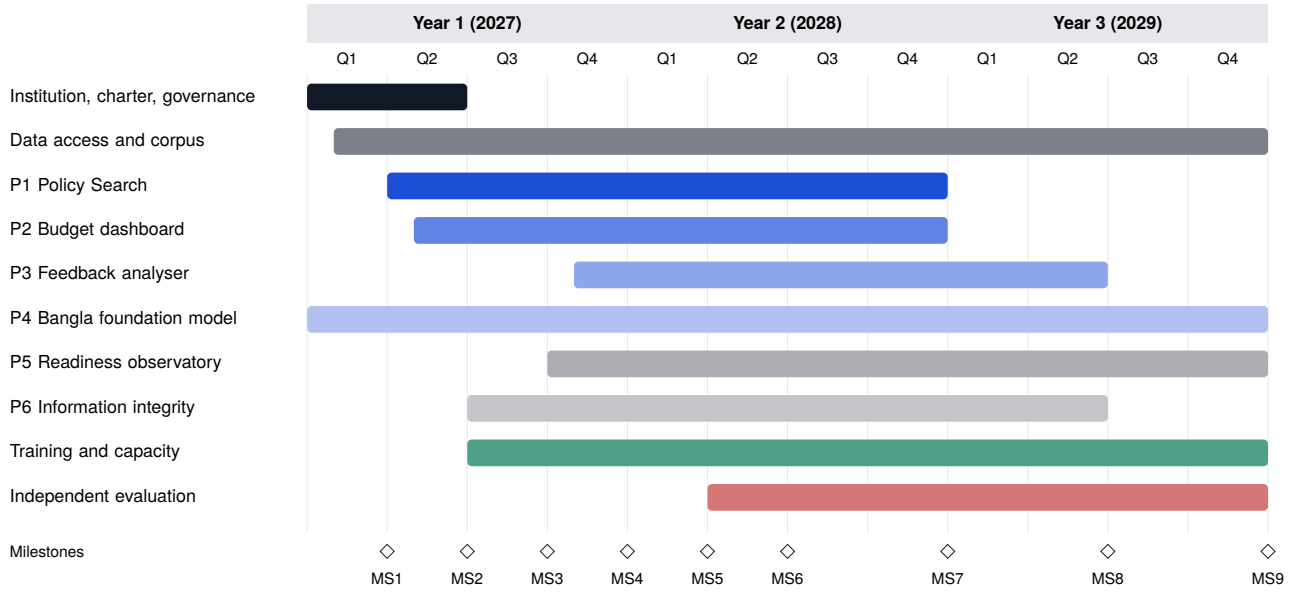


Figure 12. Indicative 36-month delivery roadmap (Month 1 is the month the MoU takes effect; indicative start January 2027).

15.2 Work Packages

The plan is organised in ten work packages, which correspond to the rows of Figure 12:

1. **Institution, charter and governance:** MoU, host agreement, governance bodies, charter, register and policies.
2. **Data access and corpus:** source connectors, ingestion, OCR correction, metadata, quality and data agreements.
3. **P1 Policy Search:** search and answer services, portal, user research, pilot and evaluation.
4. **P2 Budget dashboard:** extraction pipeline, open dataset, dashboard and methods note.
5. **P3 Feedback analyser:** DPIA, consent, de-identification, analysis and validation.
6. **P4 Bangla foundation model:** baselines, benchmark versions, Stage 1 decision and adaptation.
7. **P5 Readiness observatory:** indicator framework, data collection and annual reports.
8. **P6 Information integrity:** research protocol, method review and publication.
9. **Training and capacity:** programmes T1 to T6 and fellowships.
10. **Independent evaluation:** baseline study, mid-term and final evaluations.

15.3 Milestones and Deliverables

Table 44. Milestones and deliverables over 36 months

Milestone	Month	Achievement	Key deliverables	Gate
MS1	3	Host institution and government MoU signed; Steering Board, ERRC and TAB constituted; charter adopted	D1.1 Charter; D1.2 Risk and ethics procedures	G0
MS2	6	Feasibility and costing study complete; P0 data access confirmed; benchmark protocol agreed	D1.3 Feasibility study; D1.4 Data access plan	G1
MS3	9	P1 Policy Search alpha in internal use; BanglaPolicyQA v0 (500 questions)	D2.1 P1 alpha; D2.2 Benchmark v0	G2
MS4	12	P1 pilot with 30 or more users in two to three agencies; P2 dashboard v1 public; Year 1 report	D2.3 Pilot report; D2.4 Open budget dataset	G3
MS5	15	P3 consultation-analysis pilot after DPIA; P5 observatory framework	D3.1 DPIA; D3.2 Observatory framework	G2, G3
MS6	18	Independent mid-term evaluation; decision on BPFM Stage 1	D3.3 Mid-term evaluation; D3.4 Stage 1 decision	Review
MS7	24	Adoption decisions on P1 and P2 with host agencies; first observatory report; 300 officials trained	D3.5 Adoption memos; D3.6 Observatory report 1	G4
MS8	30	BPFM Stage 1 model with model card if approved; benchmark v1 (2,000 questions)	D4.1 Model and model card; D4.2 Benchmark v1	G2
MS9	36	Final evaluation; sustainability plan; Blueprint version 2.0	D4.3 Final evaluation; D4.4 Blueprint 2.0	Review

Note. Month 1 is the month the Phase 1 MoU takes effect; an indicative calendar start is January 2027.

15.4 The First 90, 180 and 365 Days

Days 1 to 90. Sign the MoU; appoint the Director; constitute the Steering Board, ERRC and TAB; adopt the charter; open the AI system register; publish the transparency register (initially listing P1, P2 and P4 at G0); start the legal review; begin ingestion of laws and budget documents.

Days 91 to 180. Complete the feasibility and costing study; sign data-access arrangements with the first custodians; agree the BanglaPolicyQA protocol and recruit annotators; complete G1 design reviews for P1 and P2; run the first training session for pilot users.

Days 181 to 365. Release the P1 alpha and benchmark v0; pass G2 and start the P1 pilot; publish the P2 open budget dataset and dashboard; publish the Year 1 report with evaluation results.

15.5 Phase Reviews

The Steering Board reviews progress at months 12, 24 and 36. An independent mid-term evaluation at month 18 assesses value for money, adoption and governance, and recommends whether to continue, adjust or stop each pilot. These reviews are the first three of the strategic review points in Table 48.

15.6 Reporting Calendar

Frequency	Report	Audience
Monthly	Pilot progress notes; demonstration to partner agencies	Director; host agencies
Quarterly	Progress against milestones and KPIs; risk register; financial report; ERRC decision summaries	Steering Board; public summary

Frequency	Report	Audience
Annually	Annual report in Bangla and English; transparency register update; independent audit of a sample of systems	Public; ICT Division; funders
Months 18 and 36	Independent evaluations	Public; Steering Board
At each gate	Gate evidence and decision	TAB; ERRC; Steering Board as relevant

15.7 Change Control

The plan will change as evidence arrives. Changes are handled at three levels. Minor changes within a work package (for example, re-ordering backlog items) are decided by the pilot lead. Changes to milestone dates of up to one quarter, or to the scope of a pilot, are decided by the Director and reported to the Steering Board. Changes that affect the budget by more than 10%, add or remove a pilot, or alter safeguards require a Steering Board decision and, for safeguards, ERRC consultation. Every change is logged with its reason, so that evaluators can see how the plan evolved.

15.8 Dependencies and Critical Path

The critical path runs through four dependencies: the MoU and host agreement (without which no staff can be hired); access to the core document collections (without which P1 cannot reach its corpus); the benchmark protocol and annotators (without which quality cannot be measured); and ERRC constitution (without which no T2 pilot can pass G2). Each has a fallback: interim hosting of the research team by a partner university; use of documents already online; a smaller initial benchmark; and an interim ERRC drawn from existing university ethics committees with external members.

Chapter 16: Strategic Roadmap 2026-2041 and Beyond

16.1 Why a Long-Term Roadmap

Pilots prove that something works; institutions make it last. A 36-month plan is the right unit for delivery, but decisions taken in the first years (on data standards, open publication, governance and institutional form) shape what is possible for decades. Bangladesh's national plans already look to 2041: the Perspective Plan 2041 and the ICT Masterplan 2041 set long-term goals for a knowledge-based economy and digital government [29, 30], and the draft national AI policy covers 2026 to 2030 [88]. This chapter connects BAPIL's short-term pilots to those horizons through a strategy that is ambitious in direction but cautious in commitment.

16.2 Strategic Principles for the Long Term

Five principles guide every horizon:

1. **Evidence before expansion.** Each horizon begins only when the signals of the previous horizon have been met. Targets are revised, not forced.
2. **Public goods first.** Data, benchmarks, methods and code remain open, so that value persists even if institutions change.
3. **Institution-neutral design.** Standards and tools are designed to work whether BAPIL remains a university lab, becomes a foundation, or is absorbed by government.
4. **Safeguards do not erode with scale.** Tiers, gates, the ERRC veto and prohibited uses apply at every horizon; growth that requires weaker safeguards is not pursued.
5. **Bangla and inclusion as long-term goals.** Each horizon raises the standard for Bangla-language quality, accessibility and reach beyond Dhaka.

16.3 The Horizons at a Glance

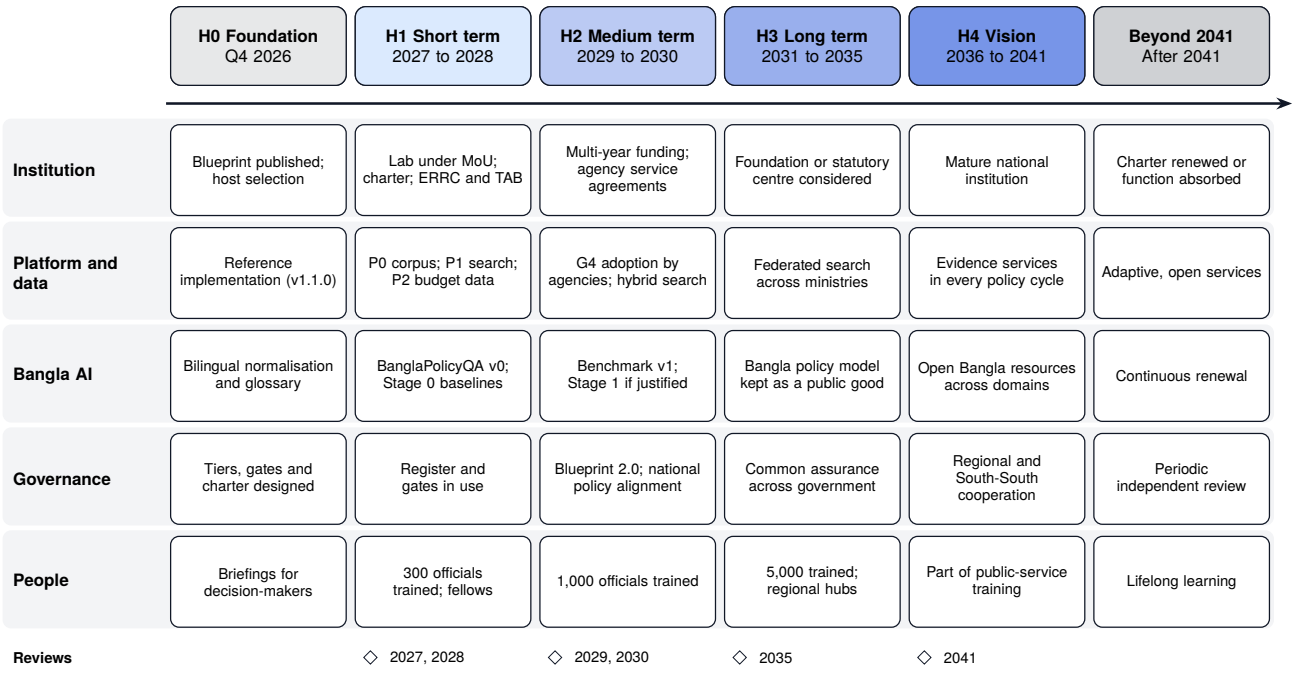


Figure 13. Strategic roadmap, 2026 to 2041 and beyond: horizons, workstreams and review points.

Table 45. Strategic horizons, 2026 to 2041 and beyond

Horizon	Period	Strategic aim	Priority actions	Institutional form	Signal to move on
H0 Foundation	Q4 2026	Prepare a safe, open start	Publish this Blueprint and code; brief the ICT Division and custodians; offer input to national AI policy drafting; select host institution	Independent research initiative	Decisions D1, D2 and D4 taken
H1 Short term	2027 to 2028	Prove value with public data	Charter, ERRC and register; P1 and P2 pilots; BanglaPolicyQA v0; 300 officials trained; mid-term evaluation	University-anchored lab under MoU (Option C)	P1 and P2 pass G3; positive mid-term evaluation
H2 Medium term	2029 to 2030	Adopt, scale and align with national policy	G4 adoption of P1 and P2 by agencies; BPFM Stage 1 if justified; benchmark v1; Blueprint 2.0; align with the national AI policy cycle ending 2030	Lab with multi-year core funding; agency service agreements	Tools in operational use; sustained funding agreed
H3 Long term	2031 to 2035	Build a national policy-intelligence network	Federated search across ministries and divisions; Bangla policy model maintained as a public good; standard assurance used across government; regional training hubs	Consider an independent public-interest foundation (Option D) or statutory centre	Assurance methods used beyond BAPIL; network in most ministries
H4 Vision	2036 to 2041	Trusted AI-enabled policy system for a knowledge economy	Evidence services embedded in policy cycles; open Bangla language resources across domains; regional and South-South cooperation; contribution to Perspective Plan 2041 and ICT Masterplan 2041 goals	Mature national institution with international partnerships	Independent review at 2041 confirms public value
Beyond 2041	After 2041	Keep the capability adaptive and accountable	Periodic renewal of mission and charter; sunset or merger if the function is fully absorbed by government; preserve open archives	Determined by the 2041 review	Charter renewal decision

Note. Horizons H1 and H2 correspond to the 36-month roadmap of Chapter 15 (indicative start January 2027). Later horizons are strategic intentions, reviewed at the points in Table 48.

16.4 Horizon 0: Foundation (Q4 2026)

The foundation horizon is already under way. Its purpose is to prepare a safe, open start. The Blueprint, the Summary, the Decision Memo and the reference implementation are published with persistent identifiers. Briefings are offered to the ICT Division, the data custodians and the national AI policy drafting team, and the Blueprint is offered as a technical input to that process. A host institution is identified through an open call. The horizon ends when decisions D1, D2 and D4 (Chapter 21) are taken.

16.5 Horizon 1: Short Term (2027 to 2028)

Strategic aim: prove value with public data and show that governance works.

The short term corresponds to the first two years of the delivery plan. BAPIL is established as a university-anchored lab under an MoU, adopts its charter and constitutes the ERRC and TAB. Policy search (P1) and the open budget

dataset (P2) are piloted and evaluated; BanglaPolicyQA v0 is released; the transparency register lists every system; and about 300 officials are trained by the end of 2028. The independent mid-term evaluation in mid-2028 decides whether BPFM Stage 1 is justified and how Phase 2 should be staffed.

Key results by the end of 2028: P1 and P2 have passed Gate G3; about 15,000 public documents are indexed with full provenance; at least three agencies have signed data or collaboration agreements; the Year 1 and mid-term evaluation reports are published.

Dependencies: an MoU and host agreement in 2027; access to already-public document collections; funding for Phase 1 and Phase 2.

16.6 Horizon 2: Medium Term (2029 to 2030)

Strategic aim: move from pilots to adoption, and align with the national AI policy cycle.

In the medium term, successful tools pass Gate G4 and are adopted by agencies as class O services under agency ownership, with BAPIL continuing to maintain benchmarks and evaluations. Hybrid retrieval is in production, BanglaPolicyQA v1 is released, and BPFM Stage 1 is completed if justified. The end of the 36-month plan in late 2029 brings the final evaluation, a sustainability plan and Blueprint 2.0. In 2030, the end of the draft national AI policy period [88], BAPIL contributes evaluation evidence and governance experience to the design of the next policy cycle.

Key results by the end of 2030: tools in operational use in at least two agencies; about 1,000 officials trained; ten agencies using BAPIL tools or methods; multi-year core funding agreed; the choice between continuing as a university lab and moving towards an independent foundation documented.

Dependencies: positive final evaluation; agency budgets for operating adopted tools; continued political sponsorship.

16.7 Horizon 3: Long Term (2031 to 2035)

Strategic aim: build a national policy-intelligence network on shared standards.

If the medium term succeeds, the long term extends the approach across government. A federated search connects document collections maintained by ministries and divisions, using the same metadata, quality and assurance standards. A maintained Bangla policy model, if evaluation supports it, is kept as a public good available to agencies and researchers. BAPIL's risk-tiering, gate and register methods, refined through practice, are offered as a common assurance approach for public-sector AI, in coordination with the authority that oversees the national AI policy. Regional training hubs at universities outside Dhaka multiply the capacity programme, with a target of about 5,000 officials trained by 2035.

At this horizon the institutional form is reconsidered. Options include an independent public-interest foundation (Option D), a statutory centre attached to government, or a smaller BAPIL focused on research and evaluation while agencies operate the services. The choice is made at the 2035 strategic review against four criteria: independence, sustainability, policy uptake and public trust.

16.8 Horizon 4: Vision (2036 to 2041)

Strategic aim: a trusted, AI-enabled policy system that supports a knowledge-based economy.

By 2041, evidence services are embedded in policy cycles: analysts across government can find, cite and compare the authoritative text of laws, plans, budgets and statistics in Bangla and English as a matter of routine, and public-sector AI systems are registered, assessed and monitored using common methods. Open Bangla language resources cover many policy domains. Bangladesh shares its methods and benchmarks through regional and South-South cooperation, contributing to and learning from other countries that face similar language and resource challenges. An independent review in 2041 assesses whether BAPIL has delivered public value aligned with the Perspective Plan 2041 and the ICT Masterplan 2041 [29, 30].

16.9 Beyond 2041

No institution should assume its own permanence. After 2041 the charter is renewed only if the 2041 review confirms that BAPIL still adds value that government, universities or the market do not provide. If its functions have been fully absorbed by agencies, BAPIL should close or merge in an orderly way, with its open archives, benchmarks and code preserved on persistent repositories. If it continues, its mission is renewed for the next planning period with the same commitment to openness, independence and safeguards.

16.10 Indicative Long-Term Targets

Table 46. Indicative long-term targets by horizon

Indicator	2026	2028	2030	2035	2041
Public documents indexed with full provenance	0	15,000	30,000	100,000	All public instruments in force
Share of national laws in force searchable in Bangla and English with citations	0%	50%	90%	100%	100%, updated within 7 days
BanglaPolicyQA questions (all versions)	0	1,000	2,500	6,000	Multi-domain, maintained
Officials trained (cumulative)	0	300	1,000	5,000	Part of public-service training
Agencies using BAPIL tools or methods	0	3	10	30	Most ministries and divisions
Monthly active users of policy search	0	100	500	3,000	Routine use across government
Bangla and English Recall@10 gap (points)	n/a	5 or less	3 or less	2 or less	2 or less
AI systems listed in a public transparency register	0	6	15	Public-sector-wide register proposed	Standard practice
Independent evaluations published	0	1	3	8	Every 5 years at least
Open releases archived with a DOI	3	8	15	40	Continuous

Note. Planning targets, not commitments. They assume the baseline scenario of Table 47 and are reset at every strategic review.

The targets express the scale of ambition at each horizon. They are deliberately indicative: they assume the baseline scenario below, they will be revised at every strategic review, and they must never be pursued at the expense of quality or safeguards.

16.11 Alignment with National and Global Plans

The roadmap is aligned with four reference points:

- **Perspective Plan 2041 and ICT Masterplan 2041** [29, 30]: long-term goals for a knowledge-based economy, digital government and human capital, to which horizons H3 and H4 contribute directly.
- **National AI policy, 2026 to 2030** [88, 91]: risk categories, oversight, a Bangla-based national LLM, national compute and an innovation fund. BAPIL's tiers, benchmark and governed corpus are designed to support these elements, and horizon H2 ends with a contribution to the next policy cycle.
- **Data protection and data management laws, 2026** [85, 86]: the legal basis for public-data-first design and for data exchange through national infrastructure.

- **Sustainable Development Goals** [23]: in particular goals on quality education, industry and innovation, and peace, justice and strong institutions, through capacity building, open innovation and transparent, accountable public institutions.

16.12 Technology Outlook

AI technology is changing quickly, and a roadmap that fixed specific technologies to 2041 would soon be out of date. The roadmap therefore commits to capabilities (cited evidence, Bangla parity, assurance) rather than to particular models. Several changes can be anticipated: models will become more capable and cheaper to run; multilingual capability, including Bangla, will improve; speech and document understanding will make scanned and spoken sources more accessible; and AI systems that carry out multi-step tasks will raise new oversight questions. Each of these changes is handled through the same mechanisms: model re-selection on the benchmark (Table 29), architecture decision records, the tier and gate procedure for new uses, and the strategic reviews. The long-term investment that retains its value whatever happens to models is the governed corpus, the benchmark, the evaluation methods and the people trained to use them.

16.13 Scenarios

The future will not follow a single path. Four scenarios, plus a governance shock, test the robustness of the strategy.

Table 47. Strategic scenarios and responses

Scenario	Description	Early signals	Strategic response
Accelerated	Strong political sponsorship, national compute and data exchange arrive early	AI policy adopted with funding; NRDEX operational; agencies request tools	Bring forward H2 adoption; expand to more ministries; keep gates unchanged so that speed does not erode safeguards
Baseline	Steady progress with normal delays in data access and funding	Milestones met within about two quarters	Follow the roadmap; use reviews to adjust scope
Constrained	Funding cuts, slow data agreements or loss of sponsorship	Missed MS2 or MS4; budget delays; staff turnover	Protect the core (P1, benchmark, governance); pause P3, P5 and P6; seek research grants; publish what exists
Technology shift	Much stronger or cheaper models change what is possible	Open models exceed thresholds without adaptation; costs fall sharply	Re-run model selection (Table 29); skip BPFM stages that are no longer needed; invest in evaluation and data
Governance shock	A serious AI incident in Bangladesh or abroad reduces public trust	Media coverage; new restrictions; complaints	Pause affected systems; independent review; publish findings; strengthen controls; communicate openly

In every scenario, the core of BAPIL is protected: the governed corpus, policy search with citations, the benchmark and the governance system. These are relatively inexpensive to maintain and provide the base from which the strategy can be accelerated or rebuilt.

16.14 Strategic Reviews and Triggers

Table 48. Strategic review points and triggers

When	Review	Key questions	Possible decisions
End 2027	Year 1 review (MS4)	Did P1 and P2 deliver value with public data? Is governance working?	Continue, adjust scope or stop pilots; confirm Phase 2 budget

When	Review	Key questions	Possible decisions
Mid 2028	Independent mid-term evaluation (MS6)	Value for money; adoption; quality of assurance	BPFM Stage 1 decision; staffing for Phase 2
End 2029	Final evaluation of the 36-month plan (MS9)	Which tools should agencies own? Is the institutional form right?	G4 transfers; Blueprint 2.0; choice between Options C and D
2030	Alignment with the next national AI policy cycle	How should BAPIL support the post-2030 policy?	Revised mission and targets for H3
2035	Long-term strategic review	Is a national network justified? Are methods used beyond BAPIL?	Network expansion, consolidation or narrowing of scope
2041	Vision review	Has BAPIL delivered public value aligned with the 2041 plans?	Renew charter, merge into government or close with archives preserved
Any time	Event triggers	New law or rules; serious incident; major technology shift; loss of funding	Extraordinary Steering Board review within 30 days

Each review produces a short published report that compares results with targets, records the scenario that best describes conditions at the time, and sets or revises the targets for the next horizon. Event triggers ensure that the strategy also responds to the unexpected.

16.15 What Could Make the Strategy Fail

A long-term strategy should state how it could fail. Five failure modes are monitored at every review: **loss of independence**, if funding or political pressure shapes evaluation results; **pilot purgatory**, if tools work but are never adopted because agencies lack budgets or ownership; **erosion of safeguards**, if pressure to scale leads to weaker gates or wider data use; **technological lock-in**, if a single model or vendor becomes indispensable; and **talent drain**, if trained staff leave faster than new ones are developed. Each has an early-warning indicator in the monitoring framework and a response in the risk register (Chapter 19).

16.16 Sustainability

Long-term sustainability rests on four pillars. **Financial:** a growing share of core public funding and service agreements with agencies, with a declining dependence on project funding (Chapter 17). **Institutional:** an institutional form that can survive political and leadership change, with independent governance. **Technical:** open, documented, replaceable components and data that does not depend on any single vendor or model. **Human:** a national network of trained officials, fellows and researchers who carry the methods beyond BAPIL.

Chapter 17: Financing, Procurement and Sustainability

17.1 Costing Approach

The Blueprint now gives an indicative budget for the 36-month plan, built from explicit assumptions rather than from a lump-sum estimate. The costing model takes the staffing plan (Table 16), the capacity plan (Table 31), the benchmark and training targets, and the evaluation schedule, applies the unit costs in Table 49, and adds 10% for host overheads and contingency. The model is published in the repository (`blueprint/src/budget_model.py`), so that the feasibility and costing study (Deliverable D1.3, month 6) can replace any assumption with a quotation or an agreed salary scale and regenerate the budget. Until then, every figure in this chapter is a planning estimate, not a commitment of funds.

Table 49. Planning assumptions for the indicative budget

Item	Assumption	Basis
Exchange rate	Tk 124 per US dollar	Bangladesh Bank weighted average rate of Tk 123.65 on 19 October 2026 [110]
Core staff	9, 15 and 19 full-time equivalents in Years 1, 2 and 3	Staffing plan (Table 16)
Staff cost	Fully loaded monthly cost per FTE: Director Tk 3.0 lakh; ML and Bangla NLP engineers Tk 2.0 lakh; security engineer Tk 1.8 lakh; policy, data and platform staff Tk 1.5 lakh; governance and UX staff Tk 1.25 lakh; coordinator Tk 0.9 lakh; training officers Tk 0.8 lakh	Above the basic pay of the Ninth National Pay Scale 2026 (Tk 20,000 to Tk 156,000 per month) to compete for scarce AI skills [109]
Language editors and fellows	Part-time editors Tk 12 to 15 lakh per year; 6, 10 and 12 fellows at Tk 40,000 per month	Fellowship programme (Section 6.6)
Equipment	General-purpose servers USD 12,000 each; GPU inference servers (L40S class) USD 25,000 each; storage USD 6,000 to 8,000 per year of growth	Single L40S card about USD 7,500 in 2025, plus chassis, memory and storage [111]
Hosting and operations	Tk 25 lakh rising to Tk 45 lakh per year	Data-centre space, power, bandwidth, licences and a development cloud
Model adaptation compute	Provided in kind from shared national compute	Table 31; draft national AI policy [88]
Benchmark annotation	2,000 questions by month 30 at Tk 2,500 each, including adjudication	BanglaPolicyQA plan (Section 9.4, Annex K)
Training	500 officials by month 36 at Tk 15,000 per participant, plus specialist courses	Training curriculum (Table 18)
Evaluation and assurance	Mid-term evaluation Tk 20 lakh; final evaluation Tk 25 lakh; annual penetration test and red team	Chapters 13 and 18
Overheads and contingency	10% of all other costs	To be confirmed against the host institution's policy

Note. Planning assumptions for the costing study (Deliverable D1.3), not quotations. The model is published as blueprint/src/budget_model.py so that any assumption can be changed and the budget regenerated.

17.2 Indicative 36-Month Budget

Table 50. Indicative 36-month budget (Tk crore)

Category	Year 1	Year 2	Year 3	Total	Share
Core staff	1.82	2.88	3.55	8.25	45%
Language editors and fellowships	0.41	0.63	0.73	1.76	10%

Category	Year 1	Year 2	Year 3	Total	Share
Compute and hosting	1.39	1.37	0.75	3.51	19%
Data and annotation	0.38	0.44	0.35	1.17	6%
Training and outreach	0.23	0.42	0.42	1.07	6%
Governance, audit and evaluation	0.28	0.36	0.41	1.05	6%
Overheads and contingency (10%)	0.45	0.61	0.62	1.68	9%
Total	**4.96**	**6.71**	**6.83**	**18.49**	**100%**

Note. Computed from the assumptions in Table 49 by blueprint/src/budget_model.py. Totals may differ slightly because of rounding. The total is equivalent to about USD 1.49 million, or under one tenth of the reported Tk 192.66 crore cost of the Bangladesh AI Hub project [35].

The indicative budget is about Tk 18.5 crore over three years: Tk 5.0 crore in Year 1, Tk 6.7 crore in Year 2 and Tk 6.8 crore in Year 3. Four features of the budget matter for decision-makers:

- **People are the main cost.** Core staff, language editors and fellows account for about 55% of the total. The largest single driver is the salary needed to recruit and keep machine learning and Bangla NLP engineers.
- **Equipment is front-loaded.** Compute and hosting take about 19% of the total, but most of it falls in Years 1 and 2 when servers and GPUs are bought. Model adaptation (BPFM Stage 1) assumes in-kind access to shared national compute; if that is not available, a separate allocation would be needed.
- **Assurance is funded explicitly.** About 6% goes to governance, audits and independent evaluation. These costs are protected in the annual plan so that safeguards are not cut when budgets are tight.
- **Scale relative to national programmes.** The total is under one tenth of the reported cost of the Bangladesh AI Hub project (Tk 192.66 crore) [35], and well below the Tk 200 to 250 crore AI Innovation Fund proposed in the draft national AI policy [88].

The budget is most sensitive to staff costs and the exchange rate. A 20% change in core staff costs changes the total by about Tk 1.8 crore; a 10% change in the exchange rate changes it by only about Tk 0.3 crore, because only equipment is priced in US dollars.

17.3 Budget Beyond 36 Months

Beyond the 36-month plan, budgets are set at each strategic review (Table 48), and Blueprint 2.0 will contain a costed plan for horizon H2. As a planning range, a lab that maintains the Year 3 team, the benchmark and the evaluation function would need about Tk 7 to 9 crore a year in 2029 and 2030. A national network in horizon H3, with regional training hubs and support for federated search, would need more, but much of the operating cost of adopted tools should then sit in the budgets of the agencies that use them, under service agreements, rather than in BAPIL's own budget.

17.4 Funding Model

Funding should be diversified, as set out in Appendix C of the White Paper: a government contribution under the MoU, research grants, development-partner support for specific pilots, and industry support for open benchmarks, all under conflict-of-interest rules that protect research independence. Over time, the mix should shift towards stable public funding and service agreements with agencies that use BAPIL's tools.

Table 51. Indicative funding mix by horizon

Source	H1 2027 to 2028	H2 2029 to 2030	H3 2031 to 2035	H4 2036 to 2041
Government core contribution (MoU, development budget)	30 to 40%	40 to 50%	45 to 55%	50 to 60%
Competitive research grants	20 to 30%	15 to 25%	10 to 20%	10 to 15%

Source	H1 2027 to 2028	H2 2029 to 2030	H3 2031 to 2035	H4 2036 to 2041
Development partners (pilot-specific)	20 to 30%	15 to 20%	5 to 10%	0 to 5%
Service agreements with agencies (cost recovery)	0%	5 to 10%	15 to 25%	20 to 30%
Industry and philanthropy (open benchmarks, fellowships)	5 to 10%	5 to 10%	5 to 10%	5 to 10%

Note. No single non-government funder should provide more than 30% of the budget in any year, and industry support never buys influence over research results or governance decisions.

17.5 Value for Money

Value for money is assessed at the mid-term and final evaluations using four measures: economy (cost of inputs compared with benchmarks), efficiency (cost per output, such as cost per indexed document or per trained official), effectiveness (time saved and quality of evidence for users) and equity (reach beyond Dhaka and to under-served groups). Time saved by analysts is the most direct benefit, but it should not be overstated. Valuing an analyst's time at about Tk 500 an hour (based on grade 9 basic pay of Tk 44,000 a month plus allowances) [109], 300 users who each save four hours a month save time worth about Tk 0.7 crore a year, which is roughly a tenth of the Year 3 budget. Time savings alone therefore justify the cost only at the scale of the long-term targets (thousands of regular users, Table 46). In the short term, the case rests as much on the quality of evidence, the open benchmark and the governance methods as on time saved. The time-to-evidence studies in P1 provide the data to test this calculation.

17.6 Procuring AI Products and Services

Where BAPIL or a partner agency buys AI products or services, contracts should include clauses that protect data, enable evaluation and avoid lock-in. The model clauses below can be adapted for any public procurement of AI.

Table 52. Model clauses for procuring AI products and services

Clause	Requirement	Purpose
Data residency and ownership	Government data stays in Bangladesh and remains government property; no use for vendor model training	Protect data and sovereignty (NFR-02)
Right to evaluate	Buyer may test the product on BanglaPolicyQA and red-team tests before and after award	Decisions based on evidence, not claims
Bangla performance acceptance	Acceptance requires meeting agreed Bangla and English thresholds	Bangla as a first-class language
Documentation	Model cards, datasheets, SBOM and known limitations delivered with each version	Transparency and audit
Portability and exit	Open formats; export of all data and configurations; transition support at contract end	Avoid lock-in (R10)
Source code	Open source where possible; otherwise escrow and the right to audit	Long-term maintainability
Security and incidents	Compliance with Table 32 controls; vulnerability disclosure; incident notification within 24 hours	Security and accountability
Change control	Notice and re-evaluation before any model or major dependency change	Stable, assessed behaviour (G5)
Accessibility	WCAG 2.2 level AA conformance with evidence	Inclusion (NFR-05)
Integrity	Declaration of conflicts of interest; no influence over evaluation or governance	Independence

17.7 Open Innovation and the Startup Ecosystem

The White Paper emphasised collaboration with Bangladesh's startup ecosystem. BAPIL supports it in three ways that do not compromise independence: by publishing open data, benchmarks and code that startups can build on; by running open challenges on well-defined problems (for example, Bangla OCR correction or table extraction from budget documents) with transparent evaluation; and by offering a clear, fair route for startups to provide services under the procurement clauses above. BAPIL does not endorse products and does not give any company privileged access to data.

17.8 Financial Governance

The host institution manages BAPIL's funds under its audited financial procedures, with a separate cost centre. The annual plan and budget are approved by the Steering Board, financial reports are presented quarterly, and an annual summary of income by source and expenditure by category is published.

Chapter 18: Monitoring, Evaluation and Learning

18.1 Theory of Change

BAPIL's theory of change is straightforward: **if** officials and researchers can find authoritative evidence faster and more reliably, in Bangla and English, through tools that are transparent and governed, **then** policy analysis becomes better grounded and more efficient, and trust in public-sector AI grows.

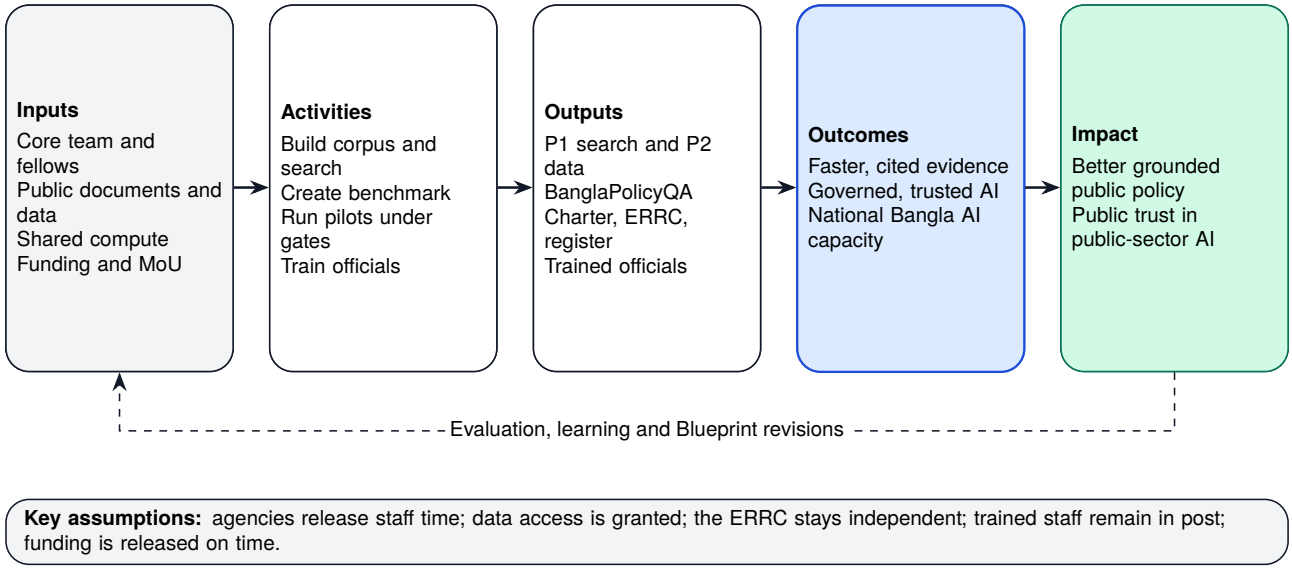


Figure 14. Theory of change: from inputs to impact, with key assumptions and a learning loop.

18.2 Logical Framework

The logical framework sets out the results chain with indicators, means of verification and assumptions for each level.

Table 53. Logical framework (36-month programme)

Level	Narrative summary	Indicators	Means of verification	Assumptions
Impact	Public policy in Bangladesh is better grounded in evidence, and AI in the public sector is trusted and well governed	Official documents citing BAPIL evidence; trust measures in user surveys	Document analysis; independent evaluation	Political commitment to evidence-based policy continues
Outcome 1	Officials and researchers find and cite authoritative evidence faster and more reliably	Time saved on evidence search; monthly active users	Time-to-evidence studies; usage logs	Agencies release staff time for pilots
Outcome 2	AI systems in BAPIL's scope are governed transparently and safely	100% registered; serious incidents handled in time	Transparency register; incident records	ERRC remains independent and resourced
Outcome 3	National capacity in Bangla policy AI grows	Officials trained; fellows; open datasets reused	Training records; download and citation counts	Trained staff stay in relevant posts
Output 1	Policy search and budget dataset piloted with users	Pilots passing G3; Recall@10	Gate records; benchmark reports	Data access granted on time

Level	Narrative summary	Indicators	Means of verification	Assumptions
Output 2	Open benchmark, corpus and evaluation methods published	Benchmark questions; releases with DOIs	Zenodo and GitHub records	Annotators available
Output 3	Governance system operational	Charter adopted; ERRC decisions published	Board minutes; register	Members nominated on time
Output 4	Officials and specialists trained	Officials trained by programme	Attendance and assessment records	Agencies nominate participants
Activities	Workplan of Chapter 15 and the work packages of the pilots	Milestones MS1 to MS9	Progress reports	Funding released as planned

18.3 Key Performance Indicators

Table 54. Monitoring and evaluation framework: key performance indicators

Level	Indicator	Baseline	Month 12	Month 36
Input	Core staff in post (FTE)	0	9	19
Input	Public documents indexed with full provenance	0	5,000	25,000
Output	BanglaPolicyQA benchmark questions released	0	500	2,000
Output	Pilots that passed Gate G3	0	2	5
Output	Open releases archived with a DOI (including this Blueprint)	3	5	10
Output	Officials trained in responsible AI and the tools	0	100	500
Outcome	Monthly active pilot users	0	30	300
Outcome	Median time saved on evidence search, user-reported	Baseline study	30%	50%
Outcome	Agencies with signed data or collaboration agreements	0	3	10
Quality	Recall@10 on Bangla and English questions	Not yet measured	0.85	0.90
Governance	Pilots listed in the transparency register with model cards	Not applicable	100%	100%
Governance	Serious incidents (all handled within agreed time limits)	Not applicable	0	0
Impact	Official documents or policy briefs citing BAPIL evidence	0	5	30

The indicators at months 12 and 36 are consistent with the long-term targets in Table 46, which extend them to 2041.

18.4 Methods

- **Benchmark evaluation** for quality indicators, on frozen benchmark versions.
- **Time-to-evidence studies** comparing matched tasks with and without the tools.
- **User surveys and interviews** each quarter during pilots.
- **Document analysis** to identify official documents and policy briefs that cite BAPIL evidence.
- **Independent evaluation** at months 18 and 36 by evaluators without a stake in BAPIL, and at each long-term review.
- **Usage statistics** from the audit logs, reported only in aggregate; individual users' queries are never reported.

18.5 Evaluation Questions

The independent evaluations at months 18 and 36, and the long-term reviews, use the OECD DAC evaluation criteria of relevance, coherence, effectiveness, efficiency, impact and sustainability [112], adapted to BAPIL:

- **Relevance:** do the pilots address real policy needs, as judged by host agencies and users?
- **Coherence:** does BAPIL complement the national AI policy, the AI Hub and other programmes, or duplicate them?
- **Effectiveness:** did the pilots meet their exit criteria, and did users find and cite evidence faster and more reliably?
- **Efficiency:** were resources used economically, and how do costs per output compare with alternatives?
- **Impact:** is there evidence that policy documents are better grounded, and that trust in public-sector AI has grown?
- **Sustainability:** will adopted tools, open datasets and trained people continue to deliver value without project funding?

Each evaluation also asks a governance question: did the tiers, gates and ERRC work as designed, and were any safeguards bypassed?

18.6 Learning and Publication

Evaluation reports are published with DOIs, including negative results. Each pilot ends with a short "what we learned" note covering what worked, what failed and what should change in the next Blueprint version. Lessons feed directly into the strategic reviews (Table 48) and into revisions of this Blueprint.

Chapter 19: Risk Management

19.1 Approach

The risk register is owned by the Director, reviewed quarterly by the Steering Board, and published in summary. Each risk has a likelihood and impact rating, mitigations and an owner. The heat map below shows the eighteen registered risks, including long-horizon risks such as technology obsolescence and climate-related disruption.

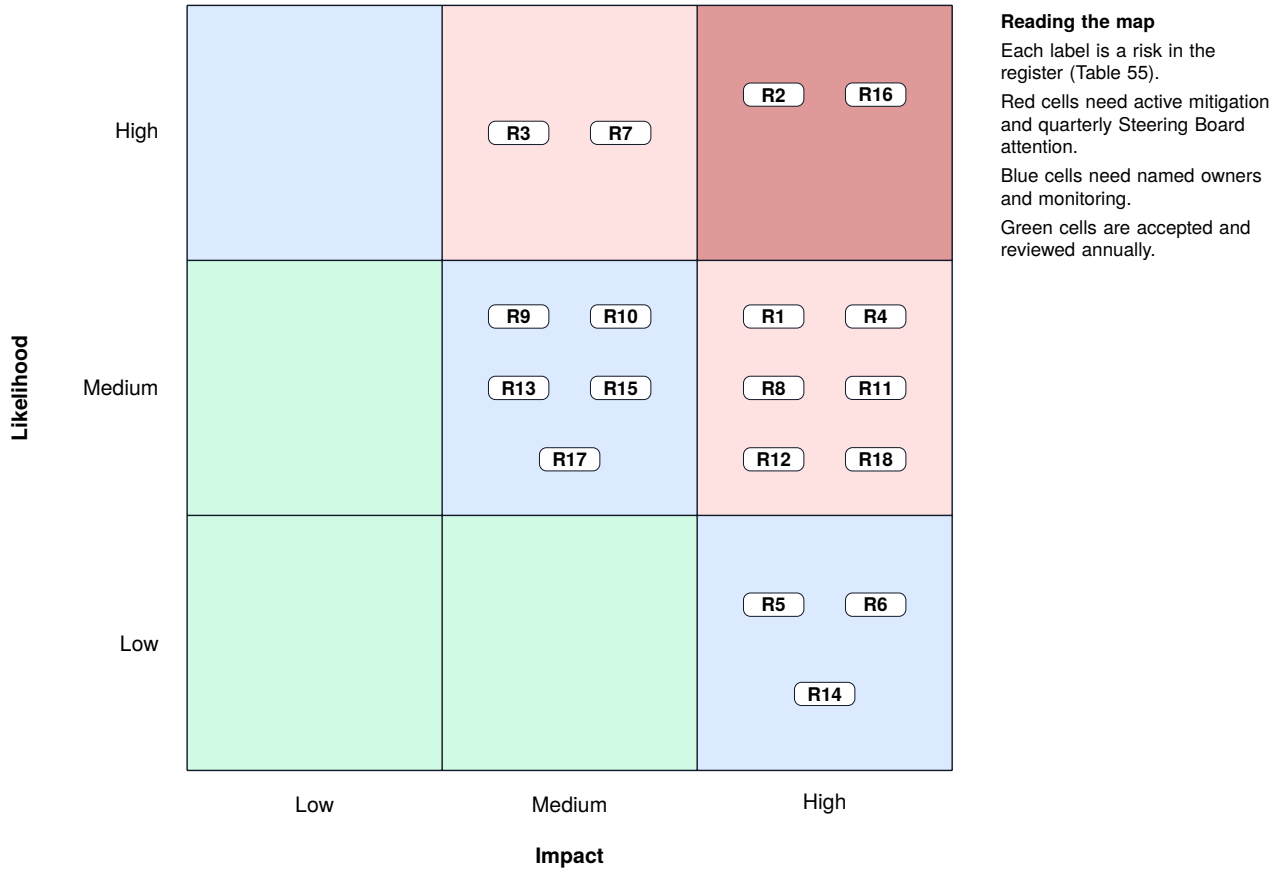


Figure 15. Risk heat map of the eighteen registered risks by likelihood and impact.

19.2 Risk Register

Table 55. Risk register

ID	Risk	Likelihood	Impact	Mitigation	Owner
R1	Political or institutional change shifts priorities	M	H	Institution-neutral design; university host; alignment with the national AI policy process; cross-party briefings	Director, Steering Board
R2	Delays in data access, including bulk document export	H	H	Start with documents already public; NRDEX agreements; RTI requests as fallback	Data lead
R3	Poor text quality from scanned Bangla PDFs	H	M	OCR evaluation; human correction loops; prioritise born-digital sources	ML lead

ID	Risk	Likelihood	Impact	Mitigation	Owner
R4	Incorrect answers mislead officials	M	H	Extractive-first answers; mandatory citations; abstention; training; classes R and A only	ML lead, ERRC
R5	Privacy breach or misuse of personal data	L	H	P0-first; DPIA; access control; independent audits	Security lead, governance analyst
R6	Mission creep toward surveillance	L	H	T0 list in charter; ERRC veto; public transparency register	ERRC
R7	Talent shortage or loss of staff	H	M	Fellowships; diaspora partnerships; competitive terms; remote collaboration	Director
R8	Funding gaps between phases	M	H	Diversified funding (White Paper Appendix C); phased commitments	Steering Board
R9	Duplication with the AI Hub or other initiatives	M	M	MoU on division of labour; shared benchmarks and data	Director
R10	Vendor or model lock-in	M	M	Open standards; open-weight models; portable containers; code in national repository	Platform lead
R11	Cyber attack on the platform	M	H	NIST CSF 2.0 controls; penetration tests; offline backups and restore drills	Security lead
R12	Low adoption by officials	M	H	Co-design with users; training; measure time saved; product owners in agencies	Policy lead
R13	Legal uncertainty while implementing rules are issued	M	M	Legal review; conservative data scope; update procedures as rules are published	Governance analyst
R14	Tools or outputs misused for misinformation or partisan purposes	L	H	Access control for official tools; no synthetic media generation; usage policy; audit logs; public register	ERRC
R15	Benchmark contamination or overfitting to the test set	M	M	Private held-out test split; versioned freezing; periodic refresh; report on several sets	ML lead
R16	Dependence on a few key people	H	H	Documentation and open code; deputies for key roles; succession plan; shared ownership with the host	Steering Board
R17	Technology obsolescence over the long horizon	M	M	Modular design and ADRs; model re-selection at reviews; investment in data and evaluation rather than one model	TAB
R18	Floods, cyclones, power or network outages affecting hosting	M	H	Disaster-recovery site; offline backups; recovery targets of Table 35; low-bandwidth design	Platform lead

Note. L = low, M = medium, H = high. The register is reviewed quarterly by the Steering Board.

19.3 Priority Risks

Four risks deserve particular attention. **Data access (R2)** is the most likely cause of delay; starting with already-published documents means that useful work can begin even if formal agreements take time. **Key-person dependency (R16)** is a serious risk for any small, new institution; documentation, deputies and shared ownership reduce it. **Incorrect answers (R4)** are the most serious technical risk; extractive answering, mandatory citations, abstention and class A use reduce it to a level that can be managed through training and monitoring. **Mission**

creep (R6) is unlikely but would be very damaging to public trust; the charter's prohibited-use list, the ERRC veto and the public transparency register are designed to make it visible and preventable.

19.4 Risk Appetite

BAPIL's risk appetite differs by area. It has a **low appetite** for risks to privacy, rights and public trust, which are controlled by strict boundaries and the ERRC veto. It has a **moderate appetite** for delivery and technical risks, accepting that some pilots will fail and treating failure as information. It has a **higher appetite** for research risk, since exploring methods that may not work is part of its mission, provided the research is safe and its results are published.

19.5 Escalation

Risks whose rating rises to high likelihood and high impact are escalated to the Steering Board within one month. Any risk involving potential harm to individuals or rights is escalated to the ERRC immediately. Event triggers in Table 48 lead to an extraordinary review of the register.

Chapter 20: Research Agenda and Knowledge Outputs

20.1 Research as a Core Function

BAPIL is a research institution as well as a builder of tools. Its research agenda serves three purposes: to answer questions that matter for the design of its own systems; to contribute open knowledge on Bangla language technology and public-sector AI governance; and to build national research capacity. Research questions are chosen where BAPIL's position (access to real policy tasks, a governed corpus and a functioning governance system) allows it to produce evidence that others cannot.

20.2 Research Questions

Table 56. Research agenda and planned knowledge outputs

ID	Research question	Method	Planned output	Timing
RQ1	How well do lexical, dense and hybrid retrieval perform on Bangla and English policy text, and what closes the gap between the two languages?	Benchmark experiments on BanglaPolicyQA; ablations of normalisation and glossary expansion	Benchmark and methods paper; open code and data	2027 to 2028
RQ2	Can abstention and citation verification be calibrated so that officials see few unsupported claims without losing useful answers?	Threshold studies; human judgement of faithfulness	Evaluation paper; calibration guidance	2027 to 2029
RQ3	Do cited-search tools change the speed and quality of policy analysis?	Matched-task studies with pilot users; interviews	Empirical study of time-to-evidence and quality	2028 to 2029
RQ4	How can risk tiering, gates and transparency registers be made to work in a public administration with new data laws?	Design science and case study of BAPIL governance	Peer-reviewed journal article on the Blueprint framework; policy brief	Manuscript 2027
RQ5	How do OCR quality and Unicode normalisation affect downstream retrieval in Bangla?	Controlled experiments on scanned and born-digital documents	Dataset of corrected text with datasheet; short paper	2028
RQ6	What institutional form sustains a public-interest AI policy lab in a lower-middle-income country?	Comparative institutional analysis; evaluation evidence	Institutional review for Blueprint 2.0	2029

Note. Outputs are published open access with DOIs. Journal choice follows scope, open-access options and peer-review quality; no outcome of peer review is assumed.

20.3 Publication Plan

Research outputs are published open access with DOIs: datasets and benchmarks on Zenodo with datasheets; code on GitHub; evaluation reports in the BAPIL series; and peer-reviewed articles in journals and conferences chosen for their scope, open-access options and quality of review. A first journal manuscript, based on this Blueprint and its reference implementation, is planned for 2027; it will present the framework, the design rationale and the demonstration results, and will be submitted to a peer-reviewed journal in the field of AI and public policy or digital government. The outcome of peer review is not assumed in any plan.

20.4 Research Ethics

All research involving people (surveys, interviews, user studies) is approved by the host institution's research ethics committee and, where relevant, the ERRC. Participation is voluntary and informed, data is minimised and de-identified, and results are reported in aggregate. Research on information integrity follows the specific protocol

required for pilot P6.

Chapter 21: Decisions Requested and Next Steps

21.1 Decisions

Table 57. Decisions requested from the relevant authorities

No.	Decision	Responsible authority	Timing
D1	Accept this Blueprint as a technical input to the national AI policy drafting process	Ministry of Posts, Telecommunications and Information Technology (ICT Division)	Q4 2026
D2	Approve a 12-month Phase 1 pilot with a named government focal point and an MoU with a host institution	ICT Division and host institution	Within 90 days
D3	Authorise structured bulk access to already-public document collections and nominate data custodians	Legislative and Parliamentary Affairs Division, Finance Division, Planning Commission, BBS, a2i	Within 120 days
D4	Nominate members to the Steering Board and an independent Ethics and Risk Review Committee	ICT Division, academia, civil society	Within 90 days
D5	Commission the feasibility and costing study that sets the Phase 2 budget	ICT Division or a development partner	By month 6
D6	Confirm guardrails: no deployment beyond class A without a G4 decision, and no use in prohibited categories	Steering Board	At charter adoption

21.2 What Happens After the Decisions

If decisions D1, D2 and D4 are taken, the first 90-day plan in Section 15.4 can start immediately. Decision D3 unlocks the full P1 corpus; without it, P1 can still begin on documents already downloadable from official websites. Decision D5 sets the budget for Phase 2. Decision D6 ensures that the safeguards in this Blueprint are binding from the first day.

21.3 Immediate Next Steps (Q4 2026 to Q1 2027)

While the decisions are considered, the following steps can proceed without commitment of public funds:

- publication of this Blueprint, the Summary, the Decision Memo and the reference implementation with persistent identifiers;
- briefings for the ICT Division, the data custodians and the national AI policy drafting team, with the Blueprint offered as a technical input;
- expert roundtables and a public comment period, with a published comment log feeding into Blueprint 1.1 (Table 8);
- preparation of the host-institution call, the draft charter and the terms of reference for consultation (Annexes E to G);
- preparation of a peer-reviewed journal manuscript presenting the framework and its reference implementation (Chapter 20).

21.4 Conclusion

Bangladesh has the data foundations, the institutions and the talent to build a trustworthy AI policy-intelligence capability, and it is now putting in place the legal framework and national AI policy within which such a capability must operate. This Blueprint shows how to start small, safely and openly: with public documents, a bilingual search and answer service that always shows its sources, a governance system with real decision rights, and pilots that must prove their value before they grow. It also shows how that start can grow, horizon by horizon, into a national capability that supports the 2041 goals, without ever relaxing the safeguards that make it trustworthy.

The reference implementation shows that the core ideas already work. The next step belongs to the authorities and partners who can decide to test them on real policy questions.

Annex A: Requirements Traceability Matrix

Table 58. Requirements traceability matrix

Requirement	Summary	Component	Code module	Verification
FR-01	Bilingual search	C2, C4	bapil.text, bapil.search	test_text, test_search
FR-02	Cited answers and abstention	C5	bapil.answer	test_answer
FR-03	Provenance at ingestion	C2, C3	bapil.corpus	test_corpus
FR-04	Dashboards	C6, C7	Specified only	Pilot P2 acceptance
FR-05	Aggregate feedback analysis	C6	Specified only	Pilot P3 DPIA and acceptance
FR-06	AI system register	C12	bapil.registry	test_registry
FR-07	Risk-tier assessment	C12	bapil.risk	test_risk
FR-08	Evaluation harness	C11	bapil.evaluate	test_evaluate
FR-09	Transparency register	C12	bapil.registry (export)	test_registry
FR-10	Versioned API	C8	bapil.api	test_api
NFR-01	Language parity	C2, C4	bapil.text glossary and normaliser	test_text; benchmark
NFR-07	Reproducibility	All	CI workflow; no external dependencies	GitHub Actions
NFR-08	Explainability	C5	bapil.answer (sources and scores)	test_answer
GR-02	Gate before build	C12	bapil.risk	test_risk
GR-04	Tamper-evident logs	C10	bapil.audit	test_audit
GR-06	Prohibited uses	C12	bapil.risk (T0 screen)	test_risk

Requirements not listed here are verified through pilot acceptance criteria (Table 38), security testing (Chapter 10) or governance review (Chapter 13).

Annex B: Risk-Tier Questionnaire

B.1 Prohibited-Use Screening

Answer *yes* or *no*. Any *yes* places the system in tier T0, and work stops.

1. Does the system score or rank individuals for general social trustworthiness or behaviour (social scoring)?
2. Does it perform mass or indiscriminate biometric identification or surveillance?
3. Does it infer or profile individuals' political opinions, religious beliefs or similar protected characteristics?
4. Does it generate deceptive synthetic media depicting real people?
5. Does it use manipulative techniques that exploit vulnerabilities such as age, disability or economic hardship?

B.2 Scored Dimensions

Table 59. Risk-tier questionnaire: scored dimensions (Annex B)

Code	Dimension	Score 0	Score 1	Score 2	Score 3
D1	Influence on decisions	Research only	Informs analysts	Informs official decisions on policies or programmes	Informs or automates decisions about specific people or entities
D2	Data sensitivity	P0 public	P1 internal, non-personal	P2 personal	P3 sensitive personal or restricted
D3	People potentially affected	None or internal	Fewer than 10,000	10,000 to 1 million	More than 1 million
D4	Reversibility of harm	Trivially reversible	Reversible with effort	Hard to reverse	Irreversible
D5	Degree of automation	Suggestions on request	Person reviews every output	Person reviews samples	Fully automated
D6	Vulnerability of affected groups	No affected individuals	General public	Includes vulnerable groups	Targets vulnerable groups
D7	Exposure	Internal researchers	Officials	Public information	Public guidance with consequences

Note. Tier rules: any prohibited-use flag gives T0. Otherwise T3 if total at least 13, or D1 = 3, or D2 = 3, or both D2 and D3 at least 2; T2 if total at least 6, or D1 = 2, or D7 at least 2; else T1. Implemented in bapil.risk.

B.3 Tiering Rules

1. Any *yes* in B.1 gives **T0 Prohibited**.
2. Otherwise, the system is **T3 High** if the total score is 13 or more, or D1 = 3, or D2 = 3, or both D2 and D3 are 2 or more.
3. Otherwise, the system is **T2 Limited** if the total score is 6 or more, or D1 = 2, or D7 is 2 or more.
4. Otherwise, the system is **T1 Low**.

The rules are implemented in `bapil.risk` and tested against the examples in `implementation/examples/risk/`, including a prohibited social-scoring example (T0) and a benefit-eligibility example (T3).

B.4 Worked Example: Pilot P1 in Class A Use

Scores: D1 = 2 (informs official decisions on policies), D2 = 0 (public documents), D3 = 0 (no individuals affected), D4 = 1, D5 = 1 (person reviews every output), D6 = 0 (no individuals affected), D7 = 1 (officials). Total =

5. No high-risk condition applies; because D1 = 2, the result is **T2 Limited**, requiring ERRC review, red-team testing, human review of every output and user training. The same system used only by BAPIL researchers (D1 = 1, D7 = 0) scores 3 and is **T1 Low**. This calculation is reproduced by `python -m bapil.cli assess examples/risk/risk_p1_policy_search.json`.

Annex C: Templates

C.1 AI System Record (summary of fields)

Field	Description
`system_id`	Unique identifier, for example `BAPIL-P1`
`name`	Name in English and Bangla
`purpose`	What the system is for and what it is not for
`owner`	Accountable person and unit; host-agency product owner if any
`maturity_class`	R, A or O
`risk_tier`	T0 to T3, with the date and reference of the latest assessment
`data_classes`	Data classes used (P0 to P3)
`models`	Models and versions, with links to model cards
`evaluation`	Benchmark version and headline results
`gate_status`	Latest gate passed and date
`oversight`	Human oversight patterns applied (Table 40)
`limitations`	Known limitations and failure modes
`contact`	How to report a problem or complaint

The full JSON schema is `implementation/schemas/ai_system_record.schema.json`.

C.2 Model Card (headings)

Model details; intended use and out-of-scope uses; training and adaptation data; evaluation data and results by language; fairness and robustness analysis; limitations and risks; ethical considerations; environmental and compute cost; versioning and contact [99].

C.3 Datasheet for a Dataset (headings)

Motivation (why the dataset was created and by whom); composition (what it contains, data classes, languages, known gaps); collection process (sources, dates, reuse terms); preprocessing (extraction, OCR, normalisation, de-identification); uses (intended and not recommended); distribution (licence, DOI); maintenance (owner, update schedule, how errors are reported) [100].

C.4 Incident Report (headings)

Reference number; date and time; system and version; reporter (optional); description; affected users or groups; severity (Table 34); immediate action; root cause; corrective action; ERRC review; public summary.

C.5 Evaluation Report (headings)

System and version; benchmark version and split; metrics by language and document type (Table 28); comparison with the previous release and with thresholds; error analysis with examples; red-team and security results; known limitations; recommendation for the next gate.

Annex D: Publication, Versioning and DOI Policy

1. **Separate records per publication.** The White Paper, Executive Summary and this Blueprint each have their own Zenodo record and DOI. The Blueprint record contains the English and Bangla Blueprint, Summary and Decision Memo, in PDF and Markdown, plus figures and tables in LaTeX, PDF and PNG.
2. **Related identifiers.** The Blueprint record declares that it *is supplement to* the White Paper (10.5281/zenodo.21363967), *references* the Executive Summary (10.5281/zenodo.21441172) and *is supplemented by* the GitHub repository release that contains the reference implementation.
3. **DOI reservation.** The DOI is reserved on Zenodo before publication so that it can be printed in the documents; the PDFs are then rebuilt and uploaded.
4. **Repository releases.** Repository releases use semantic versioning. Release v1.1.0 adds the Blueprint and the reference implementation.
5. **New versions.** Corrections produce a new version of the same Zenodo record (for example 1.0.1); substantive revisions produce Blueprint 2.0 (Milestone MS9).
6. **Licences.** Documents are CC BY 4.0; code is MIT.
7. **Language precedence.** The English edition is authoritative. Where the Bangla edition differs in interpretation, the English edition prevails.

Annex E: Draft BAPIL Charter

This draft is offered for adoption by the Steering Board at Milestone MS1, after consultation (Table 8) and legal review (Section 14.3).

Article 1. Name and status. The Bangladesh AI Policy & Innovation Lab (BAPIL) is a research, policy-intelligence and capacity-building initiative hosted by [host institution] under a Memorandum of Understanding with the ICT Division.

Article 2. Mission. BAPIL helps Bangladesh use artificial intelligence for evidence-based policymaking, responsible AI governance, Bangla-language technology and public-sector innovation, through open research, tools, methods and training.

Article 3. Boundaries. BAPIL does not make policy, regulate, perform operational security functions or operate production government systems. It works with public data first, provides advice and not decisions, and never builds or supports surveillance.

Article 4. Prohibited uses. BAPIL will not build, pilot, evaluate for deployment or otherwise support any system for social scoring; mass or indiscriminate biometric identification or surveillance; profiling of political opinions, religious beliefs or similar protected characteristics; deceptive synthetic media depicting real people; or manipulation that exploits vulnerabilities.

Article 5. Governance. BAPIL is governed by a Steering Board, an independent Ethics and Risk Review Committee (ERRC) and a Technical Architecture Board, with the composition and powers set out in their terms of reference (Annexes F and G).

Article 6. The ERRC veto. No system of tier T2 or above may pass Gates G2 or G4 without the approval of the ERRC. ERRC decisions to refuse or suspend are binding and can be reversed only by the ERRC itself.

Article 7. Registration and transparency. Every AI system is registered before work begins. A public transparency register lists each system's purpose, owner, tier, maturity class, data classes, evaluation summary, limitations and complaint contact.

Article 8. Openness. Code, benchmarks, evaluation reports and non-sensitive data are published openly with persistent identifiers. Exceptions for security or privacy are recorded in writing and reviewed annually.

Article 9. Data protection. BAPIL complies with the Personal Data Protection Act, 2026, the National Data Management Act, 2026 and related rules, and applies the data classification and safeguards of this Blueprint, which may be stricter than the law requires.

Article 10. Independence and integrity. Researchers decide methods and publish results. No funder may control research results or governance decisions. Conflicts of interest are declared and managed.

Article 11. Transfer and withdrawal. Tools that pass Gate G4 are transferred to agency ownership with documentation and support. BAPIL withdraws from any activity that conflicts with this charter.

Article 12. Review and amendment. The charter is reviewed at each strategic review (Table 48). Amendments require a two-thirds majority of the Steering Board after consultation with the ERRC; Articles 3, 4 and 6 can be weakened only after public consultation.

Annex F: Terms of Reference of the Ethics and Risk Review Committee

Purpose. To provide independent review of the ethical, legal and societal risks of BAPIL's AI systems and research, and to hold a binding veto over tier T2 and T3 work.

Composition. Five to seven members, a majority external to BAPIL and the host institution, covering law and human rights, data protection, ethics, the relevant policy domains, Bangla language and civil society, with gender balance. The chair is an external member. Members serve three-year terms, renewable once.

Responsibilities.

- Review tier T2 and T3 systems at Gates G1, G2 and G4, and any reclassification of P2 data.
- Review serious incidents and decide on suspension.
- Approve research protocols involving people or sensitive topics, including P6.
- Advise the Steering Board on charter amendments and on the risk register.
- Publish summaries of its decisions within thirty days.

Powers. The ERRC may approve, approve with conditions, refuse or suspend. Refusal and suspension are binding. The ERRC may commission independent expert advice within its budget.

Procedure. The ERRC meets at least quarterly and as needed for gate reviews. A quorum is a majority of members including at least two external members. Decisions are taken by consensus where possible and otherwise by majority; dissenting opinions are recorded. Members declare conflicts and withdraw from related decisions.

Resources. The ERRC has a secretariat provided by the governance analyst and a budget for honoraria and expert advice, protected in the annual plan.

Annex G: Terms of Reference of the Steering Board and the Technical Architecture Board

G.1 Steering Board

Purpose. To set BAPIL's strategy, approve its plans and budgets, and hold its leadership to account.

Composition. Seven to nine members: the ICT Division (co-chair), the host institution (co-chair), the Planning Commission, a data custodian agency, an academic member from another university, a civil-society member and an industry member, with at least one member from outside Dhaka and with gender balance. The Director attends without a vote.

Responsibilities. Adopt the charter; approve the annual plan and budget; approve partnerships and senior appointments; take G4 adoption decisions jointly with host agencies; review the risk register quarterly; conduct the strategic reviews in Table 48.

Procedure. Meets quarterly; quorum of a majority including both co-chairs or their delegates; minutes published in summary.

G.2 Technical Architecture Board

Purpose. To ensure that BAPIL's systems are well designed, secure, interoperable and maintainable.

Composition. BAPIL's lead engineers, the security lead and two external technical reviewers.

Responsibilities. Set architecture and security standards; approve designs at G1; approve models and major dependencies; maintain architecture decision records; review security test results.

Procedure. Meets monthly and as needed for design reviews; decisions recorded as ADRs in the repository.

Annex H: Data Protection Impact Assessment Template

A DPIA is required before any processing of P2 data and is recommended for any new use of P1 data.

1. **Description of processing.** Purpose; data subjects; categories of data; sources; recipients; retention; systems used.
2. **Lawful basis.** Basis under the Personal Data Protection Act, 2026; consent wording and records where consent is the basis.
3. **Necessity and proportionality.** Why the data is needed; why less sensitive data would not work; minimisation measures.
4. **Rights of data subjects.** How access, correction, deletion and objection are handled; confirmation that no automated decisions are made about individuals.
5. **Risks to individuals.** Re-identification, disclosure, misuse, discrimination, chilling effects; likelihood and severity of each.
6. **Measures.** De-identification method; aggregation thresholds; access control; encryption; retention and deletion; staff training.
7. **Residual risk and decision.** Remaining risks after measures; recommendation; approval by the governance analyst and the ERRC; review date.
8. **Consultation.** Views of the data custodian, the data protection officer of the host institution and, where appropriate, representatives of data subjects.

Annex I: Outline of a Data-Sharing Agreement

1. **Parties and contacts.**
2. **Purpose.** The specific research or pilot purpose; prohibition of other uses.
3. **Data description.** Datasets, fields, period, format, data class (Table 22).
4. **Legal basis.** Applicable provisions of the 2026 Acts and any rules; use of the national data exchange where available.
5. **Security.** Controls required (Table 32), hosting in Bangladesh, access list.
6. **Outputs.** Permitted outputs (for example, aggregate results only), review of outputs before publication, attribution.
7. **Retention and deletion.** Retention period (Table 26), deletion method and certificate.
8. **Audit and reporting.** Custodian's right to audit; incident notification within 24 hours.
9. **Intellectual property and licensing.** Ownership of source data and of derived outputs; open licensing of BAPIL outputs where permitted.
10. **Term, review and termination.**
11. **Transparency.** Listing of the agreement, without confidential details, in the transparency register.

Annex J: Gate Checklists

G0 Register and screen

- AI system record created in the register.
- Prohibited-use screening completed; no *yes* answers.
- Tier assessment completed (Annex B) and recorded.
- Owner and product owner named.

G1 Design review

- Data plan with data classes and sources; reuse terms recorded.
- DPIA approved where P2 data is involved.
- Architecture reviewed and approved by the TAB; threat model includes AI-specific risks (Table 33).
- Evaluation plan with benchmark version and thresholds.
- ERRC review completed for tier T2 and T3.

G2 Pre-pilot readiness

- Benchmark results meet the thresholds in Table 28.
- Red-team and security tests passed; findings resolved or accepted.
- Model card and datasheets published.
- User training materials ready; incident procedure in place.
- ERRC approval for tier T2.

G3 Pilot

- Pilot users enrolled and trained; human oversight patterns applied (Table 40).
- Monitoring and audit logging active; incidents reviewed monthly.
- User feedback and time-to-evidence data collected.
- Pilot report with results against exit criteria.

G4 Adoption decision

- Independent evaluation completed and published.
- Host agency acceptance; agency legal review completed.
- Operating, support and funding arrangements agreed; transfer plan approved.
- Joint decision by the Steering Board and the host agency; ERRC approval for tier T2.

G5 Update or retire

- Change assessed; re-evaluation completed on the current benchmark.
- Register and model card updated.
- For retirement: users informed, data retained or deleted per schedule, archive created.

Annex K: BanglaPolicyQA Annotation Guidelines (Summary)

1. **Question writing.** Questions are written by domain experts from real analyst tasks, in natural Bangla or English, including some romanised Bangla and variant spellings. Each question should have a clear answer in the corpus, except for a designated share (about 10%) of unanswerable questions.
2. **Gold passages.** A second annotator, who did not write the question, finds the passage or passages that answer it and records their identifiers. A passage is gold only if it alone supports the answer.
3. **Graded relevance.** Passages are marked 2 (fully answers), 1 (partly answers or provides necessary context) or 0 (not relevant).
4. **Adjudication.** Disagreements are resolved by a third annotator; agreement rates are reported in the datasheet.
5. **Balance.** The question set is balanced across languages, document types (laws, plans, budgets, statistics) and issuing bodies, and covers different periods.
6. **Quality control.** A random 10% of items is re-annotated each version; items with errors are corrected and logged.
7. **Ethics and licensing.** Annotators are paid fairly and credited; questions contain no personal data; the benchmark is released under CC BY 4.0 with source attribution.
8. **Splits and freezing.** Each version has a public development split and a private test split; versions are frozen and archived with a DOI.

Annex L: Glossary

Term	Meaning in this Blueprint
Abstention	The system's refusal to answer when the evidence is insufficient, with the closest sources and the reason shown
Adoption decision (G4)	The decision to transfer a tool to agency ownership as a class O system, or to stop
AI system register	The internal record of every AI system, from which the public transparency register is generated
BanglaPolicyQA	BAPIL's open benchmark of policy questions in Bangla and English with gold passages
Citation precision	The share of cited passages that actually support the statement they are attached to
Corpus version	An identifier of the exact set of documents and passages used for a result, enabling reproduction
Data class	The sensitivity class of data, from P0 (public) to P3 (sensitive personal data)
Extractive answer	An answer made of sentences selected from source passages, each with its citation
Gate	A documented decision point (G0 to G5) that a system must pass to move forward
Horizon	A period of the strategic roadmap (H0 to H4, and beyond 2041)
Maturity class	The status of a system: R (research), A (advisory) or O (operational, agency-owned)
Passage	A short section of a document, the unit of search and citation
Product owner	A named official in a host agency who defines needs and accepts or rejects a pilot
Risk tier	The level of risk of a system, from T0 (prohibited) to T3 (high), which sets the required controls
Transparency register	The public list of BAPIL's AI systems with their purpose, tier, evaluation and limitations

Annex M: Walkthrough of the Reference Implementation

This annex shows, step by step, what the reference implementation does with a question. It uses the two small test documents in `implementation/tests/fixtures/` (a sample policy note in English and its Bangla counterpart), so that the results are short, stable and can be reproduced exactly by anyone with the repository.

M.1 A Cited Answer in English

```
python -m bapil.cli ask --corpus tests/fixtures --log build/audit.jsonl \
  "When is a data protection impact assessment required?"
```

```
Q: When is a data protection impact assessment required?
Confidence: high
1. A data protection impact assessment is required before any processing
   of personal data at scale. [policy_en#p1]
2. বৃহৎ পরিসরে ব্যক্তিগত উপাত্ত প্রক্রিয়াকরণের আগে প্রভাব মূল্যায়ন আবশ্যিক। [policy_bn#p1]
3. Personal data must be processed only with consent. [policy_en#p1]
Sources:
  - [policy_en#p1] Sample Policy Note > Data Protection
  - [policy_bn#p1] নমুনা নীতি-নোট > উপাত্ত সুরক্ষা
```

Three features are visible. Every sentence carries the identifier of the passage it was taken from. The second sentence comes from the Bangla document, retrieved for an English question because the bilingual glossary links *data protection* with উপাত্ত সুরক্ষা and *impact assessment* with প্রভাব মূল্যায়ন. And the answer is extractive: each sentence appears word for word in the cited source.

M.2 A Cited Answer in Bangla

```
Q: বাজেট নথি কখন প্রকাশিত হয়?
Confidence: medium
1. প্রতি অর্থবছরে বাজেট নথি প্রকাশিত হয় যাতে নাগরিকেরা সরকারি ব্যয়
   অনুসরণ করতে পারেন। [policy_bn#p2]
2. জাতীয় বাজেটে মন্ত্রণালয়সমূহের জন্য অর্থ বরাদ্দ করা হয়। [policy_bn#p2]
3. ২০২৬ সালের বাজেটে নতুন বরাদ্দ রয়েছে। [policy_bn#p2]
Sources:
  - [policy_bn#p2] নমুনা নীতি-নোট > বাজেট স্বচ্ছতা
  - [policy_en#p2] Sample Policy Note > Budget Transparency
```

The Bangla question is answered from the Bangla passage, and the English passage on the same topic is listed among the sources. Confidence is reported as a band (high, medium or low) based on the evidence score and the share of query terms covered, not as a single number that could suggest false precision.

M.3 Abstention

```
Q: What is the boiling point of water in Kelvin?
ABSTAINED (no_matching_passages). Closest sources:
```

The question is outside the corpus, no passage matches, and the system says so instead of producing an answer. In the larger demonstration corpus, the system abstained correctly on all six out-of-scope questions (Section 11.5).

M.4 The Machine-Readable Response

The same answer can be returned as JSON (option `--json`, or the `/v1/answer` endpoint of the API). An extract:

```
{
 "question": "When is a data protection impact assessment required?",
 "abstained": false,
```

```
 "confidence": "high",
 "evidence": {"top_score": 7.881, "coverage": 1.0},
 "answer": [
  {"text": "A data protection impact assessment is required before ...",
   "citation": {"pid": "policy_en#p1", "doc_id": "policy_en",
                "heading": "Sample Policy Note > Data Protection"}},
  ...
 ],
 "sources": [
  {"pid": "policy_en#p1", "lang": "en", "score": 7.8812,
   "matched_terms": ["assessment", "data", "impact", "protection", "required"],
   "provenance": {"path": "tests/fixtures/policy_en.md",
                  "sha256": "b85002b0b751...", "data_class": "P0"}},
  ...
 ],
 "corpus_version": "3c204e06a617"
}
```

The response carries the evidence used to decide whether to answer, the terms that matched, the provenance of each source including its checksum and data class, and the corpus version. Together these allow any answer to be reproduced and audited later.

M.5 The Audit Log

Each answer is written to an append-only log in which every record contains the hash of the previous one:

```
{"hash":"760a77ff...","payload":{"abstained":false,
 "pids":["policy_en#p1","policy_bn#p1","policy_en#p1"],
 "q":"When is a data protection impact assessment required?"},
 "prev":"00000000...","ts":"...","type":"answer"}
```

```
python -m bapil.cli verify-log build/audit.jsonl
# build/audit.jsonl: 3 entries, chain intact
```

If any record is edited, deleted or reordered, verification fails and reports the first broken link. This is the mechanism behind requirement GR-04 and user story US-08.

M.6 A Risk-Tier Assessment

```
python -m bapil.cli assess examples/risk/risk_p1_policy_search.json
```

```
BAPIL-P1: T2 Limited (total 5)
  reason:   D1 = 2: informs official decisions on policies or programmes
  control:  All T1 controls
  control:  ERRC review
  control:  Red-team test
  control:  Human review of every output in use
  control:  User training
  control:  Incident procedure
  control:  Quarterly monitoring report
  approver: ERRC and Steering Board
```

The result reproduces the worked example in Annex B.4: the tier, the reason, the controls that become mandatory and the bodies that must approve.

Annex N: Contribution to the National AI Policy Elements

The draft National AI Policy 2026-2030 proposed several elements [88]. The table shows how BAPIL's design could support each of them if they are retained in the final policy.

Draft policy element	How BAPIL can contribute	Where in this Blueprint
Risk categories (prohibited, high, limited, low)	Operational tiering method with a questionnaire, rules and tested code	Chapter 13; Annex B
Algorithmic impact assessment for high-risk uses	Impact assessment template and worked examples; DPIA template	Annexes B and H
Independent oversight committee	Tested model of an ethics and risk committee with a binding veto and publication duty	Chapter 5; Annex F
Labelling of AI-generated content	Labelling and verification rules for generative features	Section 13.8
Bangla-based national LLM	Open benchmark, governed corpus, evaluation protocol and an evidence-gated adaptation path	Chapter 9
Centralised national compute	Modest pilot footprint and use of shared compute for adaptation	Chapter 10
AI Innovation Fund	Open challenges and procurement clauses that keep markets open and fair	Chapter 17
Skills and AI literacy	Competency framework, six training programmes and fellowships	Chapter 6
Public-sector AI adoption	Gate-based path from pilot to agency ownership; transparency register	Chapters 12 and 13

Annex O: Indicative First-Year Work Plan for Pilot P1

Months	Activities	Outputs	Gate
1 to 2	Appoint pilot lead and product owners; interview 15 to 20 analysts; register the system	User research summary; AI system record	G0
2 to 4	Harvest laws, plans and budget documents; extraction and OCR; metadata and quality checks	Corpus release 0.1 with quality report	
3 to 5	Design review; threat model; evaluation plan; draft user interface in Bangla and English	Approved design; evaluation plan	G1
4 to 7	Build search and answer services; bilingual glossary; portal; audit logging	Internal alpha	
5 to 8	Write and annotate BanglaPolicyQA v0 questions; baseline evaluation	Benchmark v0 (500 questions); baseline report	
8 to 9	Red-team and security tests; fix findings; train pilot users; ERRC review	Readiness evidence; model card	G2
9 to 12	Pilot with 30 or more analysts in two to three agencies; monthly problem reviews; time-to-evidence study	Pilot report; user survey; Year 1 results	G3

References

Reference numbers [1] to [83] follow the bibliography of the BAPIL White Paper (Version 1.0); new references begin at [84]. Only references cited in this document are listed.

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Comments, corrections and proposals for collaboration are welcome through the repository's issue tracker.